

Exercises**Level - 1****(Problems Based on Fundamentals)****ABC of Integration**

1. Evaluate : $\int \log_x x dx$
2. Evaluate : $\int \left(3^{\log_x 2} - 2^{\log_x 3} \right)$
3. Evaluate : $\int \left(x^m + m^x + m^m + \frac{m}{x} \right) dx$
4. Evaluate : $\int 2^x \cdot 3^x \cdot dx$
5. Evaluate : $\int \tan^2 x dx$
6. Evaluate : $\int \cot^2 x dx$
7. Evaluate : $\int \frac{dx}{\sin^2 x \cos^2 x}$

8. Evaluate: $\int \left(1 + \tan\left(x + \frac{3\pi}{8}\right)\right) \left(1 + \tan\left(\frac{\pi}{8} - x\right)\right) dx$

9. Evaluate: $\int (\tan x + \cot x)^2 dx$

10. Evaluate: $\int \frac{dx}{1 + \cos^2 x}$

11. Evaluate: $\int (3^{\log_5 x} - 2^{\log_5 x}) dx$

12. Evaluate: $\int \left(1 + \tan\left(\frac{\pi}{8} - x\right)\right) \left(1 + \tan\left(\frac{\pi}{8} + x\right)\right) dx$

13. Evaluate: $\int \left(\frac{8^{1+x} + 4^{1+x}}{2^{2x}}\right) dx$

14. Evaluate: $\int \left(\frac{x}{m} + \frac{m}{x} + x^m + m^x\right) dx$

15. Evaluate: $\int \frac{(a^x + b^x)^2}{a^x b^x} dx$

16. Evaluate: $\int \frac{(2^x + 3^x)^2}{2^x \cdot 3^x dx}$

17. If $f'(x) = \frac{1}{x} + \frac{1}{\sqrt{1-x^2}}$ and $f(1) = \frac{\pi}{2}$, find $f(x)$.

18. If $f'(x) = a \cos x + b \sin x$ and

$f'(0) = 4, f''(0) = 3, f\left(\frac{\pi}{2}\right) = 5$, find $f(x)$.

Type 1

19. Evaluate: $\int \frac{dx}{1 - \sin x}$

20. Evaluate: $\int \left(\frac{\sin^4 x + \cos^4 x}{\sin^2 x \cos^2 x}\right) dx$

21. Evaluate: $\int \left(\frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x}\right) dx$

22. Evaluate: $\int \left(\frac{\cos 2x - \cos \alpha}{\cos x - \cos \alpha}\right) dx$

23. Evaluate: $\int \left(\frac{\cos^4 x - \sin^4 x}{\sqrt{1 + \cos 4x}}\right) dx$

24. Evaluate: $\int \left(\frac{1 + \tan^2 x}{1 + \cot^2 x}\right) dx$

25. Evaluate: $\int \left(\frac{\cos x - \cos 2x}{1 - \cos x}\right) dx$

26. Evaluate: $\int \left(\frac{\sqrt{x^4 + x^{-4} + 2}}{x^3}\right) dx$

27. Evaluate: $\int \left(\frac{5\cos^3 x + 3\sin^3 x}{\cos^2 x \sin^2 x}\right) dx$

28. Evaluate: $\int \left(\frac{\cos x - \sin x}{\cos x + \sin x}\right) (1 + \sin 2x) dx$

29. Evaluate: $\int \left(\frac{\cos 5x + \cos 4x}{1 + 2\cos 3x}\right) dx$

30. Evaluate: $\int \left(\frac{\cos x - \cos 2x}{1 - \cos x}\right) dx$

31. Evaluate: $\int \frac{dx}{(\tan x + \cot x + \sec x + \cos x)}$

Type 2

32. Evaluate: $\int \frac{x}{x+1} dx$

33. Evaluate: $\int \frac{(1+x)^2}{x(1+x^2)} dx$

34. Evaluate: $\int \frac{x^2 - 2}{x^2 + 1} dx$

35. Evaluate: $\int \frac{x-1}{(x^{2/3} + x^{1/3} + 1)} dx$

36. Evaluate: $\int \left(\frac{x^4 + 2}{x^2 + 2}\right) dx$

37. Evaluate: $\int \left(\frac{x^4 - 3}{x^2 + 1}\right) dx$

38. Evaluate: $\int \left(\frac{x^6 - 1}{x^2 + 1}\right) dx$

39. Evaluate: $\int \left(\frac{x^8 + x^4 + 1}{x^4 + x^2 + 1}\right) dx$

40. Evaluate: $\int \left(\frac{x^4}{x^2 + 1}\right) dx$

41. Evaluate: $\int \left(\frac{x^4 + x^2 + 1}{x^2 + x + 1}\right) dx$

42. Evaluate: $\int \left(\frac{x^6 + 1}{x^2 + 1}\right) dx$

Type 3

43. Evaluate: $\int \sin^{-1}(\sin x) dx$

44. Evaluate: $\int \sin^{-1}(\cos x) dx$

45. Evaluate: $\int \tan^{-1}\left(\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}\right) dx$

46. Evaluate: $\int \tan^{-1}\left(\frac{\sin 2x}{1 + \cos 2x}\right) dx$

47. Evaluate: $\int \tan^{-1}\left(\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}\right) dx$

48. Evaluate: $\int \tan^{-1}\left(\frac{\sin x}{1 - \cos x}\right) dx$

49. Evaluate: $\int \tan^{-1} \left(\sqrt{\frac{1 - \sin x}{1 + \sin x}} \right) dx$

50. Evaluate: $\int \tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right) dx$

51. Evaluate: $\int \tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right) dx$

52. Evaluate: $\int \tan^{-1} \left(\frac{1 - \sin x}{\cos x} \right) dx$

53. Evaluate: $\int \tan^{-1} \left(\frac{\sqrt{(1 + \sin x)} + \sqrt{(1 - \sin x)}}{\sqrt{(1 + \sin x)} - \sqrt{(1 - \sin x)}} \right) dx$

54. Evaluate: $\int \tan^{-1} (\sec x + \tan x) dx$

55. Evaluate: $\int \tan^{-1} \left(\frac{\sin 2x}{1 + \cos 2x} \right) dx$

Type 4

56. Evaluate: $\int (3x + 2) dx$

57. Evaluate: $\int \frac{dx}{2x - 3}$

58. Evaluate: $\int \frac{dx}{5 - 2x}$

59. Evaluate: $\int e^{ax+b} dx$

60. Evaluate: $\int 3^{4x+5} dx$

61. Evaluate: $\int \cos(5x + 3) dx$

62. Evaluate: $\int \sin 2x dx$

63. Evaluate: $\int \sqrt{3x + 2} dx$

64. Evaluate: $\int \frac{dx}{\sqrt{3x + 4}}$

Type 5

65. Evaluate: $\int \frac{dx}{\sqrt{x+2} - \sqrt{x+1}}$

66. Evaluate: $\int \frac{dx}{(\sqrt{2x+5} - \sqrt{2x+3})}$

67. Evaluate: $\int \frac{dx}{\sqrt{3x+4} - \sqrt{3x+1}}$

68. Evaluate: $\int \frac{dx}{\sqrt{2x+3} + \sqrt{2x-3}}$

69. Evaluate: $\int \frac{dx}{\sqrt{x+1} + \sqrt{x}}$

70. Evaluate: $\int \frac{dx}{\sqrt{x+a} + \sqrt{x+b}}$

71. Evaluate: $\int \frac{dx}{\sqrt{2x+2014} + \sqrt{2x+3013}}$

Type 6

72. Evaluate: $\int \frac{x}{\sqrt{x-1}} dx$

73. Evaluate: $\int \frac{\sqrt{x}}{x+1} dx$

74. Evaluate: $\int \frac{x}{\sqrt{3x+1}} dx$

75. Evaluate: $\int \frac{x+1}{\sqrt{2x-1}} dx$

76. Evaluate: $\int \frac{x-1}{\sqrt{x+4}} dx$

77. Evaluate: $\int \frac{x}{x^2+1} dx$

Type 7

78. Evaluate: $\int \frac{\cos x - \sin x}{\sin x + \cos x} dx$

79. Evaluate: $\int \frac{3 \cos x}{2 \sin x + 5} dx$

80. Evaluate: $\int \frac{\cos x - \sin x}{2 + \sin 2x} dx$

81. Evaluate: $\int \frac{xe^x + e^x}{\cos^2(xe^x)} dx$

82. Evaluate: $\int \frac{dx}{x(1 + \ln x)^2}$

83. Evaluate: $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx$

84. Evaluate: $\int \frac{dx}{1 + e^x}$

85. Evaluate: $\int \frac{dx}{x(x^3 + 1)}$

86. Evaluate: $\int \frac{dx}{x(x^4 + 1)}$

87. Evaluate: $\int \frac{dx}{x(x^5 - 1)}$

88. Evaluate: $\int \frac{\sin 2x}{\sin 5x \sin 3x} dx$

89. Evaluate: $\int \frac{dx}{\sin(x-a) \sin(x-b)}$

90. Evaluate: $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$

91. Evaluate: $\int \frac{dx}{1 + e^{-x}}$
92. Evaluate: $\int \frac{dx}{1 + e^x}$
93. Evaluate: $\int \frac{\sin 2x}{a \sin^2 x + b \cos^2 x} dx$
94. Evaluate: $\int \frac{\sin(x-a)}{\sin x} dx$
95. Evaluate: $\int \frac{\sin x}{\sin(x-a)} dx$
96. Evaluate: $\int \frac{\sin(x+a)}{\sin(x+b)} dx$
97. Evaluate: $\int \frac{dx}{\sqrt{x}(\sqrt{x}+1)}$
98. Evaluate: $\int \frac{1 + \tan x}{x + \log \sec x} dx$
99. Evaluate: $\int \frac{\sin 2x}{\sin 5x \cdot \sin 3x} dx$
100. Evaluate: $\int \frac{\cos x - \sin x}{1 + \sin 2x} dx$
101. Evaluate: $\int \frac{dx}{\sin x \cdot \cos^2 x}$
102. Evaluate: $\int \frac{dx}{\sin 2x \cdot \cos^2 x}$
103. Evaluate: $\int \frac{dx}{\sin(x-a) \sin(x-b)}$
104. Evaluate: $\int \frac{dx}{\cos(x-a) \cos(x-b)}$
105. Evaluate: $\int \frac{dx}{\sin(x-a) \sin(x-b)}$
106. Evaluate: $\int \frac{x^x(1 + \ln x)}{x^x + 1} dx$
107. Evaluate: $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx$
108. Evaluate: $\int \frac{\sin^3 x}{(\cos^4 x + 3\cos^2 x + 1)\tan^{-1}(\sec x + \cos x)} dx$

Type 8

109. Evaluate: $\int 3x^2 \sin(x^3) dx$
110. Evaluate: $\int \frac{(1 + \ln x)^3}{x} dx$
111. Evaluate: $\int \frac{dx}{x^2(1 + x^4)^{3/4}}$

112. Evaluate: $\int \frac{dx}{\sqrt{x}(4 + 3\sqrt{x})^2}$
113. Evaluate: $\int 3^{3^x} \cdot 3^{3^x} dx$
114. Evaluate: $\int \tan^3 x \cdot \sec^2 x dx$
115. Evaluate: $\int \sin^3 x \cdot \cos x dx$
116. Evaluate: $\int \frac{(\log x)^3}{x} dx$
117. Evaluate: $\int \frac{\sin x}{\sqrt{3 + 2\cos x}} dx$
118. Evaluate: $\int \frac{\sqrt{2 + \log x}}{x} dx$
119. Evaluate: $\int \frac{dx}{1 + \sqrt{x}}$
120. Evaluate: $\int x^3 \sin x^4 dx$
121. Evaluate: $\int 5^{5^x} \cdot 5^{5^x} \cdot 5^x dx$
122. Evaluate: $\int \frac{\sin x - \cos x}{e^x + \sin x} dx$
123. Evaluate: $\int \frac{dx}{x(1 + x^3)}$

Type 9**Case I:**

124. Evaluate: $\int \sin^3 x \cdot \cos^4 x dx$
125. Evaluate: $\int \sin x \cdot \cos^6 x dx$
126. Evaluate: $\int \sin^5 x \cdot \cos^9 x dx$

Case II:

127. Evaluate: $\int \sin^2 x \cdot \cos^3 x dx$
128. Evaluate: $\int \sin^4 x \cdot \cos^3 x dx$
129. Evaluate: $\int \sin^6 x \cdot \cos^5 x dx$

Case III:

130. Evaluate: $\int \sin^3 x \cdot \cos^3 x dx$
131. Evaluate: $\int \sin^5 x \cdot \cos^5 x dx$
132. Evaluate: $\int \sin^5 x \cdot \cos^7 x dx$

Case IV:

133. Evaluate: $\int \sin^2 x \cdot \cos^2 x dx$
134. Evaluate: $\int \sin^2 x \cdot \cos^4 x dx$

135. Evaluate: $\int \sin^4 x \cdot \cos^2 x \, dx$

Case V:

136. Evaluate: $\int \sin^3 x \, dx$

137. Evaluate: $\int \sin^5 x \, dx$

138. Evaluate: $\int \sin^7 x \, dx$

Case VI:

139. Evaluate: $\int \cos^5 x \, dx$

140. Evaluate: $\int \cos^3 x \, dx$

141. Evaluate: $\int \cos^7 x \, dx$

Case VII:

142. Evaluate: $\int \sin^4 x \, dx$

143. Evaluate: $\int \sin^2 x \, dx$

144. Evaluate: $\int \sin^6 x \, dx$

Case VIII:

145. Evaluate: $\int \cos^6 x \, dx$

146. Evaluate: $\int \cos^2 x \, dx$

147. Evaluate: $\int \cos^4 x \, dx$

Case IX:

148. Evaluate: $\int \frac{dx}{\sin^{1/2} x \cos^{3/2} x}$

149. Evaluate: $\int \frac{dx}{\sin^{3/2} x \cos^{5/2} x}$

150. Evaluate: $\int \frac{\sqrt{\tan x}}{\sin x \cos x} \, dx$

151. Evaluate: $\int \frac{\sin x}{\cos^5 x} \, dx$

152. Evaluate: $\int \frac{dx}{\sin^3 x \cos^5 x}$

153. Evaluate: $\int \frac{\sin^2 x \, dx}{\cos^6 x}$

154. Evaluate: $\int \frac{dx}{\sin x \cos^3 x}$

155. Evaluate: $\int \frac{dx}{\sin^2 x \cos^4 x}$

156. Evaluate: $\int \frac{dx}{\sin^{1/2} x \cdot \cos^{7/2} x}$

157. Evaluate: $\int \frac{dx}{\sqrt{\sin^3 x \cdot \cos^5 x}}$

ABC of Formula 6

158. Evaluate: $\int \frac{dx}{x^2 + 4}$

159. Evaluate: $\int \frac{dx}{9x^2 + 1}$

160. Evaluate: $\int \frac{dx}{x^2 - 4}$

161. Evaluate: $\int \frac{x^4 - 1}{x^2 + 5} \, dx$

162. Evaluate: $\int \frac{dx}{\sqrt{x^4 + 4}}$

163. Evaluate: $\int \frac{dx}{\sqrt{4x^2 + 1}}$

164. Evaluate: $\int \left(\frac{x+4}{x^3 + 4x} \right) dx$

165. Evaluate: $\int \left(\frac{x^4 + 1}{x^2 + 1} \right) dx$

166. Evaluate: $\int \frac{dx}{\sqrt{(2-x)^2 + 1}}$

167. Evaluate: $\int \frac{x+9}{x^3 + 9x} \, dx$

168. Evaluate: $\int \frac{1+x}{1+x^2} \, dx$

169. Evaluate: $\int \frac{1+x}{x^3 + x} \, dx$

170. Evaluate: $\int \frac{dx}{x^4 + 1}$

171. Evaluate: $\int \frac{dx}{x^3 + x}$

Type 10

172. Evaluate: $\int \frac{dx}{x^3 + 4x + 4}$

173. Evaluate: $\int \frac{dx}{x^2 + 6x + 10}$

174. Evaluate: $\int \frac{dx}{2x^2 + 5x + 6}$

175. Evaluate: $\int \frac{dx}{x^2 + x + 1}$

176. Evaluate: $\int \frac{dx}{1 + x + x^2}$

177. Evaluate: $\int \frac{dx}{x^2 + 4x + 3}$

178. Evaluate: $\int \frac{dx}{4x^2 + 7x + 10}$

179. Evaluate: $\int \frac{dx}{x^2 - 2ax}$

180. Evaluate: $\int \frac{dx}{x^2 + 2ax}$

181. Evaluate: $\int \frac{dx}{a^2 + 2ax}$

182. Evaluate: $\int \frac{dx}{2ax - x^2}$

183. Evaluate: $\int \frac{dx}{(x^2 + 1)(x^2 + 4)}$

184. Evaluate: $\int \frac{x^2}{x^6 + 1} dx$

185. Evaluate: $\int \frac{\cos x dx}{\sin^2 x + 3\sin x + 2}$

186. Evaluate: $\int \frac{x^x(1 + \log x)}{x^{2x} + x^x + 1} dx$

187. Evaluate: $\int \frac{dx}{x(x^4 + 1)}$

188. Evaluate: $\int \frac{x dx}{x^4 + x^2 + 1}$

189. Evaluate: $\int \frac{e^x dx}{e^{2x} + 6e^x + 5}$

190. Evaluate: $\int \frac{3x^5}{1 + x^{12}} dx$

191. Evaluate: $\int \frac{dx}{x(x^4 + 1)}$

192. Evaluate: $\int \frac{dx}{x(x^3 + 1)}$

193. Evaluate: $\int \frac{dx}{x(x^n + 1)}$

Type 11

194. Evaluate: $\int \frac{2x + 3}{x^2 + 4x + 5} dx$

195. Evaluate: $\int \frac{3x + 2}{x^2 - 3x + 4} dx$

195. Evaluate: $\int \frac{x}{x^2 + x + 1} dx$

196. Evaluate: $\int \frac{4x + 1}{x^2 + 3x + 2} dx$

197. Evaluate: $\int \frac{dx}{2e^{2x} + 3e^x + 1}$

198. Evaluate: $\int \frac{(3\sin x - 2)\cos x dx}{(5 - \cos^2 x - 4\sin x)}$

199. Evaluate: $\int \frac{ax^3 + bx}{x^4 + c^2} dx$

200. Evaluate: $\int \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) \times (2 + 2\sin 2x) dx$

201. Evaluate: $\int \frac{\sin x + \cos x}{5 + 3\sin 2x} dx$

202. Evaluate: $\int \frac{\sin x - \cos x}{3 + 5\sin 2x} dx$

Type 12

203. Evaluate: $\int \frac{dx}{\sqrt{x^2 + x + 1}}$

204. Evaluate: $\int \frac{dx}{\sqrt{x^2 - 2ax}}$

205. Evaluate: $\int \frac{dx}{\sqrt{4x - x^2}}$

206. Evaluate: $\int \frac{dx}{\sqrt{6 - x - x^2}}$

207. Evaluate: $\int \frac{dx}{\sqrt{1 + x + x^2}}$

208. Evaluate: $\int \frac{dx}{\sqrt{1 + x - x^2}}$

209. Evaluate: $\int \frac{dx}{\sqrt{x^2 + 2ax}}$

210. Evaluate: $\int \frac{dx}{\sqrt{2ax - x^2}}$

Type 13

211. Evaluate: $\int \frac{x dx}{\sqrt{x^4 - x^2 + 1}}$

212. Evaluate: $\int \sqrt{\sec x - 1} dx$

213. Evaluate: $\int \frac{dx}{x^{3/4} \sqrt{x^{1/2} - 1}}$

214. Evaluate: $\int \sqrt{\frac{\sin(x - \alpha)}{\sin(x + \alpha)}} dx$

215. Evaluate: $\int \frac{e^x}{\sqrt{4 - e^{2x}}} dx$

216. Evaluate: $\int \frac{\sec^2 x}{\sqrt{16 + \tan x}} dx$

217. Evaluate: $\int \sqrt{\sec x + 1} dx$

218. Evaluate: $\int \sqrt{\operatorname{cosec} x - 1} dx$

219. Evaluate: $\int \frac{dx}{\sqrt{1 - e^{2x}}}$

220. Evaluate: $\int \sqrt{\frac{\sin(x - \alpha)}{\sin(x + \alpha)}} dx$

221. Evaluate: $\int \frac{dx}{x^{2/3} \sqrt{x^{2/3} - 4}}$

222. Evaluate: $\int \sqrt{\frac{x}{x^3 - x^3}} dx$

223. Evaluate: $\int \left(\frac{\cos \theta + \sin \theta}{\sqrt{5 + \sin 2\theta}} \right) d\theta$

224. Evaluate: $\int \frac{\sin \theta - \cos \theta}{\sqrt{2 - \sin 2\theta}} d\theta$

Type 14

225. Evaluate: $\int \frac{x - 1}{\sqrt{x^2 - 3x + 2}} dx$

226. Evaluate: $\int \frac{3x + 4}{\sqrt{x^2 + 5x + 2}} dx$

227. Evaluate: $\int \frac{x + 2}{\sqrt{x^2 + 5x + 6}} dx$

228. Evaluate: $\int \frac{6x - 5}{\sqrt{3x^2 - 5x + 1}} dx$

229. Evaluate: $\int \sqrt{\frac{a - x}{a + x}} dx$

230. Evaluate: $\int x^2 \sqrt{\frac{a^2 - x^2}{a^2 + x^2}} dx$

231. Evaluate: $\int x^2 \sqrt{\frac{4 - x^3}{4 + x^3}} dx$

Type 15

232. Evaluate: $\int \frac{dx}{3 + 4 \sin^2 x}$

233. Evaluate: $\int \frac{dx}{3 \sin^2 x + 4 \cos^2 x}$

234. Evaluate: $\int \frac{dx}{(2 \sin x + 3 \cos x)^2}$

235. Evaluate: $\int \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$

236. Evaluate: $\int \frac{dx}{(\sin x + 2 \cos x)^2}$

237. Evaluate: $\int \frac{dx}{(\sin x + 2 \sec x)^2}$

238. Evaluate: $\int \frac{\sin 2x dx}{\sin^4 x + \cos^4 x}$

239. Evaluate: $\int \frac{dx}{(2 \sin x + 3 \cos x)^2}$

240. Evaluate: $\int \frac{dx}{2 + \cos^2 x}$

Type 16

241. Evaluate: $\int \frac{\sin x}{\sin 3x} dx$

242. Evaluate: $\int \frac{\operatorname{cosec} 3x}{\operatorname{cosec} x} dx$

243. Evaluate: $\int \frac{\sec 3x}{\sec x} dx$

Type 17

244. Evaluate: $\int \frac{dx}{1 + 2 \sin x}$

245. Evaluate: $\int \frac{dx}{3 \cos x + 4}$

246. Evaluate: $\int \frac{dx}{1 + \sin x + \cos x}$

247. Evaluate: $\int \frac{1 + \sin x}{\sin x(1 + \cos x)} dx$

248. Evaluate: $\int \frac{dx}{1 + 2 \sin x}$

249. Evaluate: $\int \frac{dx}{3 \sin x + 4 \cos x + 5}$

250. Evaluate: $\int \frac{dx}{\cos x + \cos \alpha}$

251. Evaluate: $\int \frac{dx}{\tan x + 4 \cot x + 4}$

Type 18

252. Evaluate: $\int \frac{dx}{\sin x + \cos x}$

253. Evaluate: $\int \frac{dx}{\sqrt{3} \sin x + \cos x}$

254. Evaluate: $\int \frac{dx}{\sin x + \sqrt{3} \cos x}$

255. Evaluate: $\int \frac{dx}{\sqrt{3} \sin x + \cos x}$

256. Evaluate: $\int \frac{dx}{\sqrt{3} \sin x - \cos x}$

Type 19

257. Evaluate: $\int \frac{2 \sin x + \cos x}{3 \sin x + 2 \cos x} dx$

258. Evaluate: $\int \frac{\sin x}{\sin x + \cos x} dx$

259. Evaluate: $\int \frac{2 \sin x + 3 \cos x}{3 \sin x + 4 \cos x} dx$

260. Evaluate: $\int \left(\frac{\sin x}{\sin x + \cos x} \right) dx$

261. Evaluate: $\int \frac{\cos x}{\sin x + \cos x} dx$

262. Evaluate: $\int \frac{1}{1 + \tan x} dx$

263. Evaluate: $\int \frac{1}{1 - \tan x} dx$

264. Evaluate: $\int \frac{1}{1 + \tan x} dx$

Type 20

265. Evaluate: $\int \frac{3 \sin x + 2 \cos x + 4}{3 \cos x + 4 \sin x + 5} dx$

266. Evaluate: $\int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$

267. Evaluate: $\int \frac{2 \cos x + 3 \sin x}{2 \sin x + 3 \cos x + 5} dx$

Type 21

268. Evaluate: $\int \left(\frac{\sin x + 2 \cos x}{9 + 16 \sin 2x} \right) dx$

269. Evaluate: $\int (\cos x - \sin x)(2 + 3 \sin 2x) dx$

270. Evaluate: $\int \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$

271. Evaluate: $\int \frac{dx}{\cos x + \operatorname{cosec} x}$

272. Evaluate: $\int (\sin x + \cos x)(2 + 3 \sin 2x) dx$

273. Evaluate: $\int (\sin x - \cos x)(3 - 4 \sin 2x) dx$

274. Evaluate: $\int \left(\frac{\cos x - \sin x}{3 + 2 \sin 2x} \right) dx$

275. Evaluate: $\int \left(\frac{\cos x - \sin x}{5 - 7 \sin 2x} \right) dx$

276. Evaluate: $\int \left(\frac{2 \cos x - \sin x}{9 + 16 \sin 2x} \right) dx$

Type 22

277. Evaluate: $\int \frac{x^2 + 1}{x^4 + 1} dx$

278. Evaluate: $\int \frac{x^2 - 1}{x^4 + 1} dx$

279. Evaluate: $\int \frac{x^4 + 1}{x^6 + 1} dx$

280. Evaluate: $\int \frac{x^4 + 3x + 1}{x^4 + x^2 + 1} dx$

281. Evaluate: $\int \left(\frac{1 - x^2}{1 + x^2} \right) \frac{dx}{\sqrt{1 + x^2 + x^4}}$

282. Evaluate: $\int \left(\frac{x - 1}{x + 1} \right) \times \frac{dx}{\sqrt{x^3 + x^2 + x}}$

283. Evaluate: $\int \frac{x^2 - 1}{x^2 + 1} \times \frac{dx}{\sqrt{x^4 + 1}}$

284. Evaluate: $\int \frac{x^2 - 1}{x^3 \sqrt{x^4 - 2x^2 + 1}} dx$

285. Evaluate: $\int \left(\frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} \right) dx$

286. Evaluate: $\int \frac{dx}{x^2(x + \sqrt{1 + x^2})}$

287. Evaluate: $\int \frac{x^2 - 3x - 1}{x^4 + x^2 + 1} dx$

288. Evaluate: $\int \frac{dx}{\sin^4 x + \cos^4 x}$

289. Evaluate: $\int \frac{dx}{x^4 + 1}$

290. Evaluate: $\int \frac{x^2 dx}{x^4 + 1}$

291. Evaluate: $\int \frac{2dx}{x^4 + 1}$

292. Evaluate: $\int \left(\frac{x^4 + 1}{x^6 + 1} \right) dx$

293. Evaluate: $\int \frac{dx}{x^2(1 + x^4)^{3/4}}$

294. Evaluate: $\int \left(\frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} \right) dx$

295. Evaluate: $\int \frac{dx}{x^2(1+x^5)^{4/5}} dx$

296. Evaluate: $\int \left(\frac{x^2-1}{x\sqrt{1+x^4}} \right) dx$

297. Evaluate: $\int \frac{dx}{x\sqrt{x^4+3x^2+1}}$

298. Evaluate: $\int \left(\frac{x^2+1}{1-x^2} \right) \times \frac{dx}{\sqrt{x^4+x^2+1}}$

299. Evaluate: $\int \left(\frac{x^2+1}{x} \right) \times \frac{dx}{\sqrt{x^4+3x^2+1}}$

300. Evaluate: $\int \frac{x^x(x^{2x}+1)(\ln x+1)}{(x^{4x}+1)} dx$

301. Evaluate: $\int \frac{x^2-1}{x^3\sqrt{x^4-2x^2+1}} dx$

302. Evaluate: $\int \left(\frac{x^{2009}}{(1+x^2)^{1006}} \right) dx$

303. Evaluate: $\int \frac{(2x+1)dx}{(x^2+4x+1)^{3/2}}$

Type 23

304. Evaluate: $\int \left(\frac{\sqrt{\cot x} - \sqrt{\tan x}}{1+3\sin 2x} \right) dx$

305. Evaluate: $\int \sqrt{\tan x} dx$

306. Evaluate: $\int (\sqrt{\tan x} - \sqrt{\cot x}) dx$

307. Evaluate: $\int \sqrt{\cot x} dx$

308. Evaluate: $\int \left(\sqrt{\cot x} - \frac{1}{\sqrt{\cot x}} \right) dx$

309. Evaluate: $\int \left(\sqrt{\cot x} + \frac{1}{\sqrt{\cot x}} \right) dx$

310. Evaluate: $\int \left(\sqrt{\tan x} + \frac{1}{\sqrt{\tan x}} \right) dx$

311. Evaluate: $\int (\sqrt[4]{\tan x} + \sqrt[4]{\cot x})^2 dx$

312. Evaluate: $\int \frac{dx}{(\sqrt{\sin x} + \sqrt{\cos x})^4} dx$

Integration by Parts

ABC of Integration by Parts

313. Evaluate: $\int x e^x dx$

314. Evaluate: $\int x \sin x dx$

315. Evaluate: $\int x^2 \sin x dx$

316. Evaluate: $\int \log x dx$

317. Evaluate: $\int (\log x)^2 dx$

318. Evaluate: $\int \log(x^2+1) dx$

319. Evaluate: $\int \tan^{-1} x dx$

320. Evaluate: $\int \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) dx$

321. Evaluate: $\int e^{\sqrt{x}} dx$

322. Evaluate: $\int \left(\frac{x+\sin x}{1+\cos x} \right) dx$

323. Evaluate: $\int \tan^{-1} \left(\sqrt{\frac{1-x}{1+x}} \right) dx$

324. Evaluate: $\int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx$

325. Evaluate: $\int \frac{x^2}{(x \sin x + \cos x)^2} dx$

326. Evaluate: $\int \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} dx$

327. Evaluate: $\int \sin^{-1} \left(\sqrt{\frac{x}{a+x}} \right) dx$

328. Evaluate: $\int \log(1+x) dx$

329. Evaluate: $\int \frac{x - \sin x}{1 - \cos x} dx$

330. Evaluate: $\int \sin \sqrt{x} dx$

331. Evaluate: $\int x \log(1+x) dx$

332. Evaluate: $\int (\sin^{-1} x)^2 dx$

333. Evaluate: $\int \frac{x - \sin x}{1 - \cos x} dx$

334. Evaluate: $\int x \sin^3 x dx$

335. Evaluate: $\int x \left(\frac{\sec 2x - 1}{\sec 2x + 1} \right) dx$

336. Evaluate: $\int \sin^{-1} \left(\frac{2x}{1+x^2} \right) dx$

337. Evaluate: $\int \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx$

338. Evaluate: $\int \frac{x^3 \sin^{-1}(x^2)}{\sqrt{1-x^4}} dx$

339. Evaluate: $\int \frac{\sec x(2 + \sec x)}{(1 + 2\sec x)^2} dx$

340. Evaluate: $\int \frac{1-x}{e^x + x} dx$

Type 24

341. Evaluate: $\int e^x(\sin x + \cos x) dx$

342. Evaluate: $\int \frac{x e^x}{(x+1)^2} dx$

343. Evaluate: $\int e^x \left(\frac{2 + \sin 2x}{1 + \cos 2x} \right) dx$

344. Evaluate: $\int e^x \left(\frac{1+x+x^3}{(1+x^2)^{3/2}} \right) dx$

345. Evaluate: $\int e^x \left(\frac{x^2 + 1}{(x+1)^2} \right) dx$

346. Evaluate: $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

347. Evaluate: $\int e^x \left(\frac{x}{(x+1)^2} \right) dx$

348. Evaluate: $\int e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx$

349. Evaluate: $\int e^x \left(\frac{2 + \sin 2x}{1 + \cos 2x} \right) dx$

350. Evaluate: $\int e^x \left(\frac{x^2 + 1}{(x+1)^2} \right) dx$

351. Evaluate: $\int \frac{\log x}{(1 + \log x)^2} dx$

352. Evaluate: $\int e^x \left(\log x + \frac{1}{x} \right) dx$

353. Evaluate: $\int e^x \left(\frac{x-1}{(x+1)^3} \right) dx$

354. Evaluate: $\int \left(\frac{1}{\log x} - \frac{1}{(\log x)^2} \right) dx$

355. Evaluate: $\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$

356. Evaluate: $\int e^x \left(\frac{1-x}{1+x} \right)^2 dx$

357. Evaluate: $\int e^x \left(\frac{x^2 + 1}{(x+1)^2} \right) dx$

358. Evaluate: $\int e^x \left(\frac{(x+1) + \sqrt{1-x^2}}{(x+1)^2 \sqrt{1-x^2}} \right) dx$

359. Evaluate: $\int e^x(2\sec^2 x - 1)\tan x dx$

360. Evaluate: $\int e^x(\log(\sec x + \tan x) + \sec x) dx$

361. Evaluate: $\int e^{\sin x} \left(\frac{x \cos^3 x - \sin x}{\cos^2 x} \right) dx$

Type 25

362. Evaluate: $\int e^x \left(\log x + \frac{1}{x^2} \right) dx$

363. Evaluate: $\int \left(\log(\log x) + \frac{1}{(\log x)^2} \right) dx$

364. Evaluate: $\int e^x \left(\frac{x^4 + 2}{(1+x^2)^{5/2}} \right) dx$

365. Evaluate: $\int \frac{e^{\sin x}(x \cos^3 x - \sin x)}{\cos^2 x} dx$

366. Evaluate: $\int e^x \left(\frac{x^3 - x + 2}{(x^2 - 1)^2} \right) dx$

Type 26

367. Evaluate: $\int e^{3x}(3\sin x + \cos x) dx$

368. Evaluate: $\int e^{2x}(\sec^2 x + 2\tan x) dx$

369. Evaluate: $\int e^{2x} \left(\frac{2\sin 4x - 4}{1 - \cos 4x} \right) dx$

370. Evaluate: $\int e^{2x}(-\sin x + 2\cos x) dx$

371. Evaluate: $\int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx$

372. Evaluate: $\int e^{-x/2} \left(\sqrt{\frac{1 - \sin x}{1 + \cos x}} \right) dx$

373. Evaluate: $\int e^{2x}(2 \times \log(\sec x + \tan x) + \sec x) dx$

Type 27

374. Evaluate: $\int e^x \sin 3x dx$

375. Evaluate: $\int e^{4x} \cos 3x dx$

376. Evaluate: $\int e^{2x} \sin 3x dx$

377. Evaluate: $\int e^{-x} \cos x dx$

378. Evaluate: $\int e^{2x} \cos(3x + 4) dx$

379. Evaluate: $\int e^x \cos 2x dx$

380. Evaluate: $\int \frac{1}{x^3} \sin(\log x) dx$

Type 28

381. Evaluate: $\int \sqrt{4 - x^2} dx$
382. Evaluate: $\int \sqrt{1 - 9x^2} dx$
383. Evaluate: $\int \sqrt{x^2 + 1} dx$
384. Evaluate: $\int \sqrt{3x^2 + 1} dx$
385. Evaluate: $\int \sqrt{x^2 - 9} dx$
386. Evaluate: $\int \sqrt{4x^2 - 1} dx$
387. Evaluate: $\int \sqrt{x^2 + 2x + 3} dx$
388. Evaluate: $\int \sqrt{3 - 4x - x^2} dx$
389. Evaluate: $\int \sqrt{2ax - x^2} dx$
390. Evaluate: $\int \sqrt{2ax + x^2} dx$
391. Evaluate: $\int \sqrt{x - 4x^2} dx$

Type 29

392. Evaluate: $\int (2x + 1)\sqrt{x^2 + 3x + 4} dx$
393. Evaluate: $\int (x - 5)\sqrt{x^2 + x} dx$
394. Evaluate: $\int (3x - 2)\sqrt{x^2 + x + 1} dx$
395. Evaluate: $\int (4x + 1)\sqrt{x^2 - x - 2} dx$
396. Evaluate: $\int x\sqrt{1 + x - x^2} dx$

Partial Fractions**Type 1**

397. Evaluate: $\int \frac{(2x + 1)dx}{(x + 2)(x + 3)}$
398. Evaluate: $\int \frac{dx}{(x - 1)(x - 2)}$
399. Evaluate: $\int \frac{dx}{(x + 1)(x + 2)(x + 3)}$
400. Evaluate: $\int \frac{dx}{(x + 1)(x + 2)(x + 3)(x + 4)}$
401. Evaluate: $\int \frac{x - 1}{(x + 1)(x - 2)} dx$
402. Evaluate: $\int \frac{2x - 1}{(x + 1)(x + 2)(x + 3)} dx$
403. Evaluate: $\int \frac{x^3}{(x - 1)(x - 2)} dx$

404. Evaluate: $\int \frac{2x}{(x^2 + 1)(x^2 + 2)} dx$
405. Evaluate: $\int \frac{\cos \theta}{(2 + \cos \theta)(3 + \cos \theta)} d\theta$
406. Evaluate: $\int \frac{(1 - \cos x)}{\cos x(1 + \cos x)} dx$
407. Evaluate: $\int \frac{dx}{\sin x - \sin 2x}$

Type 2

408. Evaluate: $\int \frac{dx}{(x + 1)(x + 1)^2}$
409. Evaluate: $\int \frac{2x + 1}{(x + 2)(x - 3)^2} dx$
410. Evaluate: $\int \frac{3x + 1}{(x - 2)^2(x + 2)} dx$
411. Evaluate: $\int \frac{x^2 + 1}{(x - 1)^2(x + 3)} dx$
412. Evaluate: $\int \frac{x^2}{(x - 1)^3(x + 1)} dx$
413. Evaluate: $\int \frac{(x - 1)}{x^2(x + 4)} dx$
414. Evaluate: $\int \frac{(2x - 1)}{x^3(x - 2)} dx$

Type 3

415. Evaluate: $\int \frac{(2x + 3)dx}{(x + 1)(x^2 + 4)}$
416. Evaluate: $\int \frac{(3x - 2)dx}{(x - 1)(x^2 + 9)}$
417. Evaluate: $\int \frac{2x - 1}{(x + 1)(x^2 + 2)} dx$
418. Evaluate: $\int \frac{x}{(x + 1)(x^2 + 4)} dx$
419. Evaluate: $\int \frac{8}{(x + 2)(x^2 + 9)} dx$
420. Evaluate: $\int \frac{x}{(x + 1)(x^2 + 1)} dx$

Type 4

421. Evaluate: $\int \frac{x^2}{(x^2 - 1)(x^2 + 1)} dx$
422. Evaluate: $\int \frac{x^2}{(x^2 - 3)(x^2 + 4)} dx$

HINTS AND SOLUTIONS

Level 1

1. We have,

$$\int \log_x x dx = \int 1 \cdot dx = x + c$$

2. We have,

$$\begin{aligned} \int (3^{\log_x 2} - 2^{\log_x 3}) dx &= \int (2^{\log_x 3} - 2^{\log_x 3}) dx \\ &= \int 0 \cdot dx = c \end{aligned}$$

3. We have,

$$\begin{aligned} \int \left(x^m + m^x + m^m + \frac{m}{x} \right) dx \\ = \frac{x^{m+1}}{m+1} + \frac{m^x}{\log m} + m^m x + m \log |x| + c \end{aligned}$$

4. We have,

$$\begin{aligned} \int 2^x \cdot 3^x dx &= \int (2 \cdot 3)^x dx = \int 6^x dx \\ &= \frac{6^x}{\log 6} + c \end{aligned}$$

5. We have,

$$\begin{aligned} \int \tan^2 x dx &= \int (\sec^2 x - 1) dx \\ &= \tan x - x + c \end{aligned}$$

6. We have,

$$\begin{aligned} \int \cot^2 x dx &= \int (\operatorname{cosec}^2 x - 1) dx \\ &= -\cot x - x + c \end{aligned}$$

7. We have,

$$\begin{aligned} \int \frac{dx}{\sin^2 x \cos^2 x} &= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} dx \\ &= \int \left(\frac{1}{\cos^2 x} + \frac{1}{\sin^2 x} \right) dx \\ &= \int (\sec^2 x + \operatorname{cosec}^2 x) dx \\ &= \tan x - \cot x + c \end{aligned}$$

$$8. \int \left(1 + \tan \left(x + \frac{3\pi}{8} \right) \right) \left(1 + \tan \left(\frac{\pi}{8} - x \right) \right) dx$$

$$= \int 2 dx \quad [\because \text{if } A + B = \frac{\pi}{4},$$

$$\text{then } (1 + \tan A)(1 + \tan B) = 2] = 2x + c.$$

9. We have,

$$\begin{aligned} \int (\tan x + \cot x)^2 dx &= \int (\tan^2 x + \cot^2 x + 2) dx \\ &= \int (\sec^2 x + \operatorname{cosec}^2 x) dx \\ &= \tan x - \cot x + c \end{aligned}$$

10. We have,

$$\begin{aligned} \int \frac{dx}{1 + \cos 2x} &= \int \frac{dx}{2 \cos^2 x} = \int \frac{1}{2} \sec^2 x dx \\ &= \frac{1}{2} \tan x + c \end{aligned}$$

11. The given integral is

$$\begin{aligned} \int (3^{\log_5 x} - x^{\log_5 3}) dx &= \int (3^{\log_6 x} - 3^{\log_6 x}) dx \\ &= \int (0) \cdot dx = c \end{aligned}$$

12. The given integral is

$$\begin{aligned} \int \left(1 + \tan \left(\frac{\pi}{8} - x \right) \right) \left(1 + \tan \left(\frac{\pi}{8} + x \right) \right) dx \\ = \int (2) \cdot dx \\ = 2x + C \end{aligned}$$

13. The given integral is

$$\begin{aligned} \int \left(\frac{8^{1+x} + 4^{1+x}}{2^{2x}} \right) dx &= \int \left(\frac{2^{3+3x} + 2^{2+2x}}{2^{2x}} \right) dx \\ &= \int (2^{3+x} + 2^2) dx \\ &= 8 \int 2^x dx + \int 4 dx \\ &= \frac{8 \cdot 2^x}{\log 2} + 4x + c \end{aligned}$$

14. The given integral is

$$\begin{aligned} \int \left(\frac{x}{m} + \frac{m}{x} + x^m + m^x \right) dx \\ = \frac{x^2}{2m} + m \log x + \frac{x^{m+1}}{m+1} + \frac{m^x}{\log m} + c \end{aligned}$$

15. The given integral is

$$\begin{aligned}\int \frac{(a^x + b^x)^2}{a^x b^x} dx &= \int \left(\frac{a^{2x} + b^{2x} + 2a^x b^x}{a^x b^x} \right) dx \\ &= \int \left(\left(\frac{a}{b} \right)^x + \left(\frac{b}{a} \right)^x + 2 \right) dx \\ &= \frac{(a/b)^x}{\log(a/b)} + \frac{(b/a)^x}{\log(b/a)} + 2x + c\end{aligned}$$

16. The Given integral is

$$\begin{aligned}\int \frac{(2x + 3x)^2}{2^x \cdot 3^x} dx &= \int \left(\left(\frac{2}{3} \right)^x + \left(\frac{3}{2} \right)^x + 2 \right) dx \\ &= \frac{(2/3)^x}{\log(2/3)} + \frac{(3/2)^x}{\log(3/2)} + 2x + c\end{aligned}$$

17. It is given that

$$f'(x) = \frac{1}{x} + \frac{1}{\sqrt{1-x^2}}$$

On integration, we get

$$f(x) = \log|x| + \sin^{-1}x + c$$

When $x = 1$ and $f(1) = \frac{\pi}{2}$, then $c = 0$

Hence, the function is

$$f(x) = \log|x| + \sin^{-1}x$$

18. It is given that

$$f'(x) = a \cos x + b \sin x$$

On integration, we get

$$f(x) = a \sin x - b \cos x + c$$

When $x = 0$ and $f'(0) = 4$, then $a = 4$

When $x = 0$, $f(0) = 3$ and $f\left(\frac{\pi}{2}\right) = 5$

then, we get

$$b = -2 \text{ and } c = 1$$

Hence, the function is

$$f(x) = 4 \cos x + 2 \sin x + 1$$

19. We have,

$$\begin{aligned}\int \frac{dx}{1 - \sin x} &= \int \frac{(1 + \sin x)}{(1 - \sin x)(1 + \sin x)} dx \\ &= \int \frac{(1 + \sin x)}{(1 - \sin^2 x)} dx = \int \frac{1 + \sin x}{\cos^2 x} dx \\ &= \int (\sec^2 x + \sec x \cdot \tan x) dx \\ &= \tan x + \sec x + c\end{aligned}$$

20. We have,

$$\int \frac{\sin^4 x + \cos^4 x}{\sin^2 x \cos^2 x} dx$$

$$\begin{aligned}&= \int \frac{((\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x)}{\sin^2 x \cos^2 x} dx \\ &= \int \frac{(1 - 2\sin^2 x \cos^2 x)}{\sin^2 x \cos^2 x} dx \\ &= \int \frac{(\sin^2 x + \cos^2 x - 2\sin^2 x \cos^2 x)}{\sin^2 x \cos^2 x} dx \\ &= \int (\sec^2 x + \operatorname{cosec}^2 x - 2) dx \\ &= \tan x - \cot x - 2x + c\end{aligned}$$

21. We have,

$$\begin{aligned}&\int \left(\frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} \right) dx \\ &= \int \left(\frac{(\sin^2 x)^3 + (\cos^2 x)^3}{\sin^2 x \cdot \cos^2 x} \right) dx \\ &= \int \left(\frac{(\sin^4 x + \cos^4 x - \sin^2 x \cdot \cos^2 x)}{\sin^2 x \cos^2 x} \right) dx \\ &= \int \left(\frac{(1 - 3\sin^2 x \cdot \cos^2 x)}{\sin^2 x \cos^2 x} \right) dx \\ &= \int \left(\frac{(\sin^2 x + \cos^2 x - 3\sin^2 x \cos^2 x)}{\sin^2 x \cos^2 x} \right) dx \\ &= \int (\sec^2 x + \operatorname{cosec}^2 x - 3) dx \\ &= \tan x - \cot x - 3x + c\end{aligned}$$

22. We have,

$$\begin{aligned}&\int \left(\frac{\cos 2x - \cos \alpha}{\cos x - \cos \alpha} \right) dx \\ &= \int \left(\frac{2\cos^2 x - 1 - 2\cos^2 \alpha + 1}{\cos x - \cos \alpha} \right) dx \\ &= \int 2 \left(\frac{\cos^2 x - \cos^2 \alpha}{\cos x - \cos \alpha} \right) dx \\ &= \int 2(\cos x + \cos \alpha) dx \\ &= 2(\sin x + x \cos \alpha) + c\end{aligned}$$

23. We have,

$$\begin{aligned}&\int \left(\frac{\cos^4 x - \sin^4 x}{\sqrt{1 + \cos 4x}} \right) dx \\ &= \int \left(\frac{(\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x)}{\sqrt{1 + \cos 4x}} \right) dx\end{aligned}$$

$$\begin{aligned}
&= \int \left(\frac{\cos^2 x - \sin^2 x}{\sqrt{2} \cos 2x} \right) dx \\
&= \int \left(\frac{\cos 2x}{\sqrt{2} \cos 2x} \right) dx \\
&= \frac{x}{\sqrt{2}} + c
\end{aligned}$$

24. We have,

$$\begin{aligned}
\int \left(\frac{1 + \tan^2 x}{1 + \cot^2 x} \right) dx &= \int \left(\frac{1 + \tan^2 x}{1 + \frac{1}{\tan^2 x}} \right) dx \\
&= \int \left(\frac{\tan^2 x (1 + \tan^2 x)}{(1 + \tan^2 x)} \right) dx \\
&= \int (\tan^2 x) dx \\
&= \int (\sec^2 x - 1) dx \\
&= \tan x - x + c
\end{aligned}$$

25. We have,

$$\begin{aligned}
&\int \left(\frac{\cos x - \cos 2x}{1 - \cos x} \right) dx \\
&= \int \left(\frac{\cos x - 2\cos^2 x + 1}{1 - \cos x} \right) dx \\
&= -\int \left(\frac{2\cos^2 x - \cos x - 1}{1 - \cos x} \right) dx \\
&= -\int \left(\frac{(\cos x - 1)(2\cos x + 1)}{1 - \cos x} \right) dx \\
&= \int \left(\frac{(1 - \cos x)(2\cos x + 1)}{1 - \cos x} \right) dx \\
&= \int ((2\cos x + 1)) dx \\
&= 2\sin x + x + c.
\end{aligned}$$

26. We have,

$$\begin{aligned}
\int \left(\frac{\sqrt{x^4 + x^{-4} + 2}}{x^3} \right) dx &= \int \left(\frac{\sqrt{x^4 + x^{-4} + 2}}{x^6} \right) dx \\
&= \int \left(\frac{\sqrt{x^8 + 2x^4 + 1}}{x^{10}} \right) dx \\
&= \int \left(\frac{x^4 + 1}{x^5} \right) dx \\
&= \int \left(x^{-5} + \frac{1}{x} \right) dx \\
&= \log |x| - \frac{1}{4x^4} + c
\end{aligned}$$

27. We have,

$$\begin{aligned}
&\int \left(\frac{5\cos^3 x + 3\sin^3 x}{\cos^2 x \sin^2 x} \right) dx \\
&= \int \left(\frac{5\cos^3 x}{\cos^2 x \sin^2 x} + \frac{3\sin^3 x}{\cos^2 x \sin^2 x} \right) dx \\
&= 5 \int \operatorname{cosec} x \cot x dx + 3 \int \sec x \tan x dx \\
&= -5 \operatorname{cosec} x + 3 \sec x + c
\end{aligned}$$

28. We have,

$$\begin{aligned}
&\int \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) (1 + \sin 2x) dx \\
&= \int \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) (\cos x + \sin x)^2 dx \\
&= \int (\cos x - \sin x)(\cos x + \sin x) dx \\
&= \int (\cos^2 x - \sin^2 x) dx \\
&= \int \cos 2x dx \\
&= \frac{\sin 2x}{2} + c
\end{aligned}$$

29. The given integral is

$$\begin{aligned}
&\int \left(\frac{\cos 5x + \cos 4x}{1 - 2\cos 3x} \right) dx \\
&= \int \frac{\sin 3x (\cos 5x + \cos 4x)}{\sin 3x - \sin 6x} dx \\
&= \int \frac{2\sin\left(\frac{3x}{2}\right) \cos\left(\frac{3x}{2}\right) 2\cos\left(\frac{9x}{2}\right) \cos\left(\frac{x}{2}\right)}{-2\cos\left(\frac{9x}{2}\right) \sin\left(\frac{3x}{2}\right)} dx \\
&= -\int 2\cos\left(\frac{3x}{2}\right) \cos\left(\frac{x}{2}\right) dx \\
&= -\int (\cos 2x + \cos x) dx \\
&= -\left(\frac{\sin 2x}{2} + \sin x \right) + c
\end{aligned}$$

30. The given integral is

$$\begin{aligned}
&\int \left(\frac{\cos x - \cos 2x}{1 - \cos x} \right) dx \\
&= -\int \frac{2\cos^2 x - \cos x - 1}{1 - \cos x} dx \\
&= -\int \frac{(\cos x - 1)(2\cos x + 1)}{1 - \cos x} dx \\
&= \int (2\cos x + 1) dx \\
&= (2\sin x + x) + C
\end{aligned}$$

31. The given integral is

$$\begin{aligned}
 & \int \frac{dx}{(\tan x + \cot x + \sec x + \operatorname{cosec} x)} \\
 &= \int \frac{\sin x \cos x dx}{1 + \sin x + \cos x} \\
 &= \int \frac{\sin x dx}{\sec x + \tan x + 1} \\
 &= \int \frac{\sin x(1 + \tan x - \sec x) dx}{(1 + \tan x)^2 - \sec^2 x} \\
 &= \int \frac{\sin x(1 + \tan x - \sec x) dx}{2 \tan x} \\
 &= \frac{1}{2} \int \cos x(1 + \tan x - \sec x) dx \\
 &= \frac{1}{2} \int (\cos x + \sin x - 1) dx \\
 &= \frac{1}{2} (\sin x + \cos x - x) + C
 \end{aligned}$$

32. We have,

$$\begin{aligned}
 \int \frac{x}{x+1} dx &= \int \frac{(x+1-1)}{(x+1)} dx \\
 &= \int \left(1 - \frac{1}{x+1}\right) dx \\
 &= x - \log|x+1| + C
 \end{aligned}$$

33. We have,

$$\begin{aligned}
 \int \frac{(1+x)^2}{x(1+x^2)} dx &= \int \frac{(1+x^2+2x)}{x(1+x^2)} dx \\
 &= \frac{(1+x^2)+2x}{x(1+x^2)} dx \\
 &= \int \frac{dx}{x} + 2 \int \frac{dx}{1+x^2} \\
 &= \log|x| + 2 \tan^{-1} x + c
 \end{aligned}$$

34. We have,

$$\begin{aligned}
 \int \frac{x^2-2}{x^2+1} dx &= \int \left(\frac{x^2+1-3}{x^2+1} \right) dx \\
 &= \int \left(1 - \frac{3}{x^2+1} \right) dx \\
 &= x - 3 \tan^{-1} x + c
 \end{aligned}$$

35. We have,

$$\int \frac{x-1}{(x^{2/3} + x^{1/3} + 1)} dx = \int \frac{(x^{1/3})^3 - 1}{(x^{2/3} + x^{1/3} + 1)} dx$$

$$\begin{aligned}
 &= \int \frac{(x^{1/3} - 1)(x^{2/3} + x^{1/3} + 1)}{(x^{2/3} + x^{1/3} + 1)} dx \\
 &= \int (x^{1/3} - 1) dx \\
 &= -\frac{3}{2} x^{-2/3} - x + c
 \end{aligned}$$

36. We have,

$$\begin{aligned}
 \int \left(\frac{x^4 - 1 + 3}{x^2 + 1} \right) dx &= \int \left(\frac{x^4 - 1}{x^2 + 1} + \frac{3}{x^2 + 1} \right) dx \\
 &= \int \left(\frac{(x^2 - 1)(x^2 + 1)}{x^2 + 1} + \frac{3}{x^2 + 1} \right) dx \\
 &= \int \left((x^2 - 1) + \frac{3}{x^2 + 1} \right) dx \\
 &= \frac{x^3}{3} - x + 3 \tan^{-1} x + c
 \end{aligned}$$

37. We have,

$$\begin{aligned}
 \int \left(\frac{x^4 - 1 - 2}{x^2 + 1} \right) dx &= \int \left[\left(\frac{x^4 - 1}{x^2 + 1} \right) - \frac{2}{x^2 + 1} \right] dx \\
 &= \int \left[\left(\frac{(x^2 + 1)(x^2 - 1)}{x^2 + 1} \right) - \frac{2}{x^2 + 1} \right] dx \\
 &= \int \left((x^2 - 1) - \frac{2}{x^2 + 1} \right) dx \\
 &= \frac{x^3}{3} - x - 2 \tan^{-1} x + c
 \end{aligned}$$

38. We have,

$$\begin{aligned}
 \int \left(\frac{x^6 + 1 - 2}{x^2 + 1} \right) dx &= \int \left[\left(\frac{x^6 + 1}{x^2 + 1} \right) - \frac{2}{x^2 + 1} \right] dx \\
 &= \int \left[\left(\frac{(x^2 + 1)(x^4 - x^2 + 1)}{x^2 + 1} \right) - \frac{2}{x^2 + 1} \right] dx \\
 &= \int \left((x^4 - x^2 + 1) - \frac{2}{x^2 + 1} \right) dx \\
 &= \frac{x^5}{5} - \frac{x^3}{3} + x - 2 \tan^{-1} x + c
 \end{aligned}$$

39. We have,

$$\begin{aligned}
 & \int \left(\frac{x^8 + x^4 + 1}{x^4 + x^2 + 1} \right) dx \\
 &= \int \left(\frac{(x^4 - x^2 + 1)(x^4 + x^2 + 1)}{(x^4 + x^2 + 1)} \right) dx \\
 &= \int ((x^4 + x^2 + 1)) dx
 \end{aligned}$$

$$= \frac{x^5}{5} + \frac{x^3}{3} + x + c$$

40. The given integral is

$$\begin{aligned}\int \left(\frac{x^4}{x^2 + 1} \right) dx &= \int \frac{(x^4 - 1) + 1}{x^2 + 1} dx \\ &= \int \left(x^2 - 1 + \frac{1}{x^2 + 1} \right) dx \\ &= \left(\frac{x^3}{3} - x + \tan^{-1} x + c \right)\end{aligned}$$

41. The given integral is

$$\begin{aligned}\int \left(\frac{x^4 + x^2 + 1}{x^2 + x + 1} \right) dx &= \int \frac{(x^2 - x + 1)(x^2 + x + 1)}{(x^2 + x + 1)} dx \\ &= \int (x^2 - x + 1) dx \\ &= \left(\frac{x^3}{3} - \frac{x^2}{2} + x + C \right)\end{aligned}$$

42. The given integral is

$$\begin{aligned}\int \left(\frac{x^6 + 1}{x^2 + 1} \right) dx &= \int \frac{(x^2 + 1)(x^4 - x^2 + 1)}{(x^2 + 1)} dx \\ &= \int (x^4 - x^2 + 1) dx \\ &= \left(\frac{x^5}{5} - \frac{x^3}{3} + x + c \right)\end{aligned}$$

43. We have,

$$\int \sin^{-1}(\sin x) dx = \int x dx = \frac{x^2}{2} + c$$

44. We have,

$$\begin{aligned}\int \sin^{-1}(\cos x) dx &= \int \sin^{-1} \left[\sin \left(\frac{\pi}{2} - x \right) \right] dx \\ &= \int \left(\frac{\pi}{2} - x \right) dx \\ &= \frac{\pi}{2} x - \frac{x^2}{2} + c\end{aligned}$$

45. We have,

$$\begin{aligned}\int \tan^{-1} \left(\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} \right) dx &= \int \tan^{-1} \left(\sqrt{\frac{2 \sin^2 x}{2 \cos^2 x}} \right) dx \\ &= \int \tan^{-1}(\tan x) dx = \int x dx \\ &= \frac{x^2}{2} + c\end{aligned}$$

46. We have,

$$\begin{aligned}\int \tan^{-1} \left(\frac{\sin 2x}{1 + \cos 2x} \right) dx &= \int \tan^{-1} \left(\frac{2 \sin x \cos x}{2 \cos^2 x} \right) dx \\ &= \int \tan^{-1}(\tan x) dx \\ &= \int x dx \\ &= \frac{x^2}{2} + c\end{aligned}$$

47. The given integral is

$$\begin{aligned}\int \tan^{-1} \left(\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} \right) dx &= \int \tan^{-1} \left(\frac{\sin x}{\cos x} \right) dx \\ &= \int \tan^{-1}(\tan x) dx \\ &= \int x dx \\ &= \frac{x^2}{2} + c\end{aligned}$$

48. The given integral is

$$\begin{aligned}\int \tan^{-1} \left(\frac{\sin x}{1 - \cos x} \right) dx &= \int \tan^{-1} \left(\frac{2 \sin(x/2) \cos(x/2)}{2 \sin^2(x/2)} \right) dx \\ &= \int \tan^{-1} \left[\tan \left(\frac{x}{2} \right) \right] dx \\ &= \int \left(\frac{x}{2} \right) dx \\ &= \frac{x^2}{4} + c\end{aligned}$$

49. The given integral is

$$\begin{aligned}\int \tan^{-1} \left(\sqrt{\frac{1 - \sin x}{1 + \sin x}} \right) dx &= \int \tan^{-1} \left(\frac{\cos(x/2) - \sin(x/2)}{\cos(x/2) + \sin(x/2)} \right) dx \\ &= \int \tan^{-1} \left(\frac{1 - \tan(x/2)}{1 + \tan(x/2)} \right) dx \\ &= \int \tan^{-1} \left[\tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right] dx \\ &= \int \left(\frac{\pi}{4} - \frac{x}{2} \right) dx \\ &= \left(\frac{\pi x}{4} - \frac{x^2}{4} \right) + c\end{aligned}$$

50. The given integral is

$$\begin{aligned}
& \int \tan^{-1} \left(\frac{\sin x}{1 + \cos x} \right) dx \\
&= \int \tan^{-1} \left(\frac{2 \sin(x/2) \cos(x/2)}{2 \cos^2(x/2)} \right) dx \\
&= \int \tan^{-1} [\tan(x/2)] dx \\
&= \int \left(\frac{x}{2} \right) dx \\
&= \frac{x^2}{4} + c
\end{aligned}$$

51. The given integral is

$$\begin{aligned}
& \int \tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right) dx \\
&= \int \tan^{-1} \left(\frac{\sin\left(\frac{\pi}{2} - x\right)}{1 - \cos\left(\frac{\pi}{2} - x\right)} \right) dx \\
&= \int \tan^{-1} \left(\frac{2 \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \cos\left(\frac{\pi}{4} - \frac{x}{2}\right)}{2 \sin^2\left(\frac{\pi}{4} - \frac{x}{2}\right)} \right) dx \\
&= \int \tan^{-1} \left(\cot\left(\frac{\pi}{4} - \frac{x}{2}\right) \right) dx \\
&= \int \tan^{-1} \left(\tan\left(\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{x}{2}\right)\right) \right) dx \\
&= \int \tan^{-1} \left(\tan\left(\frac{\pi}{4} + \frac{x}{2}\right) \right) dx \\
&= \int \left(\frac{\pi}{4} + \frac{x}{2} \right) dx \\
&= \left(\frac{\pi x}{4} + \frac{x^2}{4} + c \right)
\end{aligned}$$

52. The given integral is

$$\begin{aligned}
& \int \tan^{-1} \left(\frac{1 - \sin x}{\cos x} \right) dx \\
&= \int \tan^{-1} \left(\frac{1 - \cos\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right)} \right) dx \\
&= \int \tan^{-1} \left(\frac{2 \sin^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}{2 \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \cos\left(\frac{\pi}{4} - \frac{x}{2}\right)} \right) dx \\
&= \int \tan^{-1} \left(\tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \right) dx \\
&= \int \left(\frac{\pi}{4} - \frac{x}{2} \right) dx \\
&= \frac{\pi x}{4} - \frac{x^2}{4} + c
\end{aligned}$$

53. The given integral is

$$\begin{aligned}
& \int \tan^{-1} \left(\frac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}} \right) dx \\
&= \int \tan^{-1} \left(\frac{\left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right) \right) + \left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right) \right)}{\left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right) \right) - \left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right) \right)} \right) dx \\
&= \int \tan^{-1} \left(\frac{\cos\left(\frac{x}{2}\right)}{\sin\left(\frac{x}{2}\right)} \right) dx \\
&= \int \tan^{-1} \left[\cot\left(\frac{x}{2}\right) \right] dx \\
&= \int \tan^{-1} \left[\tan\left(\frac{\pi}{2} - \frac{x}{2}\right) \right] dx \\
&= \int \left(\frac{\pi}{2} - \frac{x}{2} \right) dx \\
&= \frac{\pi x}{2} - \frac{x^2}{4} + c
\end{aligned}$$

54. The given integral is

$$\begin{aligned}
& \int \tan^{-1} (\sec x + \tan x) dx \\
&= \int \tan^{-1} \left(\frac{1 + \sin x}{\cos x} \right) dx \\
&= \int \tan^{-1} \left(\frac{1 + \cos\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right)} \right) dx \\
&= \int \tan^{-1} \left(\frac{2 \cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}{2 \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \cos\left(\frac{\pi}{4} - \frac{x}{2}\right)} \right) dx \\
&= \int \tan^{-1} \left[\cot\left(\frac{\pi}{4} - \frac{x}{2}\right) \right] dx \\
&= \int \tan^{-1} \left[\tan\left\{ \frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{x}{2} \right) \right\} \right] dx \\
&= \int \left(\frac{\pi}{4} + \frac{x}{2} \right) dx \\
&= \frac{\pi x}{4} + \frac{x^2}{4} + c
\end{aligned}$$

55. The given integral is

$$\begin{aligned}
& \int \tan^{-1} \left(\frac{\sin 2x}{1 + \cos 2x} \right) dx \\
&= \int \tan^{-1} \left(\frac{2 \sin x \cos x}{2 \cos^2 x} \right) dx
\end{aligned}$$

$$\begin{aligned}
 &= \int \tan^{-1}(\tan x) dx \\
 &= \int x dx \\
 &= \frac{x^2}{2} + c
 \end{aligned}$$

56. We have,

$$\begin{aligned}
 \int (3x+2) dx &= \int \frac{(3x+2)^2}{2 \cdot 3} + c \\
 &= \int \frac{(3x+2)^2}{6} + c
 \end{aligned}$$

57. We have,

$$\int \frac{dx}{2x-3} = \frac{1}{2} \log |2x-3| + c$$

58. We have,

$$\int \frac{dx}{5-2x} = -\frac{1}{2} \log |5-2x| + c$$

59. We have,

$$\int e^{ax+b} dx = \frac{e^{ax+b}}{a} + c$$

60. We have,

$$\int 3^{4x+5} dx = \frac{3^{4x+5}}{4 \log 3} + c$$

61. We have,

$$\int \cos(5x+3) dx = \frac{\sin(5x+3)}{5} + c$$

62. We have,

$$\int \sin 2x dx = -\frac{\cos 2x}{2} + c$$

63. We have,

$$\begin{aligned}
 \int \sqrt{3x+2} dx &= \frac{(3x+2)^{3/2}}{\frac{3}{2} \cdot 3} + c \\
 &= \frac{2}{9} (3x+2)^{3/2} + c
 \end{aligned}$$

64. We have,

$$\begin{aligned}
 \int (3x+4)^{-\frac{1}{2}} dx &= \frac{(3x+4)^{\frac{1}{2}}}{\frac{1}{2} \cdot 3} + c \\
 &= \frac{2}{3} (3x+4)^{\frac{1}{2}} + c
 \end{aligned}$$

65. We have,

$$\begin{aligned}
 &\int \frac{dx}{\sqrt{x+2} - \sqrt{x+1}} \\
 &= \int \frac{(\sqrt{x+2} + \sqrt{x+1}) dx}{(\sqrt{x+2} - \sqrt{x+1})(\sqrt{x+2} + \sqrt{x+1})}
 \end{aligned}$$

$$\begin{aligned}
 &= \int \frac{(\sqrt{x+2} + \sqrt{x+1}) dx}{(x+2) - (x+1)} \\
 &= \int (\sqrt{x+2} + \sqrt{x+1}) dx \\
 &= \frac{2}{3} (x+2)^{3/2} + \frac{2}{3} (x+1)^{3/2} + c
 \end{aligned}$$

66. We have,

$$\begin{aligned}
 &\int \frac{dx}{(\sqrt{2x+5} - \sqrt{2x+3})} \\
 &= \int \frac{(\sqrt{2x+5} + \sqrt{2x+3}) dx}{(\sqrt{2x+5} - \sqrt{2x+3})(\sqrt{2x+5} + \sqrt{2x+3})} \\
 &= \int \frac{(\sqrt{2x+5} + \sqrt{2x+3}) dx}{(2x+5) - (2x+3)} \\
 &= \frac{1}{2} \int (\sqrt{2x+5} + \sqrt{2x+3}) dx \\
 &= \frac{1}{2} \left(\frac{2}{3} \times \frac{1}{2} (2x+5)^{3/2} + \frac{2}{3} \times \frac{1}{2} (2x+3)^{3/2} \right) + c \\
 &= \frac{1}{6} ((2x+5)^{3/2} + (2x+3)^{3/2}) + c
 \end{aligned}$$

67. The given integral is

$$\begin{aligned}
 &\int \frac{dx}{\sqrt{3x+4} - \sqrt{3x+1}} \\
 &= \int \frac{(\sqrt{3x+4} + \sqrt{3x+1}) dx}{(\sqrt{3x+4} - \sqrt{3x+1})(\sqrt{3x+4} + \sqrt{3x+1})} \\
 &= \frac{1}{3} \int (\sqrt{3x+4} + \sqrt{3x+1}) dx \\
 &= \frac{1}{3} \left(\frac{(3x+4)^{3/2}}{9/2} + \frac{(3x+1)^{3/2}}{9/2} \right) + c
 \end{aligned}$$

68. The given integral is

$$\begin{aligned}
 &\int \frac{dx}{\sqrt{2x+3} + \sqrt{2x-3}} \\
 &= \int \frac{(\sqrt{2x+3} - \sqrt{2x-3}) dx}{(\sqrt{2x+3} + \sqrt{2x-3})(\sqrt{2x+3} - \sqrt{2x-3})} \\
 &= \frac{1}{6} \int (\sqrt{2x+3} - \sqrt{2x-3}) dx \\
 &= \frac{1}{6} \left(\frac{(2x+3)^{3/2}}{3} - \frac{(2x-3)^{3/2}}{3} \right) + c
 \end{aligned}$$

69. The given integral is

$$\begin{aligned}
 &\int \frac{dx}{\sqrt{x+1} + \sqrt{x}} = \int (\sqrt{x+1} - \sqrt{x}) dx \\
 &= \frac{2}{3} ((x+1)^{3/2} - (x)^{3/2}) + c
 \end{aligned}$$

70. The given integral is

$$\begin{aligned} & \int \frac{dx}{\sqrt{x+a} + \sqrt{x+b}} \\ &= \int \left(\frac{\sqrt{x+a} - \sqrt{x+b}}{x+a-x-b} \right) dx \\ &= \frac{1}{(a-b)} \int (\sqrt{x+a} - \sqrt{x+b}) dx \\ &= \frac{2}{3(a-b)} ((x+a)^{3/2} - (x+b)^{3/2}) + c \end{aligned}$$

71. The given integral is

$$\begin{aligned} & \int \frac{dx}{\sqrt{2x+2014} + \sqrt{2x+2013}} \\ &= \int \frac{\sqrt{2x+2014} - \sqrt{2x+2013}}{(2x+2014) - (2x+2013)} dx \\ &= \int (\sqrt{2x+2014} - \sqrt{2x+2013}) dx \\ &= \frac{1}{3} [(2x+2014)^{3/2} - (2x+2013)^{3/2}] + c \end{aligned}$$

72. We have,

$$\begin{aligned} \int \frac{x}{\sqrt{x-1}} dx &= \int \frac{(t^2+1)}{t} \times 2t dt. \\ & \text{Let } x-1 = t^2 \\ & \Rightarrow dx = 2t dt \\ &= 2 \int (t^2+1) dt \\ &= 2 \left(\frac{t^3}{3} + t \right) + c \\ &= 2 \left(\frac{(x-1)^{3/2}}{3} + \sqrt{x-1} \right) + c \end{aligned}$$

73. We have,

$$\begin{aligned} \int \frac{\sqrt{x}}{x+1} dx &= \int \frac{t}{t^2+1} \cdot 2t dt, \quad \text{Let } x = t^2 \\ & \Rightarrow dx = 2t dt \\ &= 2 \int \frac{t^2}{t^2+1} dt \\ &= 2 \int \frac{(t^2+1-1)}{t^2+1} dt \\ &= 2 \int \left(1 - \frac{1}{t^2+1} \right) dt \\ &= 2(t - \tan^{-1}t) + c \\ &= 2(\sqrt{x} - \tan^{-1}(\sqrt{x})) + c \end{aligned}$$

74. We have,

$$\begin{aligned} & \int \frac{x}{\sqrt{3x+1}} dx \\ &= \int \frac{(t^2-1)}{t} \times 2t dt, \quad \text{Let } 3x+1 = t^2 \\ & \Rightarrow dx = \frac{2t dt}{3} \\ &= \frac{2}{3} \int (t^2-1) dt \\ &= \frac{2}{3} \left(\frac{t^3}{3} - t \right) + c \\ &= \frac{2}{3} \left(\frac{(3x+1)^{3/2}}{3} - \sqrt{3x+1} \right) + c \end{aligned}$$

75. We have,

$$\begin{aligned} & \int \frac{x+1}{\sqrt{2x+1}} dx \\ & \text{Let } 2x-1 = t^2 \\ & \Rightarrow 2dx = 2t dt \\ & \Rightarrow dx = t dt \end{aligned}$$

$$\begin{aligned} \therefore &= \int \frac{(t^2+1)}{2t} t dt \\ &= \frac{1}{2} \int (t^2+1) dt \\ &= \frac{1}{2} \left(\frac{t^3}{3} + t \right) + c \\ &= \frac{1}{2} \left(\frac{(2x-1)^{3/2}}{3} + (2x-1)^{1/2} \right) + c \end{aligned}$$

76. We have,

$$\begin{aligned} & \int \frac{x-1}{\sqrt{x+4}} dx \\ & \text{Let } x+4 = t^2 \\ & \Rightarrow dx = 2t dt \\ &= \int \left(\frac{t^2-5}{t} \right) \times t dt \\ &= \int (t^2-5) dt \\ &= \left(\frac{t^3}{3} - 5t \right) + c \\ &= \frac{(x-1)^{3/2}}{3} - 5(x-1)^{1/2} + c \end{aligned}$$

77. We have,

$$\begin{aligned} & \int \frac{x}{x^2+1} dx \\ & \text{Let } x^2+1 = t \end{aligned}$$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow x dx = \frac{1}{2} dt$$

$$= \frac{1}{2} \int \frac{dt}{t}$$

$$= \frac{1}{2} \log |t| + c$$

$$= \frac{1}{2} \log |x^2 + 1| + c$$

78. We have,

$$\int \frac{\cos x - \sin x}{\sin x + \cos x} dx$$

$$\text{Let } \sin x + \cos x = t$$

$$\Rightarrow (\cos x - \sin x) dx = dt$$

$$= \int \frac{dt}{t}$$

$$= \log |t| + c$$

$$= \log |\sin x + \cos x| + c$$

79. We have,

$$\int \frac{3 \cos x}{2 \sin x + 5} dx$$

$$\text{Let } 2 \sin x + 5 = t$$

$$\Rightarrow 2 \cos x dx = dt$$

$$\Rightarrow \cos x dx = \frac{1}{2} dt$$

$$= \frac{3}{2} \int \frac{dt}{t}$$

$$= \frac{3}{2} \log |t| + c$$

$$= \frac{3}{2} \log |2 \sin x + 5| + c$$

80. We have,

$$\int \frac{\cos x - \sin x}{2 + \sin 2x} dx = \int \frac{\cos x - \sin x}{1 + (1 + \sin 2x)} dx$$

$$= \int \frac{\cos x - \sin x}{1 + (\sin x + \cos x)^2} dx$$

$$\text{Let } \sin x + \cos x = t$$

$$\Rightarrow (\cos x - \sin x) dx = dt$$

$$= \int \frac{dt}{1 + t^2}$$

$$= \tan^{-1}(t) + c$$

$$= \tan^{-1}(t) + c$$

81. We have,

$$\int \frac{xe^x + e^x}{\cos^2(xe^x)} dx = \int \frac{(x+1)e^x}{\cos^2(xe^x)} dx$$

$$\text{Let } xe^x = t$$

$$\Rightarrow (xe^x + e^x) dx = dt$$

$$\Rightarrow (x+1)e^x dx = dt$$

$$= \int \frac{dt}{\cos^2 t}$$

$$= \int \sec^2 t dt$$

$$= \tan t + c$$

$$= \tan(xe^x) + c$$

82. We have,

$$\int \frac{dx}{x(1 + \ln x)^2}$$

$$\text{Let } 1 + \ln x = t$$

$$\Rightarrow \frac{1}{x} dx = dt$$

$$= \int \frac{dt}{t^2}$$

$$= -\frac{1}{t} + c$$

$$= -\frac{1}{t} + c$$

$$= -\frac{1}{1 + \ln x} + c$$

83. We have,

$$\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx$$

$$= \int \frac{(e^x + \cos x + 1) - (e^x + \sin x + x)}{e^x + \sin x + x} dx$$

$$= \int \left(\frac{(e^x + \cos x + 1)}{e^x + \sin x + x} - \frac{(e^x + \sin x + x)}{(e^x + \sin x + x)} \right) dx$$

$$= \int \left(\frac{(e^x + \cos x + 1)}{(e^x + \sin x + x)} - \frac{(e^x + \sin x + x)}{(e^x + \sin x + x)} \right) dx$$

$$= \int \left(\frac{(e^x + \cos x + 1)}{(e^x + \sin x + x)} - 1 \right) dx$$

$$= \log |e^x + \sin x + x| - x + c$$

84. We have,

$$\int \frac{dx}{1 + e^x} = \int \frac{e^{-x}}{e^{-x} + 1} dx$$

$$\text{Let } e^{-x} + 1 = t$$

$$\Rightarrow -e^{-x} dx = dt$$

$$= -\int \frac{dt}{t+1}$$

$$= -\log |t+1| + c$$

$$= -\log |e^{-x} + 1| + c$$

85. We have,

$$\begin{aligned}\int \frac{dx}{x(x^3+1)} &= \int \frac{x^2 dx}{x^3(x^3+1)} \\ &= \frac{1}{3} \int \frac{dt}{t(t+1)} \quad \text{Put } x^3 = t \\ &= \frac{1}{3} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt \\ &= \frac{1}{3} \log \left| \frac{t}{t+1} \right| + c \\ &= \frac{1}{3} \log \left| \frac{x^3}{x^3+1} \right| + c\end{aligned}$$

86. We have,

$$\begin{aligned}\int \frac{dx}{x(x^4+1)} &= \int \frac{x^3 dx}{x^4(x^4+1)} \\ &= \frac{1}{4} \int \frac{dt}{t(t+1)} \quad \text{Let } x^4 = t \\ &\Rightarrow x^3 dx = \frac{dt}{4} \\ &= \frac{1}{4} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt \\ &= \frac{1}{4} \log \left| \frac{t}{t+1} \right| + c \\ &= \frac{1}{4} \log \left| \frac{x^4}{x^4+1} \right| + c\end{aligned}$$

87. We have,

$$\begin{aligned}\int \frac{dx}{x(x^5-1)} &= \int \frac{x^4 dx}{x^5(x^5-1)} \\ &= \frac{1}{5} \int \frac{dt}{t(t-1)} \quad \text{Let } x^5 = t \\ &\quad \frac{dx}{x} = \frac{dt}{5t} \\ &= \frac{1}{5} \int \left(\frac{1}{t-1} - \frac{1}{t} \right) dt \\ &= \frac{1}{5} \log \left| \frac{t-1}{t} \right| + c \\ &= \frac{1}{5} \log \left| \frac{x^5-1}{x^5} \right| + c\end{aligned}$$

88. We have,

$$\begin{aligned}\int \frac{\sin 2x}{\sin 5x \sin 3x} dx &= \int \frac{\sin(5x-3x)}{\sin 5x \sin 3x} dx \\ &= \int \frac{\sin 5x \cos 3x - \cos 5x \sin 3x}{\sin 5x \sin 3x} dx \\ &= \int \left(\frac{\sin 5x \cos 3x}{\sin 5x \sin 3x} - \frac{\cos 5x \sin 3x}{\sin 5x \sin 3x} \right) dx\end{aligned}$$

$$\begin{aligned}&= \int (\cot 3x - \cot 5x) dx \\ &= \frac{1}{3} \log |\sin 3x| - \frac{1}{5} \log |\sin 5x| + c\end{aligned}$$

89. We have,

$$\begin{aligned}&\int \frac{dx}{\sin(x-a) \sin(x-b)} \\ &= \frac{1}{\sin(b-a)} \int \frac{\sin(b-a) dx}{\sin(x-a) \sin(x-b)} \\ &= \frac{1}{\sin(b-a)} \int \frac{\sin((x-a)-(x-b)) dx}{\sin(x-a) \sin(x-b)} \\ &= \frac{1}{\sin(b-a)} \\ &\quad \int \left(\frac{\sin(x-a) \cos(x-b) - \cos(x-a) \sin(x-b)}{\sin(x-a) \sin(x-b)} \right) dx \\ &= \frac{1}{\sin(b-a)} \int [\cot(x-b) - \cot(x-a)] dx \\ &= \frac{1}{\sin(b-a)} [\log |\sin(x-b)| - \log |\sin(x-a)|] + c \\ &= \frac{1}{\sin(b-a)} \left(\log \left| \frac{\sin(x-b)}{\sin(x-a)} \right| \right) + c\end{aligned}$$

90. The given integral is

$$\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$$

$$\text{Let } e^x + e^{-x} = t$$

$$\Rightarrow (e^x - e^{-x}) dx = dt$$

$$\begin{aligned}&= \int \frac{dt}{t} \\ &= \log |t| + c \\ &= \log |e^x + e^{-x}| + c\end{aligned}$$

91. The given integral is

$$\int \frac{dx}{1+e^{-x}} = \int \frac{e^x}{e^x+1} dx = \log |e^x+1| + c$$

92. The given integral is

$$\begin{aligned}\int \frac{dx}{1+e^x} &= \int \frac{e^{-x} dx}{e^{-x}+1} \\ &= -\log |e^{-x}+1| + c \\ &= \log \left| \frac{e^x}{e^x+1} \right| + c\end{aligned}$$

93. The given integral is

$$\int \frac{\sin 2x}{a \sin^2 x + b \cos^2 x} dx$$

$$\text{Let } a \sin^2 x + b \cos^2 x = t$$

$$\begin{aligned}
 (a-b)\sin 2x dx &= dt \\
 \sin 2x dx &= \frac{dt}{(a-b)} \\
 &= \frac{1}{(a-b)} \int \frac{dt}{t} \\
 &= \frac{1}{(a-b)} \log|t| + c \\
 &= \frac{1}{(a-b)} \log|a\cos^2 x + b\sin^2 x| + c
 \end{aligned}$$

94. The given integral is

$$\begin{aligned}
 \int \frac{\sin(x-a)}{\sin x} dx &= \int \frac{\sin x \cos a - \cos x \sin a}{\sin x} dx \\
 &= \int (\cos a - \sin a \cot x) dx \\
 &= x \cos a - \sin a \log|\sin x| + c
 \end{aligned}$$

95. The given integral is

$$\begin{aligned}
 \int \frac{\sin x}{\sin(x-a)} dx &= \int \frac{\sin(t+a)}{\sin t} dt, \quad \text{Let } t = x - a \\
 &\Rightarrow dt = dx \\
 &= t \cos a + \sin a \log|\sin t| + c \\
 &= (x-a) \cos a + \sin a \log|\sin(x-a)| + c
 \end{aligned}$$

96. The given integral is

$$\begin{aligned}
 \int \frac{\sin(x+a)}{\sin(x+b)} dx &= \int \frac{\sin[t+(a-b)]}{\sin t} dt, \quad \text{Let } t = (x+b) \\
 &\Rightarrow dt = dx \\
 &= \int \frac{\sin(t) \cos(a-b) + \cos t \sin(a-b)}{\sin t} dt \\
 &= \int (\cos(a-b) + \sin(a-b) \cot t) dt \\
 &= t \cos(a-b) + \sin(a-b) \log|\sin t| + c \\
 &= (x+b) \cos(a-b) + \sin(a-b) \log|\sin(x+b)| + c
 \end{aligned}$$

97. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sqrt{x}(\sqrt{x}+1)} &\quad \text{Let } (\sqrt{x}+1) = t \\
 &\Rightarrow \frac{dx}{2\sqrt{x}} = dt \\
 &\Rightarrow \frac{dx}{\sqrt{x}} = 2dt \\
 &= 2 \int \frac{dt}{t} \\
 &= 2 \log|t| + c \\
 &= 2 \log|\sqrt{x}+1| + c
 \end{aligned}$$

98. The given integral is

$$\begin{aligned}
 \int \frac{1+\tan x}{x+\log \sec x} dx &\quad \text{Let } x + \log \sec x = t \\
 &\Rightarrow (1+\tan x) dx = dt \\
 &= \int \frac{dt}{t} \\
 &= \log|t| + c \\
 &= \log|x + \log(\sec x)| + c
 \end{aligned}$$

99. The given integral is

$$\begin{aligned}
 \int \frac{\sin 2x}{\sin 3x \sin 5x} dx &= \int \frac{\sin(5x-3x)}{\sin 3x \sin 5x} dx \\
 &= \int \frac{\sin 5x \cos 3x - \cos 5x \sin 3x}{\sin 3x \sin 5x} dx \\
 &= \int (\cot 3x - \cot 5x) dx \\
 &= \frac{1}{3} \log|\sin 3x| - \frac{1}{5} \log|\sin 5x| + c
 \end{aligned}$$

100. The given integral is

$$\begin{aligned}
 \int \frac{\cos x - \sin x}{1 + \sin 2x} dx &= \int \frac{(\cos x - \sin x)}{1 + (\sin x + \cos x)^2} dx \\
 &\quad \text{Put } (\sin x + \cos x) = t \\
 &\Rightarrow (\cos x - \sin x) dx = dt \\
 &= \int \frac{dt}{1+t^2} \\
 &= \tan^{-1}(t) + c \\
 &= \tan^{-1}(\sin x + \cos x) + c
 \end{aligned}$$

101. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sin x \cos^2 x} &= \int \frac{(\sin^2 x + \cos^2 x) dx}{\sin x \cos^2 x} \\
 &= \int (\sec x \tan x + \operatorname{cosec} x) dx \\
 &= \sec x + \log \left| \tan \left(\frac{x}{2} \right) \right| + c
 \end{aligned}$$

102. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sin^2 x \cos^2 x} &= \int \frac{(\sin^2 x + \cos^2 x)}{\sin^2 x \cdot \cos^2 x} dx \\
 &= \int (\sec^2 x + \operatorname{cosec}^2 x) dx \\
 &= \tan x - \cot x + c.
 \end{aligned}$$

103. The given integral is

$$\int \frac{dx}{\sin x(x-a) \sin(x-b)}$$

$$\begin{aligned}
&= \frac{1}{\sin(a-b)} \int \frac{\sin(a-b)}{\sin(x-a) \sin(x-b)} dx \\
&= \frac{1}{\sin(a-b)} \int \frac{\sin[(x-b) - (x-a)]}{\sin(x-a) \sin(x-b)} dx \\
&= \frac{1}{\sin(a-b)} \int \frac{\sin(x-b) \cos(x-a) - \cos(x-b) \sin(x-a)}{\sin(x-a) \sin(x-b)} dx \\
&= \frac{1}{\sin(a-b)} \int [\cot(x-a) - \cot(x-b)] dx \\
&= \frac{1}{\sin(a-b)} [\log |\sin(x-a)| - \log |\sin(x-b)|] + c \\
&= \frac{1}{\sin(a-b)} \left(\log \left| \frac{\sin(x-a)}{\sin(x-b)} \right| \right) + c
\end{aligned}$$

104. The given integral is

$$\begin{aligned}
&\int \frac{dx}{\cos(x-a) \cos(x-b)} \\
&= \frac{1}{\sin(a-b)} \int \frac{\sin(a-b)}{\cos(x-a) \cos(x-b)} dx \\
&= \frac{1}{\sin(a-b)} \int \frac{\sin[(x-b) - (x-a)]}{\cos(x-a) \cos(x-b)} dx \\
&= \frac{1}{\sin(a-b)} \int [\tan(x-b) - \tan(x-a)] dx \\
&= \frac{1}{\sin(a-b)} [-\log |\cos(x-b)| + \log |\cos(x-a)|] + c \\
&= \frac{1}{\sin(a-b)} \left(\log \left| \frac{\cos(x-a)}{\cos(x-b)} \right| \right) + c
\end{aligned}$$

105. The given integral is

$$\begin{aligned}
&\int \frac{dx}{\sin(x-a) \cos(x-b)} \\
&= \frac{1}{\cos(a-b)} \int \frac{\cos(a-b)}{\sin(x-a) \cos(x-b)} dx \\
&= \frac{1}{\cos(a-b)} \int \frac{\cos[(x-b) - (x-a)]}{\sin(x-a) \cos(x-b)} dx \\
&= \frac{1}{\cos(a-b)} \int [\tan(x-b) \tan(x-a)] dx \\
&= \frac{1}{\cos(a-b)} \left(\log \left| \frac{\cos(x-a)}{\cos(x-b)} \right| \right) + c
\end{aligned}$$

106. The given integral is

$$\int \frac{x^x(1+\ln x)}{x^x+1} dx = \int \frac{dt}{t}, \quad \text{Let } t = x^x$$

$$\begin{aligned}
&= \log |t| + c \\
&= \log |x^x + 1| + c
\end{aligned}$$

107. The given integral is

$$\begin{aligned}
&\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx \\
&= \int \frac{(\cos x + 1) - (\sin x + x)}{(e^x + \sin x + x)} dx \\
&= \int \frac{(e^x + \cos x + 1) - (e^x + \sin x + x)}{(e^x + \sin x + x)} dx \\
&= \int \left(\frac{(e^x + \cos x + 1)}{(e^x + \sin x + x)} - 1 \right) dx \\
&= \log |e^x + \sin x + x| - x + c
\end{aligned}$$

108. The given integral is

$$\int \frac{\sin^3 x}{(\cos^4 x + 3\cos^2 x + 1) \tan^{-1}(\sec x + \cos x)} dx$$

$$\text{Let } \tan^{-1}(\sec x + \cos x) = t$$

$$\Rightarrow \frac{\sec x \tan x - \sin x}{1 + (\sec x + \cos x)^2} dx = dt$$

$$\Rightarrow \frac{\sin^3 x}{\cos^2 x (1 + (\sec x + \cos x)^2)} dx = dt$$

$$\Rightarrow \frac{\sin^3 x}{\cos^2 x (\sec^2 x + \cos^2 x + 3)} dx = dt$$

$$\Rightarrow \frac{\sin^3 x}{(\cos^4 x + 3\cos^2 x + 1)} dx = dt$$

$$= \int \frac{dt}{t}$$

$$= \log |t| + c$$

$$= \log |\tan^{-1}(\sec x + \cos x)| + c$$

109. We have,

$$\begin{aligned}
\int 3x^3 \sin(x^3) dx &= \int \sin(t) dt \quad \text{where } x^3 = t \\
&= -\cos(t) + c \\
&= -\cos(x^3) + c
\end{aligned}$$

110. We have,

$$\begin{aligned}
\int \frac{(1+\ln x)^3}{x} dx &= \int t^3 dt \quad \text{Let } (1+\ln x) = t \\
&\Rightarrow \frac{dx}{x} = dt \\
&= \frac{t^4}{4} + c \\
&= \frac{(1+\ln x)^4}{4} + c
\end{aligned}$$

111. We have,

$$\begin{aligned}
 \int \frac{dx}{x^2(1+x^4)^{3/4}} &= \int \frac{dx}{x^2 \left(x^4 \left(\frac{1}{x^4} + 1 \right) \right)^{3/4}} \\
 &= \int \frac{dx}{x^5 \left(\frac{1}{x^4} + 1 \right)^{3/4}} \\
 &= -\int \frac{4t^3}{t^3} dt, \text{ Let } \left(\frac{1}{x^4} + 1 \right) = t^4 \\
 &\Rightarrow \frac{-4dx}{x^5} = t^3 dt \\
 &= -4 \int dt \\
 &= -4t + c \\
 &= -4 \left(\frac{1}{x^4} + 1 \right)^{1/4} + c
 \end{aligned}$$

112. We have,

$$\begin{aligned}
 \int \frac{dx}{\sqrt{x}(4+3\sqrt{x})^2} &\quad \text{Let } 4+3\sqrt{x} = t \\
 &\Rightarrow \frac{3}{2\sqrt{x}} dx = dt \\
 &\Rightarrow \frac{dx}{\sqrt{x}} = \frac{2}{3} dt \\
 &= \frac{2}{3} \int \frac{dt}{t^2} \\
 &= -\frac{2}{3} \times \frac{1}{t} + c \\
 &= \frac{2}{3} \times \frac{1}{(4+3\sqrt{x})} + c
 \end{aligned}$$

113. We have,

$$\begin{aligned}
 \int 3^{3^x} \cdot 3^{3^x} \cdot 3^x dx &\quad \text{Let } 3^x = t \\
 &\Rightarrow 3^x dx = \frac{1}{\log_e 3} dt \\
 &= \frac{1}{\log 3} \int 3^{3^x} \cdot 3^t dt \\
 &= \frac{1}{(\log 3)^2} \int 3^{3^t} \cdot 3^t dt, \quad \text{Let } 3^t = v \\
 &\Rightarrow \log_3 3 dt = dv \\
 &= \frac{3^v}{(\log 3)^3} + c \\
 &= \frac{3^{3^x}}{(\log 3)^3} + c
 \end{aligned}$$

114. The given integral is

$$\begin{aligned}
 \int \tan^3 x \cdot \sec^2 x dx &\quad \text{Let } \tan x = t \\
 &\Rightarrow \sec^2 x dx = dt \\
 &= \int t^3 dt \\
 &= \frac{t^4}{4} + c \\
 &= \frac{\tan^4 x}{4} + c
 \end{aligned}$$

115. The given integral is

$$\begin{aligned}
 \int \sin^3 x \cdot \cos x dx &\quad \text{Let } t = \sin x \\
 &\Rightarrow dt = \cos x dx \\
 &= \int t^3 dt \\
 &= \frac{t^4}{4} + c \\
 &= \frac{\sin^4 x}{4} + c
 \end{aligned}$$

116. The given integral is

$$\begin{aligned}
 \int \frac{(\log x)^3}{x} dx &\quad \text{Let } t = \log x \\
 &\Rightarrow dt = \frac{dx}{x} \\
 &= \int t^3 dt \\
 &= \frac{t^4}{4} + c \\
 &= \frac{(\log x)^4}{4} + c
 \end{aligned}$$

117. The given integral is

$$\begin{aligned}
 \int \frac{\sin x}{\sqrt{3+2\cos x}} dx &\quad \text{Let } 3+2\cos x = t^2 \\
 &\Rightarrow -2\sin x dx = 2t dt \\
 &\Rightarrow \sin x dx = -t dt \\
 &= -\int \frac{t dt}{t} \\
 &= -\int dt \\
 &= -t + c \\
 &= -\sqrt{3+2\cos x} + c
 \end{aligned}$$

118. The given integral is

$$\begin{aligned}
 \int \frac{\sqrt{2+\log x}}{x} dx &\quad \text{Let } 2+\log x = t^2 \\
 &\Rightarrow \frac{dx}{x} = 2t dt \\
 &= \int \frac{t}{t} \cdot 2t dt \\
 &= \int 2t dt \\
 &= t^2 + c \\
 &= 2+\log x + c
 \end{aligned}$$

$$\begin{aligned}
&= \int (t \cdot 2t) dt \\
&= 2 \int t^2 dt \\
&= 2 \left(\frac{t^3}{3} \right) + c \\
&= \frac{2}{3} [(2 + \log x)^{3/2}] + c
\end{aligned}$$

119. The given integral is

$$\int \frac{dx}{1 + \sqrt{x}}$$

$$\begin{aligned}
\text{Let } x &= t^2 \\
\Rightarrow dx &= 2t dt
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{2t}{1+t} dt \\
&= 2 \int \left(1 - \frac{1}{1+t} \right) dt \\
&= 2(t - \log|1+t|) + c \\
&= 2(\sqrt{x} - \log|1+\sqrt{x}|) + c
\end{aligned}$$

120. The given integral is

$$\int x^3 \sin x^4 dx$$

$$\begin{aligned}
\text{Let } x^4 &= t \\
\Rightarrow 4x^3 dx &= dt \\
\Rightarrow x^3 dx &= \frac{dt}{4}
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4} \int \sin(t) dt \\
&= -\frac{1}{4} \cos(t) + c \\
&= -\frac{1}{4} \cos(x^4) + c
\end{aligned}$$

121. The given integral is

$$\int 5^{5^x} \cdot 5^{5^x} \cdot 5^x dx$$

$$\begin{aligned}
\text{Put } 5^x &= t \\
\Rightarrow 5^x dx &= \frac{dt}{\log 5}
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\log 5} \int 5^{5^x} \cdot 5^x dx \\
&= \frac{1}{(\log 5)^2} \int 5^z \cdot dz, \quad z = 5^x \\
&= \frac{1}{(\log 5)^2} \int 5^z \cdot dz \\
&= \frac{5^z}{(\log 5)^3} + c \\
&= \frac{5^{5^x}}{(\log 5)^3} + c
\end{aligned}$$

122. The given integral is

$$\begin{aligned}
&\int \frac{\sin x - \cos x}{e^x + \sin x} dx \\
&= \int \frac{(e^x + \sin x) - (e^x + \cos x)}{(e^x + \sin x)} dx \\
&= \int \left(1 - \frac{(e^x + \cos x)}{(e^x + \sin x)} \right) dx \\
&= (x - \log|e^x + \sin x|) + c
\end{aligned}$$

123. The given integral is

$$\begin{aligned}
&\int \frac{d}{x(1+x^3)} = \int \frac{x^2 dx}{x^3(x^3+1)} \\
&= \frac{1}{3} \int \frac{dt}{t(t+1)}, \quad \text{Let } t = x^3 \\
&\quad \Rightarrow dt = 3x^2 \\
&= \frac{1}{3} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt \\
&= \frac{1}{3} \log \left| \frac{t}{t+1} \right| + C \\
&= \frac{1}{3} \log \left| \frac{x^3}{x^3+1} \right| + C
\end{aligned}$$

124. We have,

$$\begin{aligned}
&\int \sin^3 x \cos^4 x dx \\
&= \int (1 - \cos^2 x) \cos^4 x \sin x dx \\
&= -\int (1 - t^2) t^4 dt \quad \text{Let } \cos x = t \\
&\quad \Rightarrow \sin x dx = -dt \\
&= -\int (t^6 - t^4) dt \\
&= \left(-\frac{t^7}{7} + \frac{t^5}{5} \right) + c \\
&= \left(-\frac{\cos^7 x}{7} + \frac{\cos^5 x}{5} \right) + c
\end{aligned}$$

127. We have,

$$\begin{aligned}
&\int \sin^2 x \cos^3 x dx = \int \sin^2 x (1 - \sin^2 x) \cos x dx \\
&= \int t^2 (1 - t^2) dt, \quad \text{Let } \sin x = t \\
&\quad \Rightarrow \cos x dx = dt \\
&= \int (t^2 - t^4) dt \\
&= \left(\frac{t^3}{3} - \frac{t^5}{5} \right) + c \\
&= \left(\frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} \right) + c
\end{aligned}$$

130. We have,

$$\begin{aligned}
 \int \sin^3 x \cos^3 x dx &= \int \sin^3 x (1 - \sin^2 x) \cos x dx \\
 &= \int t^3 (1 - t^2) dt \quad \text{Let } \sin x = t \\
 &\quad \Rightarrow \cos x dx = dt \\
 &= \int (t^3 - t^5) dt \\
 &= \left(\frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} \right) + c
 \end{aligned}$$

134. We have,

$$\begin{aligned}
 \int \sin^2 x \cos^2 x dx &= \frac{1}{4} \int (2 \sin^2 x) (2 \cos^2 x) dx \\
 &= \frac{1}{4} \int (1 - \cos 2x) (1 - \cos 2x) dx \\
 &= \frac{1}{4} \int (1 - \cos^2 2x) dx \\
 &= \frac{1}{4} \int \sin^2 2x dx \\
 &= \frac{1}{8} \int 2 \sin^2 2x dx \\
 &= \frac{1}{8} \int (1 - \cos 4x) dx \\
 &= \frac{1}{8} \left(x - \frac{\sin 4x}{4} \right) + c
 \end{aligned}$$

135. We have,

$$\begin{aligned}
 \int \sin^3 x dx &= \int \sin^2 x \cdot \sin x dx \\
 &= \int (1 - \cos^2 x) \cdot \sin x dx \\
 &= -\int (1 - t^2) dt, \quad \text{Let } \cos x = t \\
 &\quad \Rightarrow -\sin x dx = dt \\
 &= \int (t^2 - 1) dt \\
 &= \left(\frac{t^3}{3} - t \right) + c \\
 &= \left(\frac{\cos^3 x}{3} - \cos x \right) + c
 \end{aligned}$$

139. We have,

$$\begin{aligned}
 \int \cos^5 x dx &= \int \cos^4 x \cdot \cos x dx \\
 &= \int (1 - \sin^2 x)^2 \cdot \cos x dx \\
 &= \int (1 - t^2)^2 dt, \quad \text{Let } \sin x = t \\
 &\quad \Rightarrow \cos x dx = dt
 \end{aligned}$$

$$\begin{aligned}
 &= \int (t^4 - 2t^2 + 1) dt \\
 &= \left(\frac{t^5}{5} - \frac{2}{3} t^3 + t \right) + c \\
 &= \left(\frac{\sin^5 x}{5} - \frac{2}{3} \sin^3 x + \sin x \right) + c
 \end{aligned}$$

142. We have,

$$\begin{aligned}
 \int \sin^4 x dx &= \frac{1}{4} \int (2 \sin^2 x)^2 dx \\
 &= \frac{1}{4} \int (1 - \cos 2x)^2 dx \\
 &= \frac{1}{4} \int (1 - 2 \cos 2x + \cos^2 2x) dx \\
 &= \frac{1}{4} \int (1 - 2 \cos 2x) dx + \frac{1}{8} \int (2 \cos^2 2x) dx \\
 &= \frac{1}{4} \int (1 - 2 \cos 2x) dx + \frac{1}{8} \int (1 + \cos 4x) dx \\
 &= \frac{1}{4} \int \left(x - \frac{2 \sin 2x}{2} \right) + \frac{1}{8} \left(x + \frac{\sin 4x}{4} \right) + c \\
 &= \frac{3}{8} x - \frac{1}{4} \sin 2x + \frac{1}{32} \sin 4x + c
 \end{aligned}$$

145. We have,

$$\begin{aligned}
 \int \cos^6 x dx &= \frac{1}{8} \int (2 \cos^2 x)^3 dx \\
 &= \frac{1}{8} \int (1 + \cos 2x)^3 dx \\
 &= \frac{1}{8} \int (1 + 3 \cos 2x + 3 \cos^2 2x + \cos^3 2x) dx \\
 &= \frac{1}{8} \int (1 + 3 \cos 2x) dx + \frac{3}{16} \int (2 \cos^2 2x) dx \\
 &\quad + \frac{1}{32} \int (4 \cos^3 2x) dx \\
 &= \frac{1}{8} \int (1 + 3 \cos 2x) dx + \frac{3}{16} \int (1 + \cos 4x) dx \\
 &\quad + \frac{1}{32} \int (\cos 6x + 3 \cos 2x) dx \\
 &= \frac{1}{8} \left(x + \frac{3}{2} \sin 2x \right) + \frac{3}{16} \left(x + \frac{\sin 4x}{4} \right) \\
 &\quad + \frac{1}{32} \left(\frac{\sin 6x}{6} + \frac{3}{2} \sin 2x \right) + c \\
 &= \frac{5}{16} x + \frac{15}{64} \sin 2x + \frac{3}{64} \sin 4x + \frac{1}{192} \sin 6x + c
 \end{aligned}$$

148. We have,

$$\int \frac{dx}{\sin^{1/2} x \cos^{3/2} x}$$

Divide the numerator and the denominator by $\cos^2 x$, we get

$$\begin{aligned}
 \int \frac{\sec^2 x}{\sqrt{\tan x}} dx &= \int \frac{2t dt}{t}, \quad \text{Let } \tan x = t^2 \\
 &\Rightarrow \sec^2 x dx = dt \\
 &= 2 \int dt \\
 &= 2t + c \\
 &= 2\sqrt{\tan x} + c
 \end{aligned}$$

149. We have,

$$\int \frac{dx}{\sin^{3/2} x \cos^{5/2} x}$$

Divide the numerator and the denominator by $\cos^4 x$, we get

$$\begin{aligned}
 \int \frac{\sec^4 x}{\tan^{3/2} x} dx &= \int \frac{(1 + \tan^2 x) \sec^2 x}{\tan^{3/2} x} dx \\
 &= \int \frac{(1 + t^2) dt}{t^{3/2}} \quad \text{Let } \tan x = t \\
 &\Rightarrow \sec^2 x dx = dt \\
 &= \int (t^{-3/2} + t^{1/2}) dt \\
 &= 2\sqrt{t} + \frac{2}{3} t^{3/2} + c \\
 &= 2\sqrt{\tan x} + \frac{2}{3} (\tan x)^{3/2} + c
 \end{aligned}$$

150. The given integral is

$$\begin{aligned}
 \int \frac{\sqrt{\tan x}}{\sin x \cos x} dx &= \int \frac{\sqrt{\tan x}}{\sin x \cos x} dx \\
 &= \int \frac{\sec^2 x \sqrt{\tan x}}{\tan x} dx \\
 &= \int \frac{\sec^2 x}{\sqrt{\tan x}} dx \\
 &= \int \frac{2t dt}{t}, \quad \text{Let } \tan x = t^2 \\
 &\Rightarrow \sec^2 x dx = 2t dt \\
 &= \int 2 dt \\
 &= 2t + c \\
 &= 2\sqrt{\tan x} + c
 \end{aligned}$$

151. The given integral is

$$\begin{aligned}
 \int \frac{\sin x}{\cos^5 x} dx &= \int \sec^4 x \tan x dx \\
 &= \int (1 + \tan^2 x) \tan x \sec^2 x dx \\
 &= \int (1 + t^2) t dt \quad \text{Let } \tan x = t \\
 &\Rightarrow \sec^2 x dx = dt \\
 &= \int (t + t^3) dt
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{t^2}{2} + \frac{t^4}{4} + C \\
 &= \frac{\tan^2 x}{2} + \frac{\tan^4 x}{4} + C
 \end{aligned}$$

152. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sin^3 x \cos^5 x} &= \int \frac{\sec^8 x}{\tan^3 x} dx \\
 &= \int \frac{(1 + \tan^2 x)^3 \sec^2 x}{\tan^3 x} dx \\
 &= \int \frac{(1 + t^2)^3}{t^3} dt, \quad \text{Let } t = \tan x \\
 &\Rightarrow dt = \sec^2 x dx \\
 &= \int \frac{(1 + 3t^2 + 3t^4 + t^6)}{t^3} dt \\
 &= \int \left(\frac{1}{t^3} + \frac{3}{t} + 3t + t^3 \right) dt \\
 &= -\frac{1}{2t^2} + 3 \log |t| + \frac{3t^2}{2} + \frac{t^4}{4} + c
 \end{aligned}$$

where $t = \tan x$

153. The given integral is

$$\begin{aligned}
 \int \frac{\sin^2 x dx}{\cos^6 x} &= \int \tan^2 x \sec^4 x dx \\
 &= \int \tan^2 x (1 + \tan^2 x) \sec^2 x dx \\
 &= \int t^2 (1 + t^2) dt \quad \text{Let } \tan x = t \\
 &\Rightarrow \sec^2 x dx = dt \\
 &= \int (t^4 + t^2) dt \\
 &= \frac{t^5}{5} + \frac{t^3}{3} + c \\
 &= \frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + c
 \end{aligned}$$

154. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sin x \cos^3 x} &= \int \frac{\sec^4 x}{\tan x} dx \\
 &= \int \frac{(1 + \tan^2 x) \sec^2 x}{\tan x} dx \\
 &= \int \frac{(1 + t^2)}{t} dt \quad \text{Let } \tan x = t \\
 &= \int \left(1 + \frac{1}{t} \right) dt \\
 &= \frac{t^2}{2} + \log |t| + C \\
 &= \frac{\tan^2 x}{2} + \log |\tan x| + c
 \end{aligned}$$

155. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sin^2 x \cos^4 x} &= \int \frac{\sec^6 x}{\tan^2 x} dx \\
 &= \int \frac{(1 + \tan^2 x)^2 \sec^2 x}{\tan^2 x} dx \\
 &= \int \frac{(1 + t^2)^2}{t^2} dt \quad \text{Let } \tan x = t \\
 &\quad \Rightarrow \sec^2 x dx = dt \\
 &= \int \frac{(t^4 + 2t^2 + 1)}{t^2} dt \\
 &= \int \left(t^2 + \frac{1}{t^2} + 2 \right) dt \\
 &= \frac{t^3}{3} - \frac{1}{t} + 2t + c \\
 &= \frac{\tan^3 x}{3} + 2 \tan x - \cot x + c
 \end{aligned}$$

156. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sin^2 x \cos^2 x} &= \int \frac{\sec^4 x}{\sqrt{\tan x}} dx \\
 &= \int \frac{(1 + \tan^2 x) \sec^2 x}{\sqrt{\tan x}} dx \\
 &= \int \frac{(1 + t^2) 2t}{t} dt \quad \text{Let } \tan x = t \\
 &\quad \Rightarrow \sec^2 x dx = dt \\
 &= 2 \int (1 + t^2) dt \\
 &= 2 \left(t + \frac{t^3}{3} \right) + c \\
 &= 2 \left(\tan x + \frac{\tan^3 x}{3} \right) + c
 \end{aligned}$$

157. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sqrt{\sin^3 x \cos^5 x}} &= \int \frac{\sec^4 x}{\tan^{3/2} x} dx \\
 &= \int \frac{(1 + \tan^2 x) \sec^2 x}{\tan^{3/2} x} dx \\
 &= \int \frac{(1 + t^2) 2t}{t^3} dt, \quad \text{Let } \tan x = t^2 \\
 &\quad \Rightarrow \sec^2 x dx = 2t dt \\
 &= 2 \int \frac{(1 + t^2)}{t} dt \\
 &= 2 \int \left(t + \frac{1}{t} \right) dt \\
 &= 2 \left(\frac{t^2}{2} + \log |t| \right) + C
 \end{aligned}$$

$$= 2 \left(\frac{\tan^2 x}{2} + \log |\tan x| \right) + C$$

158. We have,

$$\int \frac{dx}{x^2 + 4} = \int \frac{dx}{x^2 + 2^2} = \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$$

159. We have,

$$\int \frac{dx}{9x^2 + 1} = \frac{1}{9} \int \frac{dx}{x^2 + \left(\frac{1}{3}\right)^2} = \frac{1}{9} \times 3 \tan^{-1}(3x) + c$$

160. We have,

$$\begin{aligned}
 \int \frac{dx}{x^2 - 4} &= \int \frac{dx}{x^2 - (2)^2} \\
 &= \frac{1}{2 \cdot 2} \log \left| \frac{x-2}{x+2} \right| + c \\
 &= \frac{1}{4} \log \left| \frac{x-2}{x+2} \right| + c
 \end{aligned}$$

161. We have $\int \frac{x^4 - 1}{x^2 + 4} dx$

$$\begin{aligned}
 &= \int \frac{(x^4 - 16) + 15}{x^2 + 4} dx \\
 &= \int \frac{((x^2 + 4)(x^2 - 4)) + 15}{x^2 + 4} dx \\
 &= \int \left(x^2 - 4 + \frac{5}{x^2 + 2^2} \right) dx \\
 &= \frac{x^3}{3} - 4x + \frac{5}{2} \tan^{-1} \left(\frac{x}{2} \right) + c
 \end{aligned}$$

162. We have,

$$\begin{aligned}
 \int \frac{dx}{\sqrt{x^2 + 4}} &= \int \frac{dx}{\sqrt{x^2 + 2^2}} \\
 &= \log(x + \sqrt{x^2 + 4}) + c
 \end{aligned}$$

163. We have,

$$\begin{aligned}
 \int \frac{dx}{\sqrt{4x^2 + 1}} &= \frac{1}{2} \int \frac{dx}{\sqrt{x^2 + \left(\frac{1}{2}\right)^2}} \\
 &= \frac{1}{2} \log \left| x + \sqrt{x^2 + \frac{1}{4}} \right| + c
 \end{aligned}$$

164. We have,

$$\begin{aligned}
 \int \left(\frac{x+4}{x^3 + 4x} \right) dx &= \int \left(\frac{x}{x^3 + 4x} + \frac{4}{x^3 + 4x} \right) dx \\
 &= \int \left(\frac{1}{x^2 + 4} + \frac{4}{x(x^2 + 4)} \right) dx
 \end{aligned}$$

$$\begin{aligned}
&= \int \frac{dx}{x^2 + 4} + \int \frac{4x}{x^2(x^2 + 4)} dx \\
&= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + 2 \int \frac{dt}{t(t+4)}, \text{ Let } t = x^2 \\
&\quad \Rightarrow dt = 2x^2 dx \\
&= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + \frac{2}{4} \int \left(\frac{1}{t} - \frac{1}{t+4} \right) dt \\
&= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + \frac{1}{2} \log \left| \frac{t}{t+4} \right| + c \\
&= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + \frac{1}{2} \log \left| \frac{x^2}{x^2 + 4} \right| + c
\end{aligned}$$

165. The given integral is

$$\begin{aligned}
\int \frac{x^4 + 1}{x^2 + 1} dx &= \int \frac{(x^4 - 1) + 2}{x^2 + 1} dx \\
&= \int \left(x^2 - 1 + \frac{2}{x^2 + 1} \right) dx \\
&= \left(\frac{x^3}{3} - x + 2 \tan^{-1} x \right) + c
\end{aligned}$$

166. The given integral is

$$\begin{aligned}
\int \frac{dx}{\sqrt{(2-x)^2 + 1}} &= \int \frac{dx}{\sqrt{(x-2)^2 + 1}} \\
&= \int \frac{dt}{\sqrt{t^2 + 1}}, \quad \text{Let } t = (x-2) \\
&\quad \Rightarrow dt = dx \\
&= \log |t + \sqrt{t^2 + 1}| + c \\
&= \log |(x-2) + \sqrt{(x-2)^2 + 1}| + c
\end{aligned}$$

167. The given integral is

$$\begin{aligned}
\int \frac{x+9}{x^3+9x} dx &= \int \frac{x+9}{x(x^2+9)} dx \\
&= \int \frac{x + (9+x^2) - x^2}{x(x^2+9)} dx \\
&= \int \frac{dx}{(x^2+9)} + \int \frac{dx}{x} - \int \frac{xdx}{(x^2+9)} \\
&= \frac{1}{3} \tan^{-1} \left(\frac{x}{3} \right) + \log |x| - \frac{1}{2} \log |x^2 + 9| + c
\end{aligned}$$

168. The given integral is

$$\begin{aligned}
\int \frac{1+x}{1+x^2} dx &= \int \frac{dx}{1+x^2} + \int \frac{x}{1+x^2} dx \\
&= \tan^{-1}(x) + \frac{1}{2} \log |1+x^2| + c
\end{aligned}$$

169. The given integral is

$$\begin{aligned}
\int \frac{1+x}{x^3+x} dx &= \int \frac{(1+x)dx}{x(x^2+1)} \\
&= \int \frac{(1+x^2) + x - x^2}{x(x^2+1)} dx \\
&= \int \frac{dx}{x} + \int \frac{dx}{x^2+1} - \int \frac{x}{x^2+1} dx \\
&= \log |x| + \tan^{-1} x - \frac{1}{2} \log |x^2 + 1| + c
\end{aligned}$$

170. The given integral is

$$\begin{aligned}
\int \frac{dx}{x^4-1} &= \int \frac{dx}{(x^2-1)(x^2+1)} \\
&= \frac{1}{2} \int \left(\frac{1}{x^2-1} - \frac{1}{x^2+1} \right) dx \\
&= \frac{1}{2} \left(\frac{1}{2} \log \left| \frac{x-1}{x+1} \right| - \tan^{-1} x \right) + c
\end{aligned}$$

171. The given integral is

$$\begin{aligned}
\int \frac{dx}{x^3+x} &= \int \frac{dx}{x(x^2+1)} \\
&= \int \frac{xdx}{x^2(x^2+1)} \\
&= \frac{1}{2} \int \frac{dt}{t(t+1)}, \quad \text{Let } x^2 = t^2 \\
&\quad 2x dx = 2t dt \\
&= \frac{1}{2} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt \\
&= \frac{1}{2} \log \left| \frac{t}{t+1} \right| + c \\
&= \frac{1}{2} \log \left| \frac{x^2}{x^2+1} \right| + c
\end{aligned}$$

172. We have,

$$\begin{aligned}
\int \frac{dx}{x^2+4x+4} &= \int \frac{dx}{(x^2+2)^2} \\
&= -\frac{1}{(x+2)} + c
\end{aligned}$$

173. We have,

$$\begin{aligned}
\int \frac{dx}{x^2+6x+10} &= \int \frac{dx}{(x+3)^2 + (10-9)} \\
&= \int \frac{dx}{(x+3)^2 + 1} \\
&= \tan^{-1}(x+3) + c
\end{aligned}$$

174. We have,

$$\begin{aligned}
 \int \frac{dx}{2x^2 + 5x + 6} &= \frac{1}{2} \int \frac{dx}{x^2 + \frac{5}{2}x + 3} \\
 &= \frac{1}{2} \int \frac{dx}{\left(x + \frac{5}{4}\right)^2 + \left(3 - \frac{25}{16}\right)} \\
 &= \frac{1}{2} \int \frac{dx}{\left(x + \frac{5}{4}\right)^2 + \left(\frac{\sqrt{23}}{4}\right)^2} \\
 &= \frac{1}{2} \times \frac{4}{\sqrt{23}} \tan^{-1} \left(\frac{\left(x + \frac{5}{4}\right)}{\frac{\sqrt{23}}{4}} \right) + c \\
 &= \frac{2}{\sqrt{23}} \tan^{-1} \left(\frac{(4x + 5)}{\sqrt{23}} \right) + c
 \end{aligned}$$

175. The given integral is

$$\begin{aligned}
 \int \frac{dx}{x^2 + x + 1} &= \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\
 &= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\left(x + \frac{1}{2}\right)}{\frac{\sqrt{3}}{2}} \right) + c \\
 &= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x + 1}{\sqrt{3}} \right) + c
 \end{aligned}$$

176. The given integral is

$$\begin{aligned}
 \int \frac{dx}{1 + x - x^2} &= - \int \frac{dx}{x^2 - x - 1} \\
 &= - \int \frac{dx}{\left(x - \frac{1}{2}\right)^2 - \left(\frac{\sqrt{5}}{2}\right)^2} \\
 &= - \frac{1}{\sqrt{5}} \log \left| \frac{\left(x - \frac{1}{2}\right) - \frac{\sqrt{5}}{2}}{\left(x - \frac{1}{2}\right) + \frac{\sqrt{5}}{2}} \right| + c
 \end{aligned}$$

177. The given integral is

$$\begin{aligned}
 \int \frac{dx}{x^2 + 4x + 3} &= \int \frac{dx}{(x + 2)^2 - 1} \\
 &= \frac{1}{2} \log \left| \frac{(x + 2) - 1}{(x + 2) + 1} \right| + c \\
 &= \frac{1}{2} \log \left| \frac{(x + 1)}{(x + 3)} \right| + c
 \end{aligned}$$

178. The given integral is

$$\begin{aligned}
 \int \frac{dx}{4x^2 + 7x + 10} &= \frac{1}{4} \int \frac{dx}{\left(x^2 + \frac{7}{4}x + \frac{5}{2}\right)} \\
 &= \frac{1}{4} \int \frac{dx}{\left(x + \frac{7}{8}\right)^2 + \left(\frac{\sqrt{111}}{8}\right)^2} \\
 &= \frac{1}{4} \times \frac{8}{\sqrt{111}} \tan^{-1} \left(8x + \frac{7}{\sqrt{111}} \right) + c \\
 &= \frac{2}{\sqrt{111}} \tan^{-1} \left(8x + \frac{7}{\sqrt{111}} \right) + c
 \end{aligned}$$

179. The given integral is

$$\begin{aligned}
 \int \frac{dx}{x^2 - 2ax} &= \int \frac{dx}{(x^2 - a)^2 - a^2} \\
 &= \frac{1}{2a} \log \left| \frac{(x - a) - a}{(x - a) + a} \right| + c \\
 &= \frac{1}{2a} \log \left| \frac{(x - 2a)}{x} \right| + c
 \end{aligned}$$

180. The given integral is

$$\begin{aligned}
 \int \frac{dx}{x^2 + 2ax} &= \int \frac{dx}{(x + a)^2 - a^2} \\
 &= \frac{1}{2a} \log \left| \frac{x}{x + 2a} \right| + c
 \end{aligned}$$

181. The given integral is

$$\int \frac{dx}{a^2 + 2ax} = \frac{1}{2a} \log |a^2 + 2ax| + c$$

182. The given integral is

$$\begin{aligned}
 \int \frac{dx}{2ax - x^2} &= - \int \frac{dx}{x^2 - 2ax} \\
 &= - \int \frac{dx}{(x - a)^2 - a^2} \\
 &= - \frac{1}{2a} \log \left| \frac{x - a - a}{x - a + a} \right| + c \\
 &= \frac{1}{2a} \log \left| \frac{x}{x - 2a} \right| + c
 \end{aligned}$$

183. The given integral is

$$\begin{aligned}
 \int \frac{dx}{(x^2 + 1)(x^2 + 4)} &= \frac{1}{3a} \int \left(\frac{1}{x^2 + 1} - \frac{1}{x^2 + 4} \right) dx \\
 &= \frac{1}{3} \left(\tan^{-1} x - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \right) + c
 \end{aligned}$$

184. We have,

$$\begin{aligned}\int \frac{x^2}{x^6 + x} dx &= \frac{1}{3} \int \frac{3x^2}{x^6 + 1} dx \\ &= \frac{1}{3} \int \frac{dt}{t^2 + 1}, \quad \text{Let } t = x^3 \\ &\quad \Rightarrow dt = 3x^2 dx \\ &= \frac{1}{3} \tan^{-1}(t) + c \\ &= \frac{1}{3} \tan^{-1}(x^3) + c\end{aligned}$$

185. We have,

$$\begin{aligned}\int \frac{\cos x dx}{\sin^2 x + 3 \sin x + 2} \\ &= \int \frac{dt}{t^2 + 3t + 2} \quad \text{Let } t = \sin x \\ &\quad \Rightarrow dt = \cos x dx \\ &= \int \frac{dt}{(t+1)(t+2)} \\ &= \int \left(\frac{1}{t+1} - \frac{1}{t+2} \right) dt \\ &= \log \left| \frac{t+1}{t+2} \right| + c \\ &= \log \left| \frac{\sin x + 1}{\sin x + 2} \right| + c\end{aligned}$$

186. We have,

$$\begin{aligned}\int \frac{x^x(1 + \log x)}{x^{2x} + x^x + 1} dx &= \int \frac{dt}{t^2 + t + 1} \\ &= \int \frac{dt}{\left(t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + c \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x^x + 1}{\sqrt{3}} \right) + c\end{aligned}$$

187. We have,

$$\begin{aligned}\int \frac{dx}{x(x^4 + 1)} &= \int \frac{x^3 dx}{x^4(x^4 + 1)} \\ &= \frac{1}{4} \int \frac{4x^3 dx}{x^4(x^4 + 1)} \\ &= \frac{1}{4} \int \frac{dt}{t(t+1)}, \quad \text{Let } t = x^4 \\ &\quad \Rightarrow dt = 4x^3 dx\end{aligned}$$

$$\begin{aligned}&= \frac{1}{4} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) \\ &= \frac{1}{4} \log \left| \frac{t}{t+1} \right| + c \\ &= \frac{1}{4} \log \left| \frac{x^4}{x^4 + 1} \right| + c\end{aligned}$$

194. We have,

$$\begin{aligned}\int \frac{2x+3}{x^2+4x+5} dx &= \int \frac{(2x+4) - 1}{x^2+4x+5} dx \\ &= \int \frac{(2x+4)}{x^2+4x+5} dx - \int \frac{dx}{x^2+4x+5} \\ &= \int \frac{(2x+4)}{x^2+4x+5} dx - \int \frac{dx}{(x+2)^2+1} \\ &= \log |x^2+4x+5| - \tan^{-1}(x+2) + c\end{aligned}$$

195. We have,

$$\begin{aligned}\int \frac{3x+2}{x^2-3x+4} dx \\ &= \int \frac{\frac{3}{2}(2x-3) + \left(2 + \frac{9}{2}\right)}{x^2-3x+4} dx \\ &= \frac{3}{2} \int \frac{(2x-3)}{x^2-3x+4} dx + \frac{13}{2} \int \frac{dx}{x^2-3x+4} \\ &= \frac{3}{2} \int \frac{(2x-3)}{x^2-3x+4} dx + \frac{13}{2} \int \frac{dx}{\left(x - \frac{3}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2} \\ &= \frac{3}{2} \log |x^2-3x+4| + \frac{13}{\sqrt{7}} \tan^{-1} \left(\frac{x - \frac{3}{2}}{\frac{\sqrt{7}}{2}} \right) + c \\ &= \frac{3}{2} \log |x^2-3x+4| + \frac{13}{\sqrt{7}} \tan^{-1} \left(\frac{2x-3}{\sqrt{7}} \right) + c\end{aligned}$$

195. The given integral is

$$\begin{aligned}\int \frac{x}{x^2+x+1} dx &= \frac{1}{2} \int \frac{(2x+1) - 1}{(x^2+x+1)} dx \\ &= \frac{1}{2} \int \frac{(2x+1)}{(x^2+x+1)} dx - \frac{1}{2} \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= \frac{1}{2} \log |x^2+x+1| - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C\end{aligned}$$

196. The given integral is

$$\begin{aligned}\int \frac{4x+1}{x^2+3x+2} dx &= 2 \int \frac{2x+\frac{1}{2}}{(x^2+3x+2)} dx \\ &= 2 \int \frac{(2x+3) - (5/2)}{(x^2+3x+2)} dx \\ &= 2 \int \frac{(2x+3)}{(x^2+3x+2)} dx - 5 \int \frac{dx}{(x+1)(x+2)} \\ &= 2 \log|x^2+3x+2| - 5 \log \left| \frac{x+1}{x+2} \right| + c\end{aligned}$$

197. The given integral is

$$\begin{aligned}\int \frac{dx}{2e^{2x}+3e^x+1} &= \int \frac{dx}{(2e^x+1)(e^x+1)} \\ &= \int \left(\frac{2}{(2e^x+1)} - \frac{1}{(e^x+1)} \right) dx \\ &= \int \left(\frac{2e^{-x}}{(2+e^{-x})} - \frac{e^{-x}}{(e^{-x}+1)} \right) dx \\ &= \log|e^{-x}+1| - 2 \log|e^{-x}+2| + c\end{aligned}$$

198. The given integral is

$$\begin{aligned}\int \frac{(3\sin x - 2) \cos x dx}{(5 - \cos^2 x - 4\sin x)} \\ &= \int \frac{(3\sin x - 2) \cos x dx}{\sin^2 x - 4\sin x + 4} \\ &= \int \frac{(3t - 2)}{t^2 - 4t + 4} dt, \quad \text{Let } t = \sin x \\ &\quad \Rightarrow dt = \cos x dx \\ &= \int \frac{(3t - 2)}{(t - 2)^2} dt \\ &= \int \frac{(3t - 6 + 4)}{(t - 2)^2} dt \\ &= 3 \int \frac{dt}{(t - 2)} + 4 \int \frac{dt}{(t - 2)^2} \\ &= 3 \log|t - 2| - \frac{4}{(t - 2)} + c \\ &= 3 \log|\sin x - 2| - \frac{4}{(\sin x - 2)} + c\end{aligned}$$

199. The given integral is

$$\begin{aligned}\int \frac{ax^3 + bx}{x^4 + c^2} dx &= a \int \frac{x^3}{x^4 + c^2} dx + b \int \frac{x dx}{x^4 + c^2} \\ &= \frac{a}{4} \log|x^4 + c^2| + \frac{b}{2c} \tan^{-1} \left(\frac{x^2}{c} \right) + c\end{aligned}$$

200. The given integral is.

$$\begin{aligned}\int \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) \times (2 + 2\sin 2x) dx \\ &= \int \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right) (1 + (\sin x + \cos x)^2) dx \\ &= \int \left(\frac{1+t^2}{t} \right) dt, \quad \text{Let } \cos x + \sin x = t \\ &\quad \Rightarrow (-\sin x + \cos x) dx = dt \\ &= \int \left(t + \frac{1}{t} \right) dt \\ &= \frac{t^2}{2} + \log|t| + c \\ &= \frac{(\sin x + \cos x)^2}{2} + \log|\sin x + \cos x| + c\end{aligned}$$

201. The given integral is

$$\begin{aligned}\int \frac{\sin x + \cos x}{5 + 3\sin 2x} dx &= \int \frac{(\sin x + \cos x) dx}{5 + 3(1 - 1 + \sin 2x)} \\ &= \int \frac{(\sin x + \cos x) dx}{8 - 3(1 - \sin 2x)} \\ &= \int \frac{(\sin x + \cos x) dx}{8 - 3(\sin x - \cos x)^2} \\ &= \int \frac{dt}{8 - 3t^2}, \quad \text{Let } t = (\sin x - \cos x) \\ &\quad dt = (\cos x + \sin x) dx \\ &= \int \frac{dt}{8 - 3t^2} \\ &= -\frac{1}{3} \int \frac{dt}{t^2 - \left(\sqrt{\frac{8}{3}} \right)^2} \\ &= -\frac{1}{2\sqrt{24}} \log \left| \frac{t - (\sqrt{8/3})}{t + (\sqrt{8/3})} \right| + c\end{aligned}$$

where $t = \sin x + \cos x$

202. The given integral is

$$\begin{aligned}\int \frac{\sin x - \cos x}{3 + 5\sin 2x} dx &= \int \frac{\sin x - \cos x}{5(1 + \sin 2x) - 2} dx \\ &= \int \frac{\sin x - \cos x}{5(\sin x + \cos x)^2 - 2} dx \\ &= \int \frac{\cos x - \sin x}{2 - 5(\sin x + \cos x)^2} dx \\ &= \int \frac{dt}{2 - 5t^2}, \quad \text{Let } t = (\sin x + \cos x) \\ &\quad \Rightarrow dt = (\cos x - \sin x) dx\end{aligned}$$

$$\begin{aligned}
&= \int \frac{dt}{2-5t^2} \\
&= -\frac{1}{5} \int \frac{dt}{t^2 - (\sqrt{2/5})^2} \\
&= -\frac{1}{\sqrt{10}} \log \left| \frac{t - (\sqrt{2/5})}{t + (\sqrt{2/5})} \right| + c \\
&= -\frac{1}{\sqrt{10}} \log \left| \frac{(\sin x + \cos x) - (\sqrt{2/5})}{(\sin x + \cos x) + (\sqrt{2/5})} \right| + c
\end{aligned}$$

203. We have,

$$\begin{aligned}
\int \frac{dx}{\sqrt{x^2 + x + 1}} &= \int \frac{dx}{\sqrt{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}} \\
&= \log \left| \left(x + \frac{1}{2}\right) + \sqrt{x^2 + x + 1} \right| + c
\end{aligned}$$

204. We have,

$$\begin{aligned}
\int \frac{dx}{\sqrt{x^2 - 2ax}} &= \int \frac{dx}{\sqrt{x^2 - 2ax + a^2 - a^2}} \\
&= \int \frac{dx}{\sqrt{(x-a)^2 - a^2}} \\
&= \log |(x-a) + \sqrt{x^2 - 2ax}| + c
\end{aligned}$$

205. We have,

$$\begin{aligned}
\int \frac{dx}{\sqrt{4x - x^2}} &= \int \frac{dx}{\sqrt{4 - 4 + 4x - x^2}} \\
&= \int \frac{dx}{\sqrt{2^2 - (x^2 - 4x + 4)}} \\
&= \int \frac{dx}{\sqrt{2^2 - (x-2)^2}} \\
&= \sin^{-1} \left(\frac{x-2}{2} \right) + c
\end{aligned}$$

205. The given integral is

$$\begin{aligned}
\int \frac{dx}{\sqrt{4x - x^2}} &= \int \frac{dx}{\sqrt{-(x^2 - 4x)}} \\
&= \int \frac{dx}{\sqrt{-(x-2)^2 - 4}} \\
&= \int \frac{dx}{\sqrt{4 - (x-2)^2}} \\
&= \sin^{-1} \left(\frac{x-2}{2} \right) + c
\end{aligned}$$

206. The given integral is

$$\begin{aligned}
\int \frac{dx}{\sqrt{6-x-x^2}} &= \int \frac{dx}{\sqrt{-(x^2+x-6)}} \\
&= \int \frac{dx}{\sqrt{-\left\{\left(x+\frac{1}{2}\right)^2 - \frac{25}{4}\right\}}} \\
&= \int \frac{dx}{\sqrt{\left(\frac{5}{2}\right)^2 - \left(x+\frac{1}{2}\right)^2}} \\
&= \sin^{-1} \left(\frac{\left(x+\frac{1}{2}\right)}{\frac{5}{2}} \right) + c \\
&= \sin^{-1} \left(\frac{2x+1}{5} \right) + c
\end{aligned}$$

207. The given integral is

$$\begin{aligned}
\int \frac{dx}{\sqrt{1+x+x^2}} &= \int \frac{dx}{\sqrt{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}} \\
&= \log \left| \left(x+\frac{1}{2}\right) + \sqrt{x^2+x+1} \right| + c
\end{aligned}$$

208. The given integral is

$$\begin{aligned}
\int \frac{dx}{\sqrt{1+x-x^2}} &= \int \frac{dx}{\sqrt{-\{x^2-x-1\}}} \\
&= \int \frac{dx}{\sqrt{-\left\{\left(x-\frac{1}{2}\right)^2 - \frac{5}{4}\right\}}} \\
&= \int \frac{dx}{\sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x-\frac{1}{2}\right)^2}} \\
&= \sin^{-1} \left(\frac{\left(x-\frac{1}{2}\right)}{\frac{\sqrt{5}}{2}} \right) + c \\
&= \sin^{-1} \left(\frac{2x-1}{\sqrt{5}} \right) + c
\end{aligned}$$

209. The given integral is

$$\begin{aligned}
\int \frac{dx}{x^2+2ax} &= \int \frac{dx}{\sqrt{(x+a)^2 - a^2}} \\
&= \log |(x+a) + \sqrt{x^2+2ax}| + c
\end{aligned}$$

210. The given integral is

$$\int \frac{dx}{\sqrt{2ax-x^2}} = \int \frac{dx}{\sqrt{-\{x^2-2ax\}}}$$

$$\begin{aligned}
&= \int \frac{dx}{\sqrt{-(x-a)^2 - a^2}} \\
&= \int \frac{dx}{\sqrt{a^2 - (x-a)^2}} \\
&= \sin^{-1}\left(\frac{x-a}{a}\right) + c
\end{aligned}$$

211. We have,

$$\begin{aligned}
&\int \frac{x dx}{\sqrt{x^4 + x^2 + 1}} \\
&= \frac{1}{2} \int \frac{dt}{\sqrt{t^2 + t + 1}}, \quad \text{Let } x^2 = t \\
&\quad \Rightarrow x dx = \frac{dt}{2} \\
&= \frac{1}{2} \int \frac{dt}{\sqrt{\left(t + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}} \\
&= \frac{1}{2} \log \left| \left(t + \frac{1}{2}\right) + \sqrt{t^2 + t + 1} \right| + c \\
&= \frac{1}{2} \log \left| \left(x^2 + \frac{1}{2}\right) + \sqrt{x^4 + x^2 + 1} \right| + c
\end{aligned}$$

212. We have,

$$\begin{aligned}
&\int \sqrt{\sec x - 1} dx \\
&= \int \sqrt{\frac{1 - \cos x}{\cos x} \times \frac{(1 + \cos x)}{(1 + \cos x)}} dx \\
&= \int \sqrt{\frac{1 - \cos^2 x}{\cos x(1 + \cos x)}} dx \\
&= \int \frac{\sin x}{\sqrt{\cos x(1 + \cos x)}} dx \\
&= - \int \frac{dt}{\sqrt{t(t+1)}} \quad \text{Let } \cos x = t \\
&\quad \Rightarrow -\sin x dx = dt \\
&= - \int \frac{dt}{\sqrt{\left(t + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2}} \\
&= -\log \left| \left(t + \frac{1}{2}\right) + \sqrt{t^2 + t} \right| + c \\
&= -\log \left| \left(\cos x + \frac{1}{2}\right) + \sqrt{\cos^2 x + \cos x} \right| + c
\end{aligned}$$

213. We have,

$$\begin{aligned}
\int \frac{dx}{x^{3/4} \sqrt{x^{1/2} - 1}} &= 4 \int \frac{dt}{\sqrt{t^2 - 1}}, \quad \text{Let } x^{1/4} = t \\
&\Rightarrow \frac{1}{4} x^{-3/4} = dt
\end{aligned}$$

$$\begin{aligned}
&= 4 \log |t + \sqrt{t^2 - 1}| + c \\
&= 4 \times \log |x^{1/4} + \sqrt{x^{1/2} - 1}| + c
\end{aligned}$$

214. We have,

$$\begin{aligned}
&\int \sqrt{\frac{\sin(x-a)}{\sin(x+a)}} dx = \int \sqrt{\frac{\sin(x-a)}{\sin(x+a)}} dx \\
&= \int \sqrt{\frac{\sin(x-a)}{\sin(x+a)}} \times \frac{\sin(x-a)}{\sin(x-a)} dx \\
&= \int \frac{\sin(x-a)}{\sqrt{\sin^2 x - \sin^2 a}} dx \\
&= \int \frac{(\sin x \cos a - \cos x \sin a)}{\sqrt{\sin^2 x - \sin^2 a}} dx \\
&= \cos a \int \frac{\sin x}{\sqrt{\sin^2 x - \sin^2 a}} dx \\
&\quad - \sin a \int \frac{\cos x}{\sqrt{\sin^2 x - \sin^2 a}} dx \\
&= \cos a \int \frac{\sin x}{\sqrt{\cos^2 a - \cos^2 x}} dx \\
&\quad - \sin a \int \frac{\cos x}{\sqrt{\sin^2 x - \sin^2 a}} dx \\
&= -\cos a \sin^{-1} \left(\frac{\cos x}{\cos a} \right) \\
&\quad - \sin a \log |\sin x + \sqrt{\sin^2 x - \sin^2 a}| + c
\end{aligned}$$

215. The given integral is

$$\begin{aligned}
\int \frac{e^x}{\sqrt{4 - e^{2x}}} dx &= \int \frac{e^x dx}{\sqrt{4 - (e^x)^2}} \\
&= \int \frac{dz}{\sqrt{4 - z^2}}, \quad \text{Let } z = e^x \\
&\quad \Rightarrow dx = e^x dt \\
&= \sin^{-1} \left(\frac{z}{2} \right) + c \\
&= \sin^{-1} \left(\frac{e^x}{2} \right) + c
\end{aligned}$$

216. The given integral is

$$\begin{aligned}
&\int \frac{\sec^2 x}{\sqrt{16 + \tan x}} dx \\
&= \int \frac{dt}{\sqrt{t^2 + 4}}, \quad \text{Let } t = \tan x \\
&\quad \Rightarrow dt = \sec^2 x dx \\
&= \log |t + \sqrt{t^2 + 4}| + c \\
&= \log |\tan x + \sqrt{\tan^2 x + 4}| + c
\end{aligned}$$

217. The given integral is

$$\begin{aligned}
 \int \sqrt{\sec x - 1} \, dx &= \int \sqrt{\frac{1 - \cos x}{\cos x}} \, dx \\
 &= \int \sqrt{\frac{(1 - \cos x)(1 + \cos x)}{\cos x(1 + \cos x)}} \, dx \\
 &= \int \frac{\sin x}{\sqrt{\cos x(1 + \cos x)}} \, dx \\
 &= -\int \frac{dt}{\sqrt{t^2 + t}}, \quad \text{Let } t = \cos x \\
 &\quad \Rightarrow dx = -\sin x \, dx \\
 &= -\int \frac{dt}{\sqrt{\left(t + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2}} \\
 &= -\log \left| \left(t + \frac{1}{2}\right) + \sqrt{t^2 + t} \right| + c \\
 &= -\log \left| \left(\cos x + \frac{1}{2}\right) + \sqrt{\cos^2 x + \cos x} \right| + c
 \end{aligned}$$

218. The given integral is

$$\begin{aligned}
 \int \sqrt{\operatorname{cosec} x - 1} \, dx &= \int \sqrt{\frac{1 - \sin x}{\sin x}} \, dx \\
 &= \int \sqrt{\frac{(1 - \sin x)(1 + \sin x)}{\sin x(1 + \sin x)}} \, dx \\
 &= \int \frac{\cos x}{\sqrt{\sin x(1 + \sin x)}} \, dx \\
 &= \int \frac{dt}{\sqrt{t^2 + t}}, \quad \text{Let } t = \sin x \\
 &\quad \Rightarrow dt = \cos x \, dx \\
 &= \int \frac{dt}{\sqrt{\left(t + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2}} \\
 &= \log \left| \left(t + \frac{1}{2}\right) + \sqrt{t^2 + t} \right| + c \\
 &= \log \left| \left(\sin x + \frac{1}{2}\right) + \sqrt{\sin^2 x + \sin x} \right| + c
 \end{aligned}$$

219. The given integral is

$$\begin{aligned}
 \int \frac{dx}{\sqrt{1 - e^{2x}}} &= \int \frac{e^{-x}}{\sqrt{(e - x)^2 - 1}} \, dx \\
 &= -\int \frac{dt}{\sqrt{t^2 - 1}}, \quad \text{Let } t = e^{-x} \\
 &\quad \Rightarrow dt = -e^{-x} \, dx \\
 &= -\log |t + \sqrt{t^2 - 1}| + c \\
 &= -\log |e^{-x} + \sqrt{e^{-2x} - 1}| + c
 \end{aligned}$$

220. The given integral is

$$\begin{aligned}
 \int \sqrt{\frac{\sin(x - \alpha)}{\sin(x + \alpha)}} \, dx \\
 &= \int \sqrt{\frac{\sin(x - \alpha) \sin(x - \alpha)}{\sin(x - \alpha) \sin(x + \alpha)}} \, dx \\
 &= \int \frac{\sin(x - \alpha)}{\sqrt{\sin^2 x - \sin^2 \alpha}} \, dx \\
 &= \int \frac{\sin x \cos \alpha - \cos x \sin \alpha}{\sqrt{\sin^2 x - \sin^2 \alpha}} \, dx \\
 &= \cos \alpha \int \frac{\sin x \, dx}{\sqrt{\sin^2 \alpha - \sin^2 x - \sin^2 \alpha}} - \sin \alpha \int \frac{\cos x \, dx}{\sqrt{\sin^2 x - \sin^2 \alpha}} \\
 &= \cos \alpha \int \frac{\sin x \, dx}{\sqrt{\cos^2 \alpha - \cos^2 x}} - \sin \alpha \int \frac{\cos x \, dx}{\sqrt{\sin^2 x - \sin^2 \alpha}} \\
 &= -\cos \alpha \sin^{-1} \left(\frac{\cos x}{\cos \alpha} \right) \\
 &\quad - \sin \alpha \log |\sin x + \sqrt{\sin^2 x - \sin^2 \alpha}| + c
 \end{aligned}$$

221. The given integral is

$$\begin{aligned}
 \int \frac{dx}{x^{2/3} \sqrt{x^{2/3} - 4}} &= \int \frac{dx}{x^{2/3} \sqrt{(x^{1/3})^2 - 4}} \\
 &= \frac{1}{3} \int \frac{dt}{\sqrt{t^2 - 4}}, \quad \text{Let } t = x^{1/3} \\
 &\quad \Rightarrow dt = \frac{1}{3} x^{-2/3} \, dx \\
 &= \frac{1}{3} \int \frac{dt}{\sqrt{t^2 - 4}} \\
 &= \frac{1}{3} \log |t + \sqrt{t^2 - 4}| + c \\
 &= \frac{1}{3} \log |x^{1/3} + \sqrt{x^{2/3} - 4}| + c
 \end{aligned}$$

222. The given integral is

$$\begin{aligned}
 \int \sqrt{\frac{x}{a^3 - x^3}} \, dx \\
 \text{Let } x = a \sin^{2/3} \theta \\
 \Rightarrow dx = a \cdot \sin^{-1/3} \theta \, d\theta \\
 &= \int \frac{\sqrt{a} \sin^{1/3} \theta \sin^{-1/3} \theta \cos \theta}{a^{3/2} \cos \theta} \, d\theta \\
 &= \frac{1}{a} \int d\theta \\
 &= \frac{\theta}{a} + c \\
 &= \frac{\sin^{-1} \left\{ \left(\frac{x}{a} \right)^{3/2} \right\}}{a} + c
 \end{aligned}$$

223. The given integral is

$$\begin{aligned}
 & \int \left(\frac{\cos \theta + \sin \theta}{\sqrt{5 + \sin 2\theta}} \right) d\theta \\
 &= \int \frac{\cos \theta + \sin \theta}{\sqrt{6 - (1 - \sin 2\theta)}} d\theta \\
 &= \int \frac{\cos \theta + \sin \theta}{\sqrt{6 - (\sin \theta - \cos \theta)^2}} d\theta \\
 & \quad \text{Put } (\sin \theta - \cos \theta) = t \\
 & \quad \Rightarrow (\cos \theta + \sin \theta) d\theta = dt \\
 &= \int \frac{dt}{\sqrt{6 - t^2}} \\
 &= \sin^{-1} \left(\frac{t}{\sqrt{6}} \right) + c \\
 &= \sin^{-1} \left(\frac{\sin \theta - \cos \theta}{\sqrt{6}} \right) + c
 \end{aligned}$$

224. The given integral is

$$\begin{aligned}
 & \int \frac{\sin \theta - \cos \theta}{\sqrt{2} - \sin 2\theta} d\theta \\
 &= \int \frac{\sin \theta - \cos \theta}{\sqrt{3} - (1 + \sin 2\theta)} d\theta \\
 &= \int \frac{\sin \theta - \cos \theta}{\sqrt{3} - (\sin \theta + \cos \theta)^2} d\theta \\
 &= - \int \frac{dt}{\sqrt{3 - t^2}}, \quad \text{Let } t = (\sin \theta + \cos \theta) \\
 & \quad dt = (\cos \theta - \sin \theta) d\theta \\
 &= \cos^{-1} \left(\frac{t}{\sqrt{3}} \right) + c \\
 &= \cos^{-1} \left(\frac{\sin \theta + \cos \theta}{\sqrt{3}} \right) + c
 \end{aligned}$$

225. We have,

$$\begin{aligned}
 & \int \frac{x-1}{\sqrt{x^2-3x+2}} dx \\
 &= \frac{1}{2} \int \frac{2x-2}{\sqrt{x^2-3x+2}} dx \\
 &= \frac{1}{2} \int \frac{(2x-3)+1}{\sqrt{x^2-3x+2}} dx \\
 &= \frac{1}{2} \int \frac{(2x-3)}{\sqrt{x^2-3x+2}} dx + \frac{1}{2} \int \frac{dx}{\sqrt{x^2-3x+2}} \\
 &= \frac{1}{2} \int \frac{2t dt}{t} dx + \frac{1}{2} \int \frac{dx}{\sqrt{\left(x-\frac{3}{2}\right)^2 + \left(2-\frac{9}{4}\right)}}
 \end{aligned}$$

$$\text{Let } (x^2 - 3x + 2) = t^2$$

$$\Rightarrow (2x+3)dx = 2t dt$$

$$\begin{aligned}
 &= t + \frac{1}{2} \log \left| \left(x - \frac{1}{2} \right) + \sqrt{x^2 - 3x + 2} \right| + c \\
 &= \sqrt{x^2 - 3x + 2} + \frac{1}{2} \log \left| \left(x - \frac{1}{2} \right) + \sqrt{x^2 - 3x + 2} \right| + c
 \end{aligned}$$

226. We have,

$$\begin{aligned}
 & \int \frac{3x+4}{\sqrt{x^2+5x+2}} dx \\
 &= \int \frac{\frac{3}{2}(2x+5) + \left(4 - \frac{15}{2}\right)}{\sqrt{x^2+5x+2}} dx \\
 &= \frac{3}{2} \int \frac{(2x+5)dx}{\sqrt{x^2+5x+2}} - \frac{7}{2} \int \frac{dx}{\sqrt{x^2+5x+2}} \\
 &= \frac{3}{2} \int \frac{2t dt}{t} - \frac{7}{2} \int \frac{dx}{\sqrt{\left(x+\frac{5}{2}\right)^2 + \left(2-\frac{25}{4}\right)}}
 \end{aligned}$$

$$\text{Let } (x^2 + 5x + 2) = t^2$$

$$\Rightarrow (2x+5)dx = 2t dt$$

$$\begin{aligned}
 &= 3 \int dt + \frac{7}{2} \int \frac{dx}{\sqrt{\left(x+\frac{5}{2}\right)^2 - \left(\frac{\sqrt{17}}{2}\right)^2}} \\
 &= 3t + \frac{7}{2} \log \left| \left(x + \frac{5}{2} \right) + \sqrt{x^2 + 5x + 2} \right| + c
 \end{aligned}$$

$$= 3\sqrt{x^2+5x+2} + \frac{7}{2} \log \left| \left(x + \frac{5}{2} \right) + \sqrt{x^2+5x+2} \right| + c$$

227. We have,

$$\begin{aligned}
 & \int \frac{x+2}{\sqrt{x^2+5x+6}} dx = \frac{1}{2} \int \frac{(2x+5)-1}{\sqrt{x^2+5x+6}} dx \\
 &= \frac{1}{2} \int \frac{(2x+5)}{\sqrt{x^2+5x+6}} dx - \frac{1}{2} \int \frac{1}{\sqrt{x^2+5x+6}} dx \\
 &= \frac{1}{2} \int \frac{(2x+5)}{\sqrt{x^2+5x+6}} dx - \frac{1}{2} \int \frac{1}{\sqrt{\left(x+\frac{5}{2}\right)^2 - \left(\frac{1}{2}\right)^2}} dx \\
 &= \sqrt{x^2+5x+6} \\
 & \quad - \frac{1}{2} \log \left| \left(x + \frac{5}{2} \right) + \sqrt{x^2+5x+6} \right| + c
 \end{aligned}$$

$$228. \int \frac{6x-5}{\sqrt{3x^2-5x+1}} dx$$

$$\begin{aligned} \text{Let } 3x^2 - 5x + 1 \\ \Rightarrow (6x - 5)dx = 2t dt \end{aligned}$$

$$\begin{aligned} &= \int \frac{2t dt}{t} \\ &= 2 \int dt \\ &= 2t + c \\ &= 2\sqrt{3x^2 - 5x + 1} + c \end{aligned}$$

229. We have,

$$\begin{aligned} \int \sqrt{\frac{a-x}{a+x}} dx &= \int \sqrt{\frac{a-x}{a+x} \times \frac{a-x}{a-x}} dx \\ &= \int \frac{a-x}{\sqrt{a^2-x^2}} dx \\ &= a \int \frac{dx}{\sqrt{a^2-x^2}} - \int \frac{x}{\sqrt{a^2-x^2}} dx \\ &= a \sin^{-1} \left(\frac{x}{a} \right) + \sqrt{a^2-x^2} + c \end{aligned}$$

$$230. \int x \sqrt{\frac{a^2-x^2}{a^2+x^2}} dx$$

$$\begin{aligned} \text{Let } x^2 &= a^2 t \\ \Rightarrow 2x dx &= a^2 dt \\ \Rightarrow \frac{a^2}{2} \int \sqrt{\frac{a^2(1-t)}{a^2(1+t)}} dt \end{aligned}$$

$$\begin{aligned} &= \frac{a^2}{2} \int \sqrt{\frac{(1-t)}{(1+t)}} dt \\ &= \frac{a^2}{2} \int \sqrt{\frac{(1-t)}{(1+t)}} \times \frac{1-t}{1-t} dt \\ &= \frac{a^2}{2} \int \sqrt{\frac{(1-t)^2}{(1-t^2)}} dt \\ &= \frac{a^2}{2} \int \frac{(1-t)}{\sqrt{1-t^2}} dt \\ &= \frac{a^2}{2} \left(\int \frac{dt}{\sqrt{1-t^2}} - \int \frac{t}{\sqrt{1-t^2}} dt \right) \\ &= \frac{a^2}{2} \left(\sin^{-1}(t) + \sqrt{1-t^2} \right) + c \\ &= \frac{a^2}{2} \left(\sin^{-1} \left(\frac{x^2}{a^2} \right) + \sqrt{1 - \left(\frac{x^2}{a^2} \right)} \right) + c \end{aligned}$$

231. We have,

$$\begin{aligned} \int x^2 \sqrt{\frac{4-x^3}{4+x^3}} dx \\ \text{Let } x^3 &= 4t \\ \Rightarrow x^2 dx &= \frac{4}{3} dt \\ &= \frac{4}{3} \int \sqrt{\frac{4-4t}{4+4t}} dt \\ &= \frac{4}{3} \int \sqrt{\frac{1-t}{1+t}} dt \\ &= \frac{4}{3} \int \frac{(1-t)}{\sqrt{1-t^2}} dt \\ &= \frac{4}{3} \left(\int \frac{dt}{\sqrt{1-t^2}} - \int \frac{t}{\sqrt{1-t^2}} dt \right) \\ &= \frac{4}{3} \left(\sin^{-1}(t) + \sqrt{1-t^2} \right) + c \\ &= \frac{4}{3} \left(\sin^{-1} \left(\frac{x^3}{4} \right) + \sqrt{1 - \left(\frac{x^3}{4} \right)^2} \right) + c \end{aligned}$$

232. We have,

$$\begin{aligned} \int \frac{dx}{3+4\sin^2 x} &= \int \frac{\sec^2 x dx}{3\sec^2 x + 4\tan^2 x} \\ &= \int \frac{\sec^2 x dx}{3+7\tan^2 x} \\ &= \frac{1}{7} \int \frac{\sec^2 x dx}{\tan^2 x + \frac{3}{7}} \\ &= \frac{1}{7} \int \frac{dt}{t^2 + \left(\sqrt{\frac{3}{7}} \right)^2} \\ &= \frac{1}{7} \times \frac{\sqrt{7}}{3} \tan^{-1} \left(t \sqrt{\frac{7}{3}} \right) + c \\ &= \frac{1}{\sqrt{21}} \times \tan^{-1} \left(\tan x \sqrt{\frac{7}{3}} \right) + c \end{aligned}$$

233. We have,

$$\begin{aligned} \int \frac{dx}{3\sin^2 x + 4\cos^2 x} &= \int \frac{\sec^2 x dx}{3\tan^2 x + 4} \\ &= \int \frac{dt}{3t^2 + 4} \quad \text{Let } \tan x = t \\ &\Rightarrow \sec^2 x dx = dt \\ &= \frac{1}{3} \int \frac{dt}{t^2 + \left(\frac{2}{\sqrt{3}} \right)^2} \end{aligned}$$

$$= \frac{1}{3} \times \frac{\sqrt{3}}{2} \tan^{-1} \left(\frac{\sqrt{3}}{2} t \right) + c$$

$$= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}}{2} \tan x \right) + c$$

234. We have,

$$\int \frac{dx}{(2\sin x + \cos x)^2} = \int \frac{\sec^2 x dx}{(2\tan x + 3)^2}$$

$$= \int \frac{dt}{(2t + 3)^2}$$

$$= -\frac{1}{2(2t + 3)} + c$$

$$= -\frac{1}{2(2\tan x + 3)} + c$$

235. We have,

$$\int \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$$= \int \frac{\tan x \sec^2 x}{\tan^4 x + 1} dx$$

$$= \frac{1}{2} \int \frac{2\tan x \sec^2 x}{\tan^4 x + 1} dx$$

$$= \frac{1}{2} \int \frac{dt}{t^2 + 1} \quad \text{Let } \tan^2 x = t$$

$$\Rightarrow 2\tan x \sec^2 x dx = dt$$

$$= \frac{1}{2} \tan^{-1}(t) + c$$

$$= \frac{1}{2} \tan^{-1}(\tan^2 x) + c$$

236. We have,

$$\int \frac{dx}{(\sin x + 2\cos x)^2} = \int \frac{\sec^2 x}{(\tan x + 2)^2} dx$$

$$= \int \frac{dt}{(t + 2)^2} \quad \text{Let } \tan x = t$$

$$\Rightarrow \sec^2 x dx = dt$$

$$= -\frac{1}{(t + 2)} + c$$

$$= -\frac{1}{(\tan x + 2)} + c$$

237. We have,

$$\int \frac{dx}{(\sin x + 2\sec x)^2} = \frac{1}{4} \int \frac{dx}{\left(\frac{\sin x}{2} + \sec x\right)^2}$$

$$= \frac{1}{4} \int \frac{\sec^2 x dx}{\left(\frac{\tan x}{2} + \sec x\right)^2}$$

$$= \frac{1}{4} \int \frac{\sec^2 x dx}{\left(\tan^2 x + \frac{\tan x}{2} + 1\right)^2}$$

$$= \frac{1}{4} \int \frac{\sec^2 x dx}{\left[\left(\tan x + \frac{1}{4}\right)^2 + \left(1 - \frac{1}{16}\right)^2\right]}$$

$$= \frac{1}{4} \int \frac{\sec^2 x dx}{\left[\left(\frac{\tan x + 1}{4}\right)^2 + \frac{15^2}{16}\right]}$$

$$\text{Let } \left(\tan x + \frac{1}{4}\right) = \frac{\sqrt{15}}{4} \tan \theta$$

$$\Rightarrow \sec^2 x dx = \frac{\sqrt{15}}{4} \sec^2 \theta d\theta$$

$$= \frac{1}{4} \int \frac{\frac{\sqrt{15}}{4} \sec^2 \theta}{\left(\frac{15}{16}\right)^2} \sec^4 \theta$$

$$= \frac{1}{4} \times \frac{\sqrt{15}}{4} \times \left(\frac{16}{15}\right)^2 \int \frac{d\theta}{\sec^2 \theta}$$

$$= \frac{8}{(15)^{3/2}} \int (2\cos^2 \theta) d\theta$$

$$= \frac{8}{(15)^{3/2}} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{8}{(15)^{3/2}} \left(\theta + \frac{\sin 2\theta}{2}\right) + c$$

$$= \frac{8}{(15)^{3/2}} (\theta + \sin \theta + \cos \theta) + c,$$

$$\text{where } \theta = \tan^{-1} \left[\frac{4}{\sqrt{15}} \left(\tan x + \frac{1}{4}\right) \right]$$

238. The given integral is

$$\int \frac{\sin^2 x dx}{\sin^4 x + \cos^4 x} = \int \frac{2\sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

Divide the numerator and the denominator by $\cos^4 x$ we get

$$\int \frac{2\tan x \sec^2 x}{\tan^4 x + 1} dx$$

$$\text{Let } \tan^2 x = t$$

$$\Rightarrow 2\tan x \sec^2 x dx = dt$$

$$= \int \frac{dt}{t^2} + 1$$

$$= \tan^{-1}(t) + c$$

$$= \tan^{-1}(\tan^2 x) + c$$

239. The given integral is

$$\int \frac{dx}{(2\sin x + 3\cos x)^2}$$

Divide the numerator and the denominator by $\cos^2 x$, we get

$$\int \frac{\sec^2 x \, dx}{(2\tan x + 3)^2}$$

$$\begin{aligned}\text{Let } 2\tan x + 3 &= t \\ \Rightarrow 2\sec^2 x \, dx &= dt\end{aligned}$$

$$\begin{aligned}&= \frac{1}{2} \int \frac{dt}{t^2} \\&= -\frac{1}{2t} + c \\&= -\frac{1}{2(2\tan x + 3)} + c\end{aligned}$$

240. The given integral is

$$\int \frac{dx}{2 + \cos^2 x} = \int \frac{\sec^2 x \, dx}{1 + 2\sec^2 x} = \int \frac{\sec^2 x \, dx}{2\tan^2 x + 3}$$

$$\begin{aligned}\text{Let } \tan x &= t \\ \Rightarrow \sec^2 x \, dx &= dt\end{aligned}$$

$$\begin{aligned}&= \int \frac{dt}{2t^2 + 3} \\&= \int \frac{dt}{t^2 + \left(\sqrt{\frac{3}{2}}\right)^2} \\&= \sqrt{\frac{2}{3}} \tan^{-1}\left(\sqrt{\frac{2}{3}} t\right) + c \\&= \sqrt{\frac{2}{3}} \tan^{-1}\left(\sqrt{\frac{2}{3}} \tan x\right) + c\end{aligned}$$

241. We have,

$$\int \frac{\sin x}{\sin 3x} \, dx = \int \frac{\sin x}{3\sin x - 4\sin^3 x} \, dx$$

$$= \int \frac{dx}{3 - 4\sin^2 x}$$

$$= \int \frac{\sec^2 x \, dx}{3\sec^2 x - 4\tan^2 x}$$

$$= \int \frac{\sec^2 x \, dx}{3 - \tan^2 x}$$

$$= -\int \frac{dt}{t^2 - 3} \quad \begin{aligned}\text{Let } \tan x &= t \\ \sec^2 x \, dx &= dt\end{aligned}$$

$$= -\frac{1}{2\sqrt{3}} \log \left| \frac{t - \sqrt{3}}{t + \sqrt{3}} \right| + c$$

$$= -\frac{1}{2\sqrt{3}} \log \left| \frac{\tan x - \sqrt{3}}{\tan x + \sqrt{3}} \right| + c$$

242. The given integral is

$$\begin{aligned}\int \frac{\operatorname{cosec}^3 x}{\operatorname{cosec} x} \, dx &= \int \frac{\sin x}{\sin^3 x} \, dx \\&= \int \frac{\sin x}{3\sin x - 4\sin^3 x} \, dx \\&= \int \frac{dx}{3 - 4\sin^2 x} \\&= \int \frac{\sec^2 x \, dx}{3\sec^2 x - 4\tan^2 x} \\&= \int \frac{\sec^2 x \, dx}{3 + 3\tan^2 x - 4\tan^2 x} \\&= \int \frac{\sec^2 x \, dx}{3 - \tan^2 x}\end{aligned}$$

$$\begin{aligned}\text{Let } \tan x &= t \\ \Rightarrow \sec^2 x \, dx &= dt\end{aligned}$$

$$\begin{aligned}&= -\int \frac{dt}{t^2 - 3} \\&= \frac{-1}{2\sqrt{3}} \log \left| \frac{t - \sqrt{3}}{t + \sqrt{3}} \right| + c \\&= \frac{1}{2\sqrt{3}} \log \left| \frac{t + \sqrt{3}}{t - \sqrt{3}} \right| \\&= \frac{1}{2\sqrt{3}} \log \left| \frac{\tan x + \sqrt{3}}{\tan x - \sqrt{3}} \right| + c\end{aligned}$$

243. The given integral is

$$\begin{aligned}\int \frac{\sec 3x}{\sec x} \, dx &= \int \frac{\cos x}{\cos^3 x} \, dx \\&= \int \frac{\cos x}{4\cos^3 x - 3\cos x} \, dx \\&= \int \frac{dx}{4\cos^2 x - 3}\end{aligned}$$

Divide the numerator and the denominator by $\cos^2 x$ we get,

$$\begin{aligned}&= \int \frac{\sec^2 x \, dx}{4 - 3\sec^2 x} \\&= \int \frac{\sec^2 x \, dx}{4 - 3 - 3\tan^2 x} \\&= \int \frac{\sec^2 x \, dx}{1 - 3\tan^2 x}\end{aligned}$$

$$\text{Let } \tan x = t \Rightarrow \sec^2 x \, dx = dt$$

$$= \int \frac{dt}{1 - 3t^2}$$

$$\begin{aligned}
&= \frac{1}{3} \int \frac{dt}{t^2 - (1/\sqrt{3})^2} \\
&= \frac{1}{2\sqrt{3}} \log \left| \frac{t - \frac{1}{\sqrt{3}}}{t + \frac{1}{\sqrt{3}}} \right| + c \\
&= \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} \tan x - 1}{\sqrt{3} \tan x + 1} \right| + c
\end{aligned}$$

244. We have,

$$\begin{aligned}
\int \frac{dx}{1 + 2 \sin x} &= \int \frac{dx}{1 + 2 \left(\frac{2 \tan(\frac{x}{2})}{1 + \tan^2(\frac{x}{2})} \right)} \\
&= \int \frac{\sec^2(\frac{x}{2})}{1 + \tan^2(\frac{x}{2}) + 4 \tan(\frac{x}{2})} dx \\
&\quad \text{Let } \tan\left(\frac{x}{2}\right) = t \\
&\quad \Rightarrow \sec^2\left(\frac{x}{2}\right) dx = 2dt \\
&= \int \frac{2dt}{t^2 + 4t + 1} \\
&= 2 \int \frac{dt}{(t + 2)^2 - (\sqrt{3})^2} \\
&= \frac{2}{2\sqrt{3}} \log \left| \frac{(t + 2) - \sqrt{3}}{(t + 2) + \sqrt{3}} \right| + c \\
&= \frac{1}{\sqrt{3}} \log \left| \frac{\left(\tan\left(\frac{x}{2}\right) + 2\right) - \sqrt{3}}{\left(\tan\left(\frac{x}{2}\right) + 2\right) + \sqrt{3}} \right| + c
\end{aligned}$$

245. We have,

$$\begin{aligned}
&\int \frac{dx}{\sin x + \cos x + 1} \\
&= \int \frac{dx}{\left(\frac{2 \tan(\frac{x}{2})}{1 + \tan^2(\frac{x}{2})} \right) + \left(\frac{1 - \tan^2(\frac{x}{2})}{1 + \tan^2(\frac{x}{2})} \right) + 1} \\
&= \int \frac{\sec^2(\frac{x}{2}) dx}{\left(2 \tan\left(\frac{x}{2}\right) - \tan^2\left(\frac{x}{2}\right) + 2 \right)} \\
&= \frac{1}{2} \int \frac{dt}{2t - t^2 + 2} \quad \text{Let } \left(\frac{x}{2}\right) = t
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \int \frac{-dt}{t^2 - 2t - 2} \Rightarrow \frac{1}{2} \sec^2\left(\frac{3}{2}\right) dx = dt \\
&= -\frac{1}{2} \int \frac{dt}{(t - 1)^2 - (\sqrt{3})^2} \\
&= -\frac{1}{2\sqrt{3}} \log \left| \frac{(t - 1) - \sqrt{3}}{(t - 1) + \sqrt{3}} \right| + c \\
&= -\frac{1}{2\sqrt{3}} \log \left| \frac{\left(\tan\left(\frac{x}{2}\right) - 1\right) - \sqrt{3}}{\left(\tan\left(\frac{x}{2}\right) - 1\right) + \sqrt{3}} \right| + c
\end{aligned}$$

244. The given integral is

$$\begin{aligned}
&\int \frac{dx}{1 + 2 \sin x} \\
&= \int \frac{dx}{1 + 2 \left(\frac{2 \tan(x/2)}{1 + \tan^2(x/2)} \right)} \\
&= \int \frac{(1 + \tan^2(x/2)) dx}{1 + \tan^2(x/2) + 4 \tan(x/2)} \\
&= \int \frac{\sec^2(x/2) dx}{\tan^2(x/2) + 4 \tan(x/2) + 1} \\
&\quad \text{Let } \tan(x/2) = t \\
&\quad \Rightarrow \frac{1}{2} \sec^2\left(\frac{x}{2}\right) dx = dt \\
&= \int \frac{2dt}{t^2 + 4t + 1} \\
&= 2 \int \frac{dt}{(t + 2)^2 - (\sqrt{3})^2} \\
&= \frac{1}{\sqrt{3}} \log \left| \frac{(t + 2) - \sqrt{3}}{(t + 2) + \sqrt{3}} \right| + c \\
&= \frac{1}{\sqrt{3}} \log \left| \frac{(\tan(x/2) + 2) - \sqrt{3}}{(\tan(x/2) + 2) + \sqrt{3}} \right| + c
\end{aligned}$$

245. The given integral is

$$\begin{aligned}
&\int \frac{dx}{3 \cos x + 4} = \int \frac{dx}{3 \left(\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} \right) + 4} \\
&= \int \frac{(1 + \tan^2(x/2))}{4 + 4 \tan^2(x/2) + 3 - 3 \tan^2(x/2)} dx \\
&= \int \frac{\sec^2(x/2)}{\tan^2(x/2) + 7} dx \\
&\quad \text{Let } t = \tan(x/2) \\
&\quad \Rightarrow 2dt = \sec^2\left(\frac{x}{2}\right) dx
\end{aligned}$$

$$\begin{aligned}
&= \frac{2dt}{t^2 + 7} \\
&= \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{t}{\sqrt{7}} \right) + c \\
&= \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{\tan(x/2)}{\sqrt{7}} \right) + c
\end{aligned}$$

246. The given integral is

$$\begin{aligned}
&\int \frac{dx}{\sin x + \cos x + 1} \\
&= \int \frac{dx}{\frac{2 \tan(x/2)}{1 + \tan^2(x/2)} + \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} + 1} \\
&= \int \frac{(1 + \tan^2(x/2))dx}{1 + \tan^2(x/2) + 2 \tan(x/2) + 1 - \tan^2(x/2)} \\
&= \int \frac{\sec^2(x/2)dx}{2 \tan(x/2) + 2}
\end{aligned}$$

$$\begin{aligned}
&\text{Let } \tan\left(\frac{x}{2}\right) = t \\
&\Rightarrow \frac{1}{2} \sec^2\left(\frac{x}{2}\right) dx = dt \\
&\Rightarrow \sec^2\left(\frac{x}{2}\right) dx = 2dt
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{2dt}{2t + 2} \\
&= \int \frac{dt}{t + 1} \\
&= \log|t + 1| + c \\
&= \log\left|\tan\left(\frac{x}{2}\right) + 1\right| + c
\end{aligned}$$

247. The given integral is

$$\begin{aligned}
&\int \frac{1 + \sin x}{\sin x(1 + \cos x)} dx \\
&= \int \frac{dx}{\sin x(1 + \cos x)} + \int \frac{dx}{(1 + \cos x)} \\
&= \int \frac{\sin x dx}{(1 - \cos^2 x)(1 + \cos x)} + \frac{1}{2} \int \sec^2\left(\frac{x}{2}\right) dx \\
&= \int \frac{dt}{(t^2 - 1)(t + 1)} + \frac{1}{2} \int \sec^2\left(\frac{x}{2}\right) dx
\end{aligned}$$

$$\begin{aligned}
&\text{Let } t = \cos x \\
&\Rightarrow dt = -\sin x dx
\end{aligned}$$

$$= \int \frac{dt}{(t-1)(t+1)^2} + \frac{1}{2} \int \sec^2\left(\frac{x}{2}\right) dx$$

$$\begin{aligned}
&= \frac{1}{2} \int \left(\frac{1}{t^2 - 1} - \frac{1}{(t+1)^2} \right) dt + \frac{1}{2} \int \sec^2(x/2) dx \\
&= \frac{1}{2} \left(\frac{1}{2} \log \left| \frac{t-1}{t+1} \right| + \frac{1}{(t+1)} \right) + \tan(x/2) + c \\
&= \frac{1}{2} \left(\frac{1}{2} \log \left| \frac{\cos x - 1}{\cos x + 1} \right| + \frac{1}{(\cos x + 1)} \right) \\
&\quad + \tan(x/2) + c
\end{aligned}$$

248. The given integral is

$$\begin{aligned}
&\int \frac{dx}{1 - 2 \sin x} \\
&= \int \frac{dx}{1 - 2 \left(\frac{2 \tan(x/2)}{1 + \tan^2(x/2)} \right)} \\
&= \int \frac{(1 + \tan^2(x/2))dx}{\tan^2(x/2) - 4 \tan(x/2) + 1} \\
&= \int \frac{\sec^2(x/2)dx}{\tan^2(x/2) - 4 \tan(x/2) + 1}
\end{aligned}$$

$$\begin{aligned}
&\text{Let } \tan(x/2) = t \\
&\Rightarrow \sec^2\left(\frac{x}{2}\right) dx = 2dt
\end{aligned}$$

$$\begin{aligned}
&= 2 \int \frac{dt}{t^2 - 4t + 1} \\
&= 2 \int \frac{dt}{(t-2)^2 - (\sqrt{3})^2} \\
&= 2 \times \frac{1}{2} \log \left| \frac{(t-2) - \sqrt{3}}{(t-2) + \sqrt{3}} \right| + c \\
&= \log \left| \frac{(\tan(x/2) - 2) - \sqrt{3}}{(\tan(x/2) - 2) + \sqrt{3}} \right| + c
\end{aligned}$$

249. The given integral is

$$\begin{aligned}
&\int \frac{dx}{3 \sin x + 4 \cos x + 5} \\
&= \int \frac{dx}{3 \left(\frac{2 \tan(x/2)}{1 + \tan^2(x/2)} \right) + 4 \left(\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} \right) + 5} \\
&= \int \frac{(1 + \tan^2(x/2))dx}{\tan^2(x/2) + 6 \tan(x/2) + 9}
\end{aligned}$$

$$= \int \frac{2dt}{t^2 + 6t + 9}$$

$$\begin{aligned}
&\text{Let } \tan(x/2) = t \\
&\Rightarrow \sec^2\left(\frac{x}{2}\right) dx = 2dt
\end{aligned}$$

$$= \int \frac{2dt}{(t+3)^2}$$

$$= -\frac{2}{(t+3)} + c$$

$$= -\frac{2}{(\tan(x/2) + 3)} + c$$

250. The given integral is

$$\int \frac{dx}{\cos x + \cos \alpha}$$

$$= \int \frac{dx}{\left(\frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} \right) + k}, \text{ where } k = \cos \alpha$$

$$= \int \frac{[1 + \tan^2(x/2)]dx}{1 - \tan^2(x/2) + k[1 + \tan^2(x/2)]}$$

$$= \int \frac{\sec^2(x/2)dx}{1 - \tan^2(x/2) + k[1 + \tan^2(x/2)]}$$

$$= \int \frac{2dt}{(1 - t^2) + k(1 + t^2)} \quad \text{Let } \tan\left(\frac{x}{2}\right) = t$$

$$\Rightarrow \sec^2\left(\frac{x}{2}\right)dx = 2dt$$

$$= \int \frac{2dt}{(1 + k) + (k - 1)t^2}$$

$$= \frac{1}{(1 - k)} \int \frac{2dt}{\frac{(1 + k)}{(1 - k)} - t^2}$$

$$= \frac{1}{(1 - k)} \int \frac{2dt}{\left(\sqrt{\frac{(1 + k)}{(1 - k)}} \right)^2 - t^2}$$

$$= \frac{1}{(1 - k)} \times \frac{1}{2\sqrt{\frac{(1 + k)}{(1 - k)}}} \log \left| \frac{\sqrt{\frac{(1 + k)}{(1 - k)}} + t}{\sqrt{\frac{(1 + k)}{(1 - k)}} - t} \right| + c$$

$$= \frac{1}{\sin \alpha} \log \left| \frac{\cot(\alpha/2) + \tan(x/2)}{\cot(\alpha/2) - \tan(x/2)} \right| + c$$

251. The given integral is

$$\int \frac{dx}{\tan x + 4\cot x + 4}$$

$$= \int \frac{4\sin x \cos x}{\sin^2 x + 4\cos^2 x + 4\sin x \cos x} dx$$

252. We have,

$$\int \frac{dx}{\sin x + \cos x}$$

$$= \frac{1}{\sqrt{2}} \int \frac{dx}{\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x}$$

$$= \frac{1}{\sqrt{2}} \int \frac{dx}{\sin\left(x + \frac{\pi}{4}\right)}$$

$$= \frac{1}{\sqrt{2}} \int \operatorname{cosec}\left(x + \frac{\pi}{4}\right) dx$$

$$= \frac{1}{\sqrt{2}} \log \left| \tan\left(\frac{x}{2} + \frac{\pi}{8}\right) \right| + c$$

253. We have,

$$\int \frac{dx}{\sqrt{3} \sin x + \cos x}$$

$$= \frac{1}{2} \int \frac{dx}{\frac{\sqrt{3}}{2} \sin x + \frac{1}{2} \cos x}$$

$$= \frac{1}{2} \int \frac{dx}{\sin\left(x + \frac{\pi}{6}\right)}$$

$$= \frac{1}{2} \int \operatorname{cosec}\left(x + \frac{\pi}{6}\right) dx$$

$$= \frac{1}{2} \log \left| \tan\left(\frac{x}{2} + \frac{\pi}{12}\right) \right| + c$$

257. We have,

$$2\sin x + \cos x$$

$$= l(3\sin x + 2\cos x) + m(3\cos x - 2\sin x)$$

$$= (3l - 2m)\sin x + (2l + 3m)\cos x$$

Comparing the co-efficients of $\sin x$ and $\cos x$, we get

$$3l - 2m = 2 \text{ and } 2l + 3m = 1$$

On solving, we get,

$$l = 4/5 \text{ and } m = -3/5$$

$$\therefore \frac{4}{5} \int \frac{3\sin x + 2\cos x}{3\sin x + 2\cos x} dx$$

$$+ \frac{3}{5} \int \frac{3\cos x - 2\sin x}{3\sin x + 2\cos x} dx$$

$$= \frac{4}{5} \int dx - \frac{3}{5} \int \left(\frac{3\cos x - 2\sin x}{3\sin x + 2\cos x} \right) dx$$

$$= \frac{4}{5} x - \frac{3}{5} \log |3\sin x + 2\cos x| + c$$

258. We have,

$$\int \frac{\sin x}{\sin x + \cos x} dx$$

$$= \frac{1}{2} \int \frac{2\sin x}{\sin x + \cos x} dx$$

$$= \frac{1}{2} \int \frac{(\sin x + \cos x) + (\sin x - \cos x)}{\sin x + \cos x} dx$$

$$\begin{aligned}
&= \frac{1}{2} \int \frac{(1 + \cos x - \sin x)}{\sin x + \cos x} dx \\
&= \frac{1}{2}x + \frac{1}{2} \log |\sin x + \cos x| + c
\end{aligned}$$

259. The given integral is

$$\int \frac{2 \sin x + 3 \cos x}{3 \sin x + 4 \cos x} dx$$

we have $2 \sin x + 3 \cos x$

$$\begin{aligned}
&= l(3 \sin x + 4 \cos x) + m(3 \cos x - 4 \sin x) \\
&= (3l - 4m) \sin x + (4l + 3m) \cos x
\end{aligned}$$

Comparing the co-efficients of $\sin x$ and $\cos x$, we get

$$(3l - 4m) = 2, (4l + 3m) = 3$$

On solving, we get

$$l = \frac{18}{25}, m = \frac{1}{25}$$

The given integral reduces to

$$\begin{aligned}
&\int \frac{\frac{18}{25}(3 \sin x + 4 \cos x) + \frac{1}{25}(3 \cos x - 4 \sin x)}{(3 \sin x + 4 \cos x)} dx \\
&= \int \left(\frac{18}{25} + \frac{1}{25} \frac{(3 \cos x - 4 \sin x)}{(3 \sin x + 4 \cos x)} \right) dx \\
&= \frac{18}{25}x + \frac{1}{25} \log |3 \sin x + 4 \cos x| + C
\end{aligned}$$

260. The given integral is

$$\begin{aligned}
&\int \frac{\sin x}{\sin x - \cos x} dx \\
&= \frac{1}{2} \int \frac{2 \sin x}{\sin x - \cos x} dx \\
&= \frac{1}{2} \int \frac{(\sin x - \cos x) + (\sin x + \cos x)}{\sin x - \cos x} dx \\
&= \frac{1}{2} \int \left(1 + \frac{\cos x + \sin x}{\sin x - \cos x} \right) dx \\
&= \frac{1}{2}x + \frac{1}{2} \log |\sin x - \cos x| + c
\end{aligned}$$

261. The given integral is

$$\begin{aligned}
&\int \frac{\cos x}{\sin x + \cos x} dx \\
&= \frac{1}{2} \int \frac{2 \cos x}{\sin x + \cos x} dx \\
&= \frac{1}{2} \int \frac{(\cos x + \sin x) + (\cos x - \sin x)}{\sin x + \cos x} dx \\
&= \frac{1}{2} \int \left(1 + \frac{(\cos x - \sin x)}{\sin x + \cos x} \right) dx
\end{aligned}$$

$$= \frac{1}{2}(x + \log |\sin x + \cos x|) + c$$

262. The given integral is

$$\begin{aligned}
&\int \frac{1}{1 + \tan x} dx = \int \frac{\cos x}{\sin x + \cos x} dx \\
&= \frac{1}{2} \int \frac{2 \cos x}{\sin x + \cos x} dx \\
&= \frac{1}{2}(x + \log |\sin x + \cos x|) + c
\end{aligned}$$

263. The given integral is

$$\begin{aligned}
&\int \frac{1}{1 - \tan x} dx \\
&= \int \frac{\cos x}{\cos x - \sin x} dx \\
&= \frac{1}{2} \int \frac{2 \cos x}{\cos x - \sin x} dx \\
&= \frac{1}{2} \int \frac{(\cos x - \sin x) + (\cos x + \sin x)}{\cos x - \sin x} dx \\
&= \frac{1}{2} \int \left(1 + \frac{(\cos x + \sin x)}{\cos x - \sin x} \right) dx \\
&= \frac{1}{2}(x - \log |\cos x - \sin x|) + c
\end{aligned}$$

264. The given integral is

$$\begin{aligned}
&\int \frac{1}{1 + \cot x} dx \\
&= \int \frac{\cos x}{\sin x + \cos x} dx \\
&= \frac{1}{2} \int \frac{2 \cos x}{\sin x + \cos x} dx \\
&= \frac{1}{2} \int \frac{(\cos x + \sin x) + (\cos x - \sin x)}{\sin x + \cos x} dx \\
&= \frac{1}{2} \int \left(1 + \frac{(\cos x - \sin x)}{\sin x + \cos x} \right) dx \\
&= \frac{1}{2}(x + \log |\sin x + \cos x|) + c
\end{aligned}$$

265. We have,

$$\begin{aligned}
&3 \sin x + 2 \cos x + 4 \\
&= l(3 \cos x + 4 \sin x + 5) \\
&\quad + m(-3 \sin x + 4 \cos x) + n \\
&= (4l - 3m) \sin x + (3l + 4m) \cos x \\
&\quad \quad \quad + (5l + n)
\end{aligned}$$

Comparing the co-efficients of $\sin x$ and $\cos x$ we get

$$4l - 3m = 3, 3l + 4m = 2$$

and

$$5l + n = 4$$

Solving, we get

$$l = \frac{18}{25}, m = -\frac{1}{25}, n = \frac{1}{5}$$

Thus,
$$\int \frac{3 \sin x + 2 \cos x + 4}{3 \cos x + 4 \sin x + 5} dx$$

$$= \int \left(\frac{18}{25} + \frac{1}{25} \left(\frac{4 \cos x - 3 \sin x}{4 \sin x + 3 \cos x + 5} \right) \right) dx$$

$$+ \frac{2}{5} \int \frac{dx}{4 \sin x + 3 \cos x + 5}$$

$$= \frac{18}{25}x - \frac{1}{25} \log |4 \sin x + 3 \cos x + 5|$$

$$+ \frac{2}{25} \tan^{-1} \left(\frac{x - \tan^{-1}(4/3)}{2} \right) + c.$$

269. We have,

$$\int (\cos x - \sin x)(2 + 3 \sin 2x) dx$$

$$= \int (\cos x - \sin x)(2 + 3(-1 + 1 + \sin 2x)) dx$$

$$= \int (\cos x - \sin x)(-1 + 3(\sin x + \cos x)^2) dx$$

$$= \int (3t^2 - 1) dt \quad \text{Let } t = \sin x + \cos x$$

$$\Rightarrow dt = (\cos x - \sin x) dx$$

$$= (t^3 - t) + c$$

$$= (\sin x + \cos x)^3 - (\sin x + \cos x) + c$$

270. We have,

$$\int \frac{\sin x + \cos x}{9 + 16 \sin 2x} dx$$

$$= \int \frac{\sin x + \cos x}{9 + 16(1 - 1 + \sin 2x)} dx$$

$$= \int \frac{\sin x + \cos x}{25 - 16(\sin x - \cos x)^2} dx$$

Let $\sin x - \cos x = t$

$$\Rightarrow (\cos x + \sin x) dx = dt$$

$$= \int \frac{dt}{25 - 16t^2}$$

$$= -\frac{1}{16} \int \frac{dt}{t^2 - \left(\frac{5}{4}\right)^2}$$

$$= -\frac{1}{16} \times \frac{1}{2 \times \frac{5}{4}} \log \left| \frac{t - \frac{5}{4}}{t + \frac{5}{4}} \right| + c$$

$$= -\frac{1}{40} \log \left| \frac{4t - 5}{4t + 5} \right| + c$$

$$= -\frac{1}{40} \log \left| \frac{49(\sin x - \cos x) - 5}{4(\sin x - \cos x) + 5} \right| + c$$

271. We have,

$$\int \frac{dx}{\cos x + \operatorname{cosec} x}$$

$$= \int \frac{dx}{\cos x + \frac{1}{\sin x}}$$

$$= \int \frac{\sin x dx}{\cos x \sin x + 1}$$

$$= \int \frac{2 \sin x dx}{2 \cos x \sin x + 2}$$

$$= \int \frac{(\sin x + \cos x) + (\sin x - \cos x)}{2 \cos x \sin x + 2}$$

$$= \int \frac{(\sin x + \cos x)}{2 \cos x \sin x + 2} dx$$

$$+ \int \frac{(\sin x - \cos x)}{2 \cos x \sin x + 2} dx$$

$$= \int \frac{(\sin x + \cos x)}{2 + \sin 2x} dx + \int \frac{(\sin x - \cos x)}{2 + \sin 2x} dx$$

$$= \int \frac{(\sin x + \cos x)}{3 - (\sin x - \cos x)^2} dx$$

$$+ \int \frac{(\sin x - \cos x)}{1 + (\sin x + \cos x)^2} dx$$

$$= -\frac{1}{2\sqrt{3}} \log \left| \frac{(\sin x - \cos x) - \sqrt{3}}{(\sin x - \cos x) + \sqrt{3}} \right|$$

$$- \tan^{-1}(\sin x + \cos x) + c$$

271. The given integral is

$$\int \frac{dx}{\cos x + \operatorname{cosec} x}$$

$$= \int \frac{\sin x \cos x}{\sin x + \cos x} dx$$

$$= \frac{1}{4} \int \frac{2 \sin 2x}{\sin x + \cos x} dx$$

$$= \frac{1}{4} \int \frac{(\sin x + \cos x)^2 - (\sin x - \cos x)^2}{\sin x + \cos x} dx$$

$$= \frac{1}{4} \int \left[(\sin x + \cos x) - \frac{(\sin x - \cos x)}{\sin x + \cos x} \right] dx$$

$$= \frac{1}{4} \left[(\cos x - \sin x) - \frac{1}{3} (\sin x - \cos x)^3 \right] + c$$

272. The given integral is

$$\int (\sin x + \cos x)(2 + 3 \sin 2x) dx$$

$$= \int (\sin x + \cos x)[2 + 3\{1 - (1 - \sin 2x)\}] dx$$

$$\begin{aligned}
&= \int (\sin x + \cos x)(5 - 3(\sin x - \cos x)^2) dx \\
&\quad \text{Let } \sin x - \cos x = t \\
&\quad \Rightarrow (\cos x + \sin x) dx = dt \\
&= \int (5 - 3t^2) dt \\
&= (5t - t^3) + c \\
&= (5(\sin x - \cos x) - (\sin x - \cos x)^3) + c
\end{aligned}$$

273. The given integral is

$$\begin{aligned}
&\int (\sin x - \cos x)(3 - 4 \sin 2x) dx \\
&= \int (\sin x - \cos x)(7 - 4(1 + \sin 2x)) \\
&= \int (\sin x - \cos x)(7 - 4(\sin x + \cos x)^2) dx \\
&= \int (\cos x - \sin x)(4(\sin x + \cos x)^2 - 7) dx \\
&\quad \text{Let } \sin x + \cos x = t \\
&\quad \Rightarrow (\cos x + \sin x) dx = dt \\
&= \int (4t^2 - 7) dt \\
&= \left(\frac{4t^3}{3} - 7t \right) + c \\
&= \left(\frac{4(\sin x + \cos x)^3}{3} - 7(\sin x + \cos x) \right) + c
\end{aligned}$$

274. The given integral is

$$\begin{aligned}
&\int \left(\frac{\cos x - \sin x}{3 + 2 \sin 2x} \right) dx \\
&= \int \frac{(\cos x - \sin x)}{1 + 2(1 + \sin 2x)} dx \\
&= \int \frac{(\cos x - \sin x)}{1 + 2(\sin x + \cos x)^2} dx \\
&\quad \text{Let } \sin x + \cos x = t \\
&\quad \Rightarrow (\cos x + \sin x) dx = dt \\
&= \int \frac{dt}{1 + 2t^2} \\
&= \frac{1}{2} \int \frac{dt}{\left(\frac{1}{\sqrt{2}} \right)^2 + t^2} \\
&= \frac{1}{\sqrt{2}} \tan^{-1}(t\sqrt{2}) + c \\
&= \frac{1}{\sqrt{2}} \tan^{-1}(\sqrt{2}(\sin x + \cos x)) + c
\end{aligned}$$

275. The given integral is

$$\begin{aligned}
&\int \left(\frac{\cos x + \sin x}{5 - 4 \sin 2x} \right) dx \\
&= \int \left(\frac{\cos x + \sin x}{5 - 4(1 - (\sin x - \cos x)^2)} \right) dx
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{\cos x + \sin x}{1 - 4(\sin x - \cos x)^2} dx \\
&\quad \text{Let } \sin x - \cos x = t \\
&\quad \Rightarrow (\cos x + \sin x) dx = dt \\
&= \int \frac{dt}{1 - 4t^2} \\
&= -\frac{1}{4} \int \frac{dt}{t^2 - (1/2)^2} \\
&= -\frac{1}{2} \log \left| \frac{2t - 1}{2t + 1} \right| + c \\
&= \frac{1}{2} \log \left| \frac{2(\sin x - \cos x) - 1}{2(\sin x - \cos x) + 1} \right| + c
\end{aligned}$$

276. The given integral is

$$\begin{aligned}
&\int \left(\frac{2 \cos x + \sin x}{9 + 16 \sin 2x} \right) dx \\
&= \int \frac{\frac{3}{2}(\cos x + \sin x) + \frac{1}{2}(\cos x - \sin x)}{9 + 6 \sin 2x} dx \\
&= \frac{3}{2} \int \frac{(\cos x + \sin x)}{9 + 6 \sin 2x} dx + \frac{1}{2} \int \frac{(\cos x - \sin x)}{(9 + 6 \sin 2x)} dx \\
&= \frac{3}{2} \int \frac{(\cos x + \sin x)}{15 - 6(\sin x - \cos x)^2} dx \\
&\quad + \frac{1}{2} \int \frac{(\cos x - \sin x)}{(3 + 6(\sin x + \cos x)^2)} dx
\end{aligned}$$

$$\begin{aligned}
&\text{Let } \sin x - \cos x = t \text{ and } \sin x + \cos x = z \\
&\Rightarrow (\cos x + \sin x) dx = dt \text{ and } (\cos x - \sin x) dx = dz
\end{aligned}$$

$$\begin{aligned}
&= \frac{3}{2} \int \frac{dt}{15 - 6t^2} + \frac{1}{2} \int \frac{dz}{6z^2 + 3} \\
&= \frac{1}{2} \int \frac{dt}{5 - 2t^2} + \frac{1}{6} \int \frac{dz}{2z^2 + 1} \\
&= -\frac{1}{4} \int \frac{dt}{t^2 - (\sqrt{5}/2)^2} + \frac{1}{6} \int \frac{dz}{2z^2 + 1} \\
&= -\frac{1}{4} \log \left| \frac{t - (\sqrt{5}/2)}{t + (\sqrt{5}/2)} \right| + \frac{1}{6\sqrt{2}} \tan^{-1}(z\sqrt{2}) + C
\end{aligned}$$

$$\text{where } \sin x - \cos x = t \text{ and } \sin x + \cos x = z$$

277. We have,

$$\begin{aligned}
&\int \frac{x^2 + 1}{x^4 + 1} dx \\
&= \int \left(\frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} \right) dx
\end{aligned}$$

$$= \int \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x - \frac{1}{x}\right)^2 + 2 \cdot x \cdot \frac{1}{x}} dx$$

$$\text{Let } \left(x - \frac{1}{x}\right) = t$$

$$\Rightarrow \left(1 + \frac{1}{x^2}\right) dx = dt$$

$$= \int \frac{dt}{t^2 + (\sqrt{2})^2}$$

$$= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{t}{\sqrt{2}}\right) + c$$

$$= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{1}{\sqrt{2}}\left(x - \frac{1}{x}\right)\right) + c$$

278. We have,

$$\int \frac{x^2 - 1}{x^4 + 1} dx$$

$$= \int \frac{\left(1 - \frac{1}{x^2}\right)}{x^2 + \frac{1}{x^2}} dx$$

$$= \int \frac{\left(1 - \frac{1}{x^2}\right)}{\left(x + \frac{1}{x}\right)^2 - 2 \cdot x \cdot \frac{1}{x}} dx$$

$$\text{Let } \left(x + \frac{1}{x}\right) = t$$

$$\Rightarrow \left(1 - \frac{1}{x^2}\right) dx = dt$$

$$= \int \frac{dt}{t^2 - (\sqrt{2})^2}$$

$$= \frac{1}{2\sqrt{2}} \log \left| \frac{t - \sqrt{2}}{t + \sqrt{2}} \right| + c$$

$$= \frac{1}{2\sqrt{2}} \log \left| \frac{\left(x + \frac{1}{x}\right) - \sqrt{2}}{\left(x + \frac{1}{x}\right) + \sqrt{2}} \right| + c$$

279. We have,

$$\int \frac{x^4 + 1}{x^6 + 1} dx$$

$$= \int \frac{(x^2 + 1)^2 - 2x^2}{x^6 + 1} dx$$

$$= \int \frac{(x^2 + 1)^2 - 2x^2}{(x^2 + 1)(x^4 - x^2 + 1)} dx$$

$$= \int \frac{(x^2 + 1)}{(x^4 - x^2 + 1)} dx - 2 \int \frac{x^2}{x^6 + 1} dx$$

$$= \int \frac{(x^2 + 1)}{(x^4 - x^2 + 1)} dx - \frac{2}{3} \int \frac{3x^2}{x^6 + 1} dx$$

$$= \int \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x^2 + \frac{1}{x^2} - 1\right)} dx - \frac{2}{3} \int \frac{3x^2}{x^6 + 1} dx$$

$$= \int \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x - \frac{1}{x}\right)^2 + 1} dx - \frac{2}{3} \int \frac{3x^2}{x^6 + 1} dx$$

$$= \int \tan^{-1}\left(x - \frac{1}{x}\right) - \frac{2}{3} \tan^{-1}(x^3) + c$$

280. We have,

$$\int \frac{x^2 - 3x + 1}{x^4 + x^2 + 1} dx$$

$$= \int \frac{(x^2 - x + 1) - 2x}{x^4 + x^2 + 1} dx$$

$$= \int \frac{(x^2 - x + 1)}{x^4 + x^2 + 1} dx - \int \frac{2x}{x^4 + x^2 + 1} dx$$

$$= \int \frac{dx}{x^2 + x + 1} - \int \frac{2x}{x^4 + x^2 + 1} dx$$

$$= \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} - \int \frac{dt}{t^2 + t + 1} \quad \text{Let } x^2 = t$$

$$\Rightarrow 2x dx = dt$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{t + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right) + c$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x + 1}{\sqrt{3}}\right) + \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2t + 1}{\sqrt{3}}\right) + c$$

$$= \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x + 1}{\sqrt{3}}\right) + \frac{2}{\sqrt{3}} \tan^{-1}\left(\frac{2x^2 + 1}{\sqrt{3}}\right) + c$$

281. We have,

$$\int \frac{\left(1 - x^2\right)}{\left(1 + x^2\right)\sqrt{1 + x^2 + x^4}} dx$$

$$= \int \frac{-x^2\left(1 - \frac{1}{x^2}\right) dx}{x\left(x + \frac{1}{x}\right)\sqrt{x^2\left(x^2 + \frac{1}{x^2} + 1\right)}}$$

$$= - \int \frac{\left(1 - \frac{1}{x^2}\right) dx}{\left(x + \frac{1}{x}\right)\sqrt{\left(x^2 + \frac{1}{x^2} + 1\right)}}$$

$$\begin{aligned}
&= -\int \frac{dt}{t\sqrt{t^2-1}} \quad \text{Let } \left(x + \frac{1}{x}\right) = t \\
&\quad \Rightarrow \left(1 - \frac{1}{x^2}\right)dx = dt \\
&= \operatorname{cosec}^{-1}(t) + c \\
&= \operatorname{cosec}^{-1}\left(x + \frac{1}{x}\right) + c \\
&= \sin^{-1}\left(\frac{x}{x^2+1}\right) + c
\end{aligned}$$

282. We have,

$$\begin{aligned}
&\int \left(\frac{x-1}{x+1}\right) \times \frac{dx}{\sqrt{x^3+x^2+x}} \\
&= \int \left(\frac{x^2-1}{(x+1)^2}\right) \times \frac{dx}{\sqrt{x^2\left(x+\frac{1}{x}+1\right)}} \\
&= \int \left(\frac{x^2-1}{x(x+1)^2}\right) \times \frac{dx}{\sqrt{\left(x+\frac{1}{x}+1\right)}} \\
&= \int \left(\frac{x^2-1}{x(x^2+2x+1)}\right) \times \frac{dx}{\sqrt{\left(x+\frac{1}{x}+1\right)}} \\
&= \int \left(\frac{x^2-1}{x^2\left(x+\frac{1}{x}+2\right)}\right) \times \frac{dx}{\sqrt{\left(x+\frac{1}{x}+1\right)}} \\
&= \int \left(\frac{\left(1-\frac{1}{x^2}\right)}{\left(x+\frac{1}{x}+2\right)}\right) \times \frac{dx}{\sqrt{\left(x+\frac{1}{x}+1\right)}} \\
&\quad \text{Let } \left(x+\frac{1}{x}+1\right) = t^2 \\
&\quad \Rightarrow \left(1-\frac{1}{x^2}\right)dx = dt \\
&= \int \frac{2tdt}{(t^2+1)t} \\
&= \int \frac{2dt}{(t^2+1)} \\
&= 2\tan^{-1}t + c \\
&= 2\tan^{-1}\left(x+\frac{1}{x}+1\right)^{1/2} + c
\end{aligned}$$

283. We have,

$$\begin{aligned}
&\int \frac{x^2-1}{x^2+1} \times \frac{dx}{\sqrt{x^4+1}} \\
&= \int \frac{x^2-1}{x\left(x+\frac{1}{x}\right)} \times \frac{dx}{\sqrt{x^2\left(x^2+\frac{1}{x^2}\right)}} \\
&= \int \frac{x^2-1}{x^2\left(x+\frac{1}{x}\right)} \times \frac{dx}{\sqrt{x^2\left(x^2+\frac{1}{x^2}\right)}}
\end{aligned}$$

$$\begin{aligned}
&= \int \frac{\left(1-\frac{1}{x^2}\right)}{\left(x+\frac{1}{x}\right)} \times \frac{dx}{\sqrt{\left(x+\frac{1}{x}\right)^2-2}} \\
&= \int \frac{dt}{t\sqrt{t^2-2}} \quad \text{Let } \left(x+\frac{1}{x}\right) = t \\
&\quad \Rightarrow \left(1-\frac{1}{x^2}\right)dx = dt \\
&= \int \frac{tdt}{t^2\sqrt{t^2-2}} \\
&= \int \frac{v dv}{(v^2+2)v} \quad \text{Let } t^2-2 = v^2 \\
&\quad \Rightarrow 2t dt = 2v dv \\
&= \int \frac{dv}{(v^2+2)} \\
&= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{v}{\sqrt{2}}\right) + c \\
&= \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\sqrt{t^2-2}}{\sqrt{2}}\right) + c \\
&= \frac{1}{\sqrt{2}} \tan^{-1}\left(\sqrt{\frac{1}{2}\left(x^2+\frac{1}{x^2}\right)}\right) + c
\end{aligned}$$

284. We have,

$$\begin{aligned}
&\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}} dx \\
&= \int \frac{x^2-1}{\sqrt{x^3\left(2-\frac{2}{x^2}+\frac{1}{x^4}\right)}} dx \\
&= \int \frac{x^2-1}{x^5\sqrt{\left(2-\frac{2}{x^2}+\frac{1}{x^4}\right)}} dx \\
&= \int \frac{\left(\frac{1}{x^3}-\frac{1}{x^5}\right)}{\sqrt{\left(2-\frac{2}{x^2}+\frac{1}{x^4}\right)}} dx \\
&\quad \text{Let } \left(2-\frac{2}{x^2}+\frac{1}{x^4}\right) = t^2 \\
&\quad \Rightarrow \left(\frac{4}{x^3}-\frac{4}{x^5}\right)dx = 2t dt \\
&= \frac{1}{4} \int \frac{2tdt}{t} \\
&= \frac{1}{2} \times t + c \\
&= \frac{1}{2} \times \left(2-\frac{2}{x^2}-\frac{1}{x^4}\right)^{1/2} + c
\end{aligned}$$

285. We have,

$$\int \left(\frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} \right) dx$$

$$= \int \left(\frac{x^4 - 1}{x^2 \sqrt{x^2 \left(x^2 + \frac{1}{x^2} + 1 \right)}} \right) dx$$

$$= \int \left(\frac{\left(\frac{x^4 - 1}{x^3} \right)}{\sqrt{\left(x^2 + \frac{1}{x^2} + 1 \right)}} \right) dx$$

$$= \int \left(\frac{\left(x - \frac{1}{x^3} \right)}{\sqrt{\left(x^2 + \frac{1}{x^2} + 1 \right)}} \right) dx$$

$$\text{Let } \left(x^2 + \frac{1}{x^2} + 1 \right) = t^2$$

$$\Rightarrow \left(2x - \frac{2}{x^3} \right) dx = 2t dt$$

$$\Rightarrow \left(x - \frac{1}{x^3} \right) dx = t dt$$

$$= \int \frac{t dt}{t}$$

$$= \int dt$$

$$= t + c$$

$$= \left(x^2 + \frac{1}{x^2} + 1 \right)^{1/2} + c$$

286. We have,

$$\int \frac{dx}{x^2(x + \sqrt{1 + x^2})}$$

$$= \int \frac{dx}{x^2 \left(x + \sqrt{x^2 \left(1 + \frac{1}{x^2} \right)} \right)}$$

$$= \int \frac{dx}{x^2 \left(x + x \sqrt{\left(1 + \frac{1}{x^2} \right)} \right)}$$

$$= \left(1 + \frac{1}{x^2} \right) = t^2$$

$$\text{Put } \left(1 + \frac{1}{x^2} \right) = t^2$$

$$\Rightarrow -\frac{2}{x^3} dx = 2t dt$$

$$\Rightarrow \frac{dx}{x^3} = -t dt$$

$$= \int \frac{-t dt}{1 + t}$$

$$= \int \left(\frac{1}{t + 1} - 1 \right) dt$$

$$= \log|t + 1| - t + c$$

$$= \log \left| 1 + \left(1 + \frac{1}{x^2} \right)^{1/2} \right| - \left(1 + \frac{1}{x^2} \right)^{1/2} + c$$

287. We have,

$$\int \frac{x^2 - 3x - 1}{x^4 + x^2 + 1} dx$$

$$= \int \frac{x^2 - 1}{x^4 + x^2 + 1} dx - \frac{3}{2} \int \frac{2x}{x^4 + x^2 + 1} dx$$

$$= \int \frac{\left(1 - \frac{1}{x^2} \right)}{x^2 + \frac{1}{x^2} + 1} dx - \frac{3}{2} \int \frac{2x}{x^4 + x^2 + 1} dx$$

$$= \int \frac{\left(1 - \frac{1}{x^2} \right)}{\left(1 - \frac{1}{x} \right)^2 - 1} dx - \frac{3}{2} \int \frac{2x}{x^4 + x^2 + 1} dx$$

$$\text{Put } \left(x + \frac{1}{x} \right) = t \text{ and } x^2 = z$$

$$\Rightarrow \left(1 + \frac{1}{x^2} \right) dx = dt \text{ and } 2x dx = dz$$

$$= \int \frac{dt}{t^2 - 1} - \frac{3}{2} \int \frac{dz}{z^2 + z + 1}$$

$$= \frac{1}{2} \log \left| \frac{t - 1}{t + 1} \right| - \frac{3}{2} \times \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2z + 1}{\sqrt{3}} \right) + c$$

$$= \frac{1}{2} \log \left| \frac{\left(x + \frac{1}{x} \right) - 1}{\left(x + \frac{1}{x} \right) + 1} \right| - \sqrt{3} \tan^{-1} \left(\frac{2(x^2 + 1)}{\sqrt{3}} \right) + c$$

288. The given integral is

$$\int \frac{dx}{\sin^4 x + \cos^4 x} = \int \frac{\sec^4 x}{\tan^4 x + 1} dx$$

$$= \int \frac{(1 + \tan^2 x) \sec^2 x}{\tan^4 x + 1} dx$$

$$\text{Put } \tan x = t$$

$$\Rightarrow \sec^2 x dx = dt$$

$$= \int \frac{t^2 + 1}{t^4 + 1} dt$$

and then solve it.

289. The given integral is

$$\int \frac{dx}{x^4 + 1} = \frac{1}{2} \int \frac{2}{(x^4 + 1)} dx$$

$$\begin{aligned}
&= \frac{1}{2} \int \frac{(x^2 + 1) - (x^2 - 1)}{(x^4 + 1)} dx \\
&= \frac{1}{2} \int \frac{(x^2 + 1)}{(x^4 + 1)} dx - \frac{1}{2} \int \frac{(x^2 - 1)}{(x^4 + 1)} dx
\end{aligned}$$

and then you do it.

290. The given integral is

$$\begin{aligned}
\int \frac{x^2 dx}{x^4 + 1} &= \frac{1}{2} \int \frac{2x^2}{x^4 + 1} dx \\
&= \frac{1}{2} \int \frac{(x^2 + 1) + (x^2 - 1)}{x^4 + 1} dx \\
&= \frac{1}{2} \int \frac{(x^2 + 1)}{x^4 + 1} dx + \frac{1}{2} \int \frac{(x^2 - 1)}{x^4 + 1} dx
\end{aligned}$$

and then you do it.

291. The given integral is

$$\begin{aligned}
\int \frac{2dx}{x^4 + 1} &= \int \frac{(x^2 + 1) - (x^2 - 1)}{x^4 + 1} dx \\
&= \int \frac{(x^2 + 1)}{x^4 + 1} dx - \int \frac{(x^2 - 1)}{x^4 + 1} dx
\end{aligned}$$

and then you do it.

292. The given integral is

$$\begin{aligned}
&\int \left(\frac{x^4 + 1}{x^6 + 1} \right) dx \\
&= \int \frac{(x^2 + 1)^2 - 2x^2}{(x^6 + 1)} dx \\
&= \int \frac{(x^2 + 1)^2 - 2x^2}{(x^2 + 1)(x^4 - x^2 + 1)} dx \\
&= \int \frac{(x^2 + 1)}{(x^4 - x^2 + 1)} dx - \frac{2}{3} \int \frac{3x^2}{(x^3)^2 + 1} dx \\
&= \int \frac{(x^2 + 1)}{(x^4 - x^2 + 1)} dx - \frac{2}{3} \tan^{-1}(x^3) + c
\end{aligned}$$

and then you do it.

293. The given integral is

$$\int \frac{dx}{x^2(1 + x^4)^{3/4}} = \int \frac{dx}{x^5 \left(1 + \frac{1}{x^4}\right)^{3/4}}$$

$$\text{Let } \left(1 + \frac{1}{x^4}\right) = t^4$$

$$\frac{-1}{x^5} dx = t^3 dt$$

$$= -\int dt$$

$$= -t + c$$

$$= -\left(1 + \frac{1}{x^4}\right)^{1/4} + c$$

294. The given integral is

$$\begin{aligned}
&\int \left(\frac{x^4 - 1}{x^2 \sqrt{x^4 + x^2 + 1}} \right) dx \\
&= \int \frac{x^4 - 1}{x^2 \sqrt{x^2 \left(x^2 + \frac{1}{x^2} + 1\right)}} dx \\
&= \int \frac{x^4 - 1}{x^3 \sqrt{\left(x^2 + \frac{1}{x^2} + 1\right)}} dx \\
&= \int \frac{x - \frac{1}{x^3}}{\sqrt{\left(x^2 + \frac{1}{x^2} + 1\right)}} dx
\end{aligned}$$

$$\text{Put } \left(x^2 + \frac{1}{x^2} + 1\right) = t^2$$

$$\left(x - \frac{1}{x^3}\right) dx = t dt$$

$$= \int dt$$

$$= t + c$$

$$= \sqrt{x^2 + \frac{1}{x^2} + 1} + c$$

295. The given integral is

$$\int \frac{dx}{x^2(1 + x^5)^{4/5}} = \int \frac{dx}{x^6 \left(1 + \frac{1}{x^5}\right)^{4/5}}$$

$$\text{Put } \left(1 + \frac{1}{x^5}\right) = t^5$$

$$\Rightarrow -\frac{5}{x^6} dx = 5t^4 dt$$

$$\Rightarrow -\frac{1}{x^6} dx = t^4 dt$$

$$= -\int dt$$

$$= -t + c$$

$$= -\left(1 + \frac{1}{x^5}\right)^{1/5} + c$$

296. The given integral is

$$\begin{aligned}
&\int \left(\frac{x^2 - 1}{x \sqrt{1 + x^4}} \right) dx \\
&= \int \frac{(x^2 - 1)}{x^2 \sqrt{x^2 + \frac{1}{x^2}}} dx
\end{aligned}$$

$$= \int \frac{\left(1 - \frac{1}{x^2}\right)}{\sqrt{\left(x + \frac{1}{x}\right)^2 - 2}} dx$$

$$\text{Let } \left(x + \frac{1}{x}\right) = t$$

$$\Rightarrow \left(1 - \frac{1}{x^2}\right) dx = t^4 dt$$

$$= \int \frac{dt}{\sqrt{t^2 - 2}}$$

$$= \log \left| t + \sqrt{t^2 - 2} \right| + c$$

$$= \log \left| \left(x + \frac{1}{x}\right) + \sqrt{x^2 + \frac{1}{x^2}} \right| + c$$

297. The given integral is

$$\int \frac{dx}{x\sqrt{x^4 + 3x^2 + 1}}$$

$$= \int \frac{dx}{x^2 \sqrt{x^2 + \frac{1}{x^2} + 3}}$$

$$= \frac{1}{2} \int \frac{\frac{2}{x^2}}{\sqrt{x^2 + \frac{1}{x^2} + 3}} dx$$

$$= \frac{1}{2} \int \frac{\left(1 + \frac{1}{x^2}\right) - \left(1 - \frac{1}{x^2}\right)}{\sqrt{x^2 + \frac{1}{x^2} + 3}} dx$$

$$= \frac{1}{2} \int \frac{\left(1 + \frac{1}{x^2}\right)}{\sqrt{x^2 + \frac{1}{x^2} + 3}} dx - \frac{1}{2} \int \frac{\left(1 - \frac{1}{x^2}\right)}{\sqrt{x^2 + \frac{1}{x^2} + 3}} dx$$

$$= \frac{1}{2} \int \frac{\left(1 + \frac{1}{x^2}\right)}{\sqrt{\left(x - \frac{1}{x}\right)^2 + 5}} dx - \frac{1}{2} \int \frac{\left(1 - \frac{1}{x^2}\right)}{\sqrt{\left(x + \frac{1}{x}\right)^2 + 1}} dx$$

$$= \frac{1}{2} \log \left| \left(x - \frac{1}{x}\right) + \sqrt{x^2 + \frac{1}{x^2} + 3} \right|$$

$$- \frac{1}{2} \tan^{-1} \left(x + \frac{1}{x}\right) + c$$

298. The given integral is

$$\int \left(\frac{x^2 + 1}{1 - x^2}\right) \times \frac{dx}{\sqrt{x^4 + x^2 + 1}}$$

$$= - \int \frac{\left(1 + \frac{1}{x^2}\right)}{\left(x - \frac{1}{x}\right)} \times \frac{dx}{\sqrt{\left(x - \frac{1}{x}\right)^2 + 3}}$$

$$\text{Let } \left(x - \frac{1}{x}\right) = t$$

$$\Rightarrow \left(1 - \frac{1}{x^2}\right) dx = dt$$

$$= - \int \frac{dt}{t\sqrt{t^2 + 3}}$$

$$= - \int \frac{t dt}{t^2 \sqrt{t^2 + 3}}$$

$$\text{Let } t^2 + 3 = v^2$$

$$\Rightarrow 2t dt = 2v dv$$

$$= - \int \frac{v dv}{(v^2 - 3)v}$$

$$= - \int \frac{dv}{(v^2 - 3)}$$

$$= - \frac{1}{2\sqrt{3}} \log \left| \frac{v - \sqrt{3}}{v + \sqrt{3}} \right| + C$$

$$= - \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{t^2 + 3} - \sqrt{3}}{\sqrt{t^2 + 3} + \sqrt{3}} \right| + c$$

$$= - \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{x^2 + \frac{1}{x^2} + 1} - \sqrt{3}}{\sqrt{x^2 + \frac{1}{x^2} + 1} + \sqrt{3}} \right| + c$$

299. The given integral is

$$\int \left(\frac{x^2 - 1}{x}\right) \times \frac{dx}{\sqrt{x^4 + 3x^2 + 1}}$$

$$= \int \left(1 - \frac{1}{x^2}\right) \times \frac{dx}{\sqrt{x^2 + \frac{1}{x^2} + 3}}$$

$$= \int \left(1 - \frac{1}{x^2}\right) \times \frac{dx}{\sqrt{\left(x + \frac{1}{x}\right)^2 + 1}}$$

$$= \log \left| \left(x + \frac{1}{x}\right) + \sqrt{x^2 + \frac{1}{x^2} + 3} \right| + c$$

300. The given integral is

$$\int \frac{x^x(x^{2x} + 1)(\ln x + 1)}{(x^{4x} + 1)} dx$$

$$\text{Let } x^x = t$$

$$\Rightarrow x^x(1 + \ln x) dx = dt \quad 50$$

$$\begin{aligned}
&= \int \frac{t^2 + 1}{t^4 + 1} dt \\
&= \int \frac{\left(1 + \frac{1}{t^2}\right)}{t^2 + \frac{1}{t^2}} dt \\
&= \int \frac{\left(1 + \frac{1}{t^2}\right)}{\left(t - \frac{1}{t}\right)^2 + 2} dt \\
&= \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{1}{\sqrt{2}} \left(t - \frac{1}{t}\right) \right) + c \\
&= \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{1}{\sqrt{2}} (x^x - x^{-x}) \right) + c
\end{aligned}$$

301. The given integral is

$$\begin{aligned}
&\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx \\
&= \int \frac{(x^2 - 1)}{x^5 \sqrt{\left(2 - \frac{2}{x^2} + \frac{1}{x^4}\right)}} dx \\
&= \int \frac{\left(\frac{1}{x^3} - \frac{1}{x^5}\right)}{\sqrt{\left(2 - \frac{2}{x^2} + \frac{1}{x^4}\right)}} dx \\
&\quad \text{Let } \left(2 - \frac{2}{x^2} + \frac{1}{x^4}\right) = t^2 \\
&\quad \Rightarrow \left(\frac{4}{x^2} - \frac{4}{x^4}\right) dx = 2t dt \\
&\quad \Rightarrow \left(\frac{1}{x^3} - \frac{1}{x^5}\right) dx = \frac{1}{2} t dt \\
&= \frac{1}{2} \int \frac{t dt}{t} \\
&= \frac{1}{2} \int dt \\
&= \frac{1}{2} t + c \\
&= \frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + c \\
&= \sqrt{\frac{2x^4 - 2x^2 + 1}{2x^2}} + c
\end{aligned}$$

302. The given integral is

$$\int \left(\frac{x^{2009}}{(1+x^2)^{1006}} \right) dx$$

$$\begin{aligned}
&= \int \frac{x^{2009}}{(1+x^2)^{1006}} dx \\
&= \int \frac{x^{2009}}{x^{2012} \left(1 + \frac{1}{x^2}\right)^{1006}} dx \\
&= \int \frac{dx}{x^3 \left(1 + \frac{1}{x^2}\right)^{1006}} \\
&\quad \text{Let } \left(1 + \frac{1}{x^2}\right) = t \\
&\quad \Rightarrow -\frac{2}{x^3} dx = dt \\
&\quad \Rightarrow \frac{dx}{x^3} = \frac{dt}{-2} \\
&= -\frac{1}{2} \int \frac{dt}{t^{1006}} \\
&= -\frac{1}{2} \times \frac{t^{-1005}}{-1005} + C \\
&= \frac{1}{2010 \times t^{1005}} + C \\
&= \frac{1}{2010 \times \left(1 + \frac{1}{x^2}\right)^{1005}} + C
\end{aligned}$$

303. We have,

$$\begin{aligned}
&\int \frac{(2x+1)dx}{(x^2+4x+1)^{3/2}} \\
&= \int \frac{(2x+1)dx}{x^3 \left(1 + \frac{4}{x} + \frac{1}{x^2}\right)^{3/2}} \\
&= \int \frac{\frac{2}{x^2} + \frac{1}{x^3}}{\left(1 + \frac{4}{x} + \frac{1}{x^2}\right)^{3/2}} dx \\
&\quad \text{Let } \left(1 + \frac{4}{x} + \frac{1}{x^2}\right) = t^2 \\
&\quad \Rightarrow -\frac{4}{x^2} - \frac{2}{x^3} dx = 2t dt \\
&\quad \Rightarrow \left(\frac{2}{x^2} + \frac{1}{x^3}\right) dx = -t dt \\
&= -\int \frac{t dt}{t^3} \\
&= -\int \frac{dt}{t^2} \\
&= \frac{1}{t} + c
\end{aligned}$$

$$= \frac{1}{\sqrt{1 + \frac{4}{x} + \frac{2}{x^2}}} + c$$

304. We have,

$$\begin{aligned} & \int \left(\frac{\sqrt{\cot x} - \sqrt{\tan x}}{1 + 3 \sin 2x} \right) dx \\ &= \int \frac{1}{\sqrt{\tan x}} \left(\frac{1 - \tan x}{1 + 6 \sin x \cos x} \right) dx \\ &= \int \frac{1}{\sqrt{\tan x}} \left(\frac{1 - \tan x}{\sec^2 x + 6 \tan x} \right) \sec^2 x dx \\ &= \int \frac{1}{\sqrt{\tan x}} \left(\frac{1 - \tan x}{\tan^2 x + 6 \tan x + 1} \right) \sec^2 x dx \\ &= \int \frac{1}{t} \left(\frac{1 - t^2}{t^4 + 6t^2 + 1} \right) 2t dt \\ &= 2 \int \left(\frac{1 - t^2}{t^4 + 6t^2 + 1} \right) dt \\ &= 2 \int \left(\frac{\frac{1}{t^2} - 1}{t^2 + \frac{1}{t^2} + 6} \right) dt \\ &= 2 \int \left(\frac{\frac{1}{t^2} - 1}{\left(t + \frac{1}{t}\right)^2 + 4} \right) dt \\ &= 2 \int \left(\frac{dz}{z^2 + 4} \right) \quad \text{Let } z = \left(t + \frac{1}{t}\right) \\ &\quad \Rightarrow dx^2 = \left(1 - \frac{1}{t^2}\right) dx \\ &= 2 \times \frac{1}{2} \tan^{-1} \left(\frac{z}{2} \right) + c \\ &= \tan^{-1} \left(\frac{1}{2} \left(t + \frac{1}{t}\right) \right) + c \\ &= \tan^{-1} \left(\frac{1}{2} \sqrt{\tan x} + \sqrt{\cot x} \right) + c \end{aligned}$$

305. We have,

$$\int \sqrt{\tan x} dx$$

$$\begin{aligned} & \text{Put } \tan x = t^2 \\ & \Rightarrow \sec^2 x dx = 2t dt \\ & \Rightarrow dx = \frac{2t dt}{\sec^2 x} \\ &= \frac{2t dt}{1 + \tan^2 x} \\ &= \frac{2t dt}{1 + t^4} \end{aligned}$$

$$\begin{aligned} &= \int \frac{t \cdot 2t dt}{t^4 + 1} \\ &= \int \frac{2t^2 dt}{t^4 + 1} \\ &= \int \frac{(t^2 - 1) + (t^2 + 1) dt}{t^4 + 1} \\ &= \int \frac{(t^2 - 1) dt}{t^4 + 1} + \int \frac{(t^2 + 1) dt}{t^4 + 1} \\ &= \int \frac{\left(1 - \frac{1}{t^2}\right) dt}{t^2 + \frac{1}{t^2}} + \int \frac{\left(1 + \frac{1}{t^2}\right) dt}{t^2 + \frac{1}{t^2}} \\ &= \int \frac{\left(1 - \frac{1}{t^2}\right) dt}{\left(t + \frac{1}{t}\right)^2 - 2} + \int \frac{\left(1 + \frac{1}{t^2}\right) dt}{\left(t - \frac{1}{t}\right)^2 + 2} \\ &= \frac{1}{2\sqrt{2}} \log \left| \frac{\left(t + \frac{1}{t}\right) - \sqrt{2}}{\left(t + \frac{1}{t}\right) + \sqrt{2}} \right| + \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\left(t - \frac{1}{t}\right)}{\sqrt{2}} \right) + c \\ &= \frac{1}{2\sqrt{2}} \log \left| \frac{\left(x^2 + \frac{1}{x^2}\right) - \sqrt{2}}{\left(x^2 + \frac{1}{x^2}\right) + \sqrt{2}} \right| \\ &\quad + \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\left(x^2 - \frac{1}{x^2}\right)}{\sqrt{2}} \right) + c \end{aligned}$$

306. We have,

$$\begin{aligned} & \int (\sqrt{\tan x} - \sqrt{\cot x}) dx \\ &= \int \left(\sqrt{\tan x} - \frac{1}{\sqrt{\tan x}} \right) dx \\ &= \int \left(\frac{\tan x - 1}{\sqrt{\tan x}} \right) dx \end{aligned}$$

$$\begin{aligned} & \text{Put } \tan x = t^2 \\ & \Rightarrow \sec^2 x dx = 2t dt \\ & \Rightarrow dx = \frac{2t dt}{\sec^2 x} \\ &= \frac{2t dt}{1 + \tan^2 x} \\ &= \frac{2t dt}{1 + t^4} \end{aligned}$$

$$\begin{aligned}
&= \int \left(\frac{t^2 - 1}{t^4 + 1} \right) 2t dt \\
&= \int \left(\frac{t^2 - 1}{t^4 + 1} \right) \frac{2t dt}{t} \\
&= \int \left(\frac{t^2 - 1}{t^4 + 1} \right) dt \\
&= \int \left(\frac{1 - \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} \right) dt \\
&= \int \left(\frac{1 - \frac{1}{t^2}}{\left(t + \frac{1}{t}\right)^2 - 2} \right) dt \\
&= \frac{1}{2\sqrt{2}} \log \left| \frac{\left(t + \frac{1}{t}\right) - \sqrt{2}}{\left(t + \frac{1}{t}\right) + \sqrt{2}} \right| + c \\
&= \frac{1}{2\sqrt{2}} \log \left| \frac{(\sqrt{\tan x} + \sqrt{\cot x}) - \sqrt{2}}{(\sqrt{\tan x} + \sqrt{\cot x}) + \sqrt{2}} \right| + c
\end{aligned}$$

307. Do yourself

$$308. \int \left(\sqrt{\cot x} - \frac{1}{\sqrt{\cot x}} \right) dx = \int \left(\frac{\cot x - 1}{\sqrt{\cot x}} \right) dx$$

$$\text{Let } \cot x = t^2$$

$$-\operatorname{cosec}^2 x dx = 2t dt$$

$$dx = -\frac{2t}{1+t^4} dt$$

$$= \int \left(t^2 - \frac{1}{t} \right) \times \frac{2t}{1+t^4} dt$$

$$= 2 \int \left(\frac{t^2 - 1}{t^4 + 1} \right) dt$$

$$= 2 \int \left(\frac{1 - \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} \right) dt$$

$$= 2 \int \left(\frac{1 - \frac{1}{t^2}}{\left(t + \frac{1}{t}\right)^2 - 2} \right) dt$$

$$= 2 \times \frac{1}{\sqrt{2}} \log \left| \frac{\left(t + \frac{1}{t}\right) - \sqrt{2}}{t + \frac{1}{t} + \sqrt{2}} \right| + c$$

$$\begin{aligned}
&= \sqrt{2} \log \left| \frac{\left(t + \frac{1}{t}\right) - \sqrt{2}}{t + \frac{1}{t} + \sqrt{2}} \right| + c \\
&= \sqrt{2} \log \left| \frac{(\sqrt{\cot x} + \sqrt{\tan x}) - \sqrt{2}}{(\sqrt{\cot x} + \sqrt{\tan x}) + \sqrt{2}} \right|
\end{aligned}$$

309. Do yourself

310. Do yourself

$$311. \int (\sqrt[4]{\tan x} + \sqrt[4]{\cot x})^2 dx$$

$$= \int (\sqrt{\tan x} + \sqrt{\cot x} + 2) dx$$

$$= \int \left(\frac{\tan x + 1}{\sqrt{\tan x}} + 2 \right) dx$$

$$= 2x + \int \left(\frac{\tan x + 1}{\sqrt{\tan x}} \right) dx$$

$$\begin{aligned}
&= 2x + 2 \int \left(\frac{t^2 + 1}{t^4 + 1} \right) dt \quad \text{Let } \tan x = t^2 \\
&\Rightarrow \sec^2 x dx = 2t dt \\
&\Rightarrow dx = \frac{2t}{1+t^4} dt
\end{aligned}$$

$$= 2x + 2 \int \left(\frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} \right) dt$$

$$= 2x + 2 \int \left(\frac{1 + \frac{1}{t^2}}{\left(t - \frac{1}{t}\right)^2 + 2} \right) dt$$

$$= 2x + 2 \times \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{1}{\sqrt{2}} \left(t - \frac{1}{t} \right) \right) + c$$

$$= 2x + \sqrt{2} \tan^{-1} \left(\frac{1}{\sqrt{2}} (\tan x - \cot x) \right) + c$$

312. We have,

$$\int \frac{dx}{(\sqrt{\sin x} + \sqrt{\cot x})^4} = \int \frac{\sec^2 x}{(\sqrt{\tan x} + 1)^4} dx$$

$$\text{Put } \tan x = t^2$$

$$\Rightarrow \sec^2 x dx = 2t dt$$

$$= \int \frac{dt}{(t+1)^4}$$

$$= -\frac{4}{(t+1)^3} + c$$

$$= -\frac{4}{(\sqrt{\tan x} + 1)^3} + c$$

Integration by Parts

313. We have,

$$\begin{aligned}\int x e^x dx &= x \int e^x dx - \int (1 \cdot e^x) dx \\ &= x e^x - e^x + c\end{aligned}$$

314. We have,

$$\begin{aligned}\int x \sin x dx &= x \int \sin x dx - \int (1 - \cos x) dx \\ &= x \int \sin x dx + \int \cos x dx \\ &= -x \cos x + \sin x + c\end{aligned}$$

315. We have,

$$\begin{aligned}\int x^2 \sin x dx &= x^2 \int \sin x dx - \int (2x - \cos x) dx \\ &= -x^2 \cos x + 2 \int \cos x dx \\ &= -x^2 \cos x + 2x \int \cos x dx - 2 \int (1 \sin x) dx \\ &= -x^2 \cos x + 2x \sin x + 2 \cos x + c\end{aligned}$$

316. We have,

$$\begin{aligned}\int \log x dx &= \int (\log x \cdot 1) dx \\ &= \log x \int dx - \int \left(\frac{1}{x} \cdot x\right) dx \\ &= \log x \int dx - \int 1 \cdot dx \\ &= x \log x - x + c\end{aligned}$$

317. We have,

$$\begin{aligned}\int (\log x)^2 dx &= \int \{(\log x)^2 \cdot 1\} dx \\ &= (\log x)^2 \int dx - \int \left(2 \log x \cdot \frac{1}{x} \cdot x\right) dx \\ &= x(\log x)^2 - 2 \int \log x dx + c \\ &= x(\log x)^2 - 2(x \log x - x) + c\end{aligned}$$

318. We have,

$$\begin{aligned}\int \log(x^2 + 1) dx &= \int (\log(x^2 + 1) \cdot 1) dx \\ &= \log(x^2 + 1) \int dx - \int \left(\frac{1}{x^2 + 1} \cdot 2x \cdot x\right) dx \\ &= x \log(x^2 + 1) - 2 \int \left(\frac{x^2}{x^2 + 1}\right) dx\end{aligned}$$

$$= x \log(x^2 + 1) - 2 \int \left(1 - \frac{1}{1 + x^2}\right) dx$$

$$= x \log(x^2 + 1) - 2(x - \tan^{-1} x) + c$$

319. We have,

$$\begin{aligned}\int \tan^{-1} x dx &= \int \{\tan^{-1} x \cdot 1\} dx \\ &= \tan^{-1}(x) \int dx - \int \frac{x}{1 + x^2} dx \\ &= x \tan^{-1}(x) - \frac{1}{2} \log |1 + x^2| + c\end{aligned}$$

320. We have,

$$\begin{aligned}\int \cos^{-1} \left(\frac{1-x^2}{1+x^2}\right) dx &= \int (2 \tan^{-1} x) dx \\ &= 2 \int \tan^{-1} x dx \\ &= 2 \left(\tan^{-1} x \int dx - \int \left(\frac{-1}{1+x^2} \cdot x\right) dx \right) \\ &= 2 \left(x \tan^{-1} x - \frac{1}{2} \int \left(\frac{-2x}{1+x^2}\right) dx \right) + c \\ &= 2 \left(x \tan^{-1} x - \frac{1}{2} \log |x^2 + 1| \right) + c\end{aligned}$$

321. We have,

$$\begin{aligned}\int e^{\sqrt{x}} dx &\quad \text{Let } x = t^2 \\ &\Rightarrow dx = 2t dt \\ &= 2 \int t e^t dt \\ &= 2 \left[t \int e^t dt - \int (1 \cdot e^t) dt \right] \\ &= 2(t e^t - e^t) + c \\ &= 2(\sqrt{x} e^{\sqrt{x}} - e^{\sqrt{x}}) + c\end{aligned}$$

322. We have,

$$\begin{aligned}\int \left(\frac{x + \sin x}{1 + \cos x}\right) dx &= \int \frac{x}{1 + \cos x} dx + \int \frac{\sin x}{1 + \cos x} dx \\ &= \int \frac{x}{2 \cos^2 \left(\frac{x}{2}\right)} dx + \int \frac{2 \sin \left(\frac{x}{2}\right) \cos \left(\frac{x}{2}\right)}{2 \cos^2 \left(\frac{x}{2}\right)} dx \\ &= \frac{1}{2} \int x \sec^2 \left(\frac{x}{2}\right) dx + \int \tan \left(\frac{x}{2}\right) dx\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \left[x \int \sec^2\left(\frac{x}{2}\right) dx - \int \left(2 \tan\left(\frac{x}{2}\right)\right) dx \right] \\
&\quad + \int \tan\left(\frac{x}{2}\right) dx \\
&= \frac{1}{2} \left[2x \tan\left(\frac{x}{2}\right) \right] - \int \tan\left(\frac{x}{2}\right) dx + \int \tan\left(\frac{x}{2}\right) dx + c \\
&= x \tan\left(\frac{x}{2}\right) + c
\end{aligned}$$

323. We have,

$$\int \tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right) dx$$

$$\text{Let } x = \cos \theta$$

$$\Rightarrow dx = -\sin \theta d\theta$$

$$\begin{aligned}
&= -\int \tan^{-1}\left(\sqrt{\frac{1-\cos \theta}{1+\cos \theta}}\right) \sin \theta d\theta \\
&= -\int \tan^{-1}\left(\sqrt{\frac{2\sin^2(\theta/2)}{2\cos^2(\theta/2)}}\right) \sin \theta d\theta \\
&= -\int \tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right) \sin \theta d\theta \\
&= -\frac{1}{2} \int \theta \sin \theta d\theta \\
&= -\frac{1}{2} \left[\theta \int \sin \theta d\theta - \int (1 - \cos \theta) d\theta \right] \\
&= -\frac{1}{2} [-\theta \cos \theta + \sin \theta] + c \\
&= \frac{1}{2} [\theta \cos \theta - \sin \theta] + c \\
&= \frac{1}{2} [x \cos^{-1} x - \sqrt{1-x^2}] + c
\end{aligned}$$

324. We have,

$$\begin{aligned}
&\int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}} dx \\
&= \int \frac{\sin^{-1} \sqrt{x} - \left(\frac{\pi}{2} - \sin^{-1} \sqrt{x}\right)}{\frac{\pi}{2}} dx \\
&= \frac{2}{\pi} \int \left(2 \sin^{-1} \sqrt{x} - \frac{\pi}{2}\right) dx \\
&= \frac{4}{\pi} \int \sin^{-1} \sqrt{x} dx - \int dx
\end{aligned}$$

$$\text{Let } x = \sin^2 \theta$$

$$\Rightarrow dx = \sin 2\theta d\theta$$

$$\begin{aligned}
&= \frac{4}{\pi} \int \sin^{-1}(\sin \theta) \sin 2\theta d\theta - x + c \\
&= \frac{4}{\pi} \int \theta \sin 2\theta d\theta - x + c
\end{aligned}$$

$$\begin{aligned}
&= \frac{4}{\pi} \left[\theta \int \sin 2\theta d\theta - \int \left(1 - \frac{\cos 2\theta}{2}\right) d\theta \right] - x \\
&= \frac{4}{\pi} \left[\theta \int \sin 2\theta d\theta + \frac{1}{2} \int \cos 2\theta d\theta \right] - x \\
&= \frac{4}{\pi} \left[\left(\frac{-\theta \cos 2\theta}{2}\right) + \frac{1}{4} \sin 2\theta \right] - x + c \\
&= \frac{4}{\pi} \left[-\frac{1}{2} (\theta (1 - 2\sin^2 \theta)) + \frac{1}{2} \sin \theta \cos \theta \right] - x \\
&= \frac{4}{\pi} \left[-\frac{1}{2} (\sin^{-1} x (1 - 2x^2)) + \frac{1}{2} x \sqrt{1-x^2} \right] - x
\end{aligned}$$

325. We have,

$$\begin{aligned}
&\int \frac{x^2}{(x \sin x + \cos x)^2} dx \\
&= \int \frac{x \cos x}{x \sin x + \cos x} \cdot x \sec x dx \\
&= x \sec x \int \frac{x \cos x}{x \sin x + \cos x} dx \\
&\quad - \int \left(-\frac{(\sec x + x \sec x \tan x)}{x \sin x + \cos x} \right) \\
&= \frac{x \sec x}{x \sin x + \cos x} + \int \sec x dx + c \\
&= \frac{x \sec x}{x \sin x + \cos x} + \log |\sec x + \tan x|
\end{aligned}$$

326. We have,

$$\int \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} dx$$

$$\text{Let } x = \tan \theta$$

$$\Rightarrow dx = 2 \sec^2 \theta d\theta$$

$$\begin{aligned}
&= \int \frac{\tan \theta \tan^{-1}(\tan \theta)}{\sec^3 \theta} \cdot \sec^2 \theta d\theta \\
&= \int \theta \cos \theta d\theta \\
&= \theta \int \cos \theta d\theta - \int (1 \cdot \sin \theta) d\theta \\
&= \theta \sin \theta + \cos \theta + c \\
&= \frac{x}{\sqrt{1+x^2}} \cdot \tan^{-1} x + \frac{1}{\sqrt{1-x^2}} + c
\end{aligned}$$

327. We have,

$$\int \sin^{-1}\left(\sqrt{\frac{x}{a+x}}\right) dx$$

$$\text{Put } x = a \tan^2 \theta$$

$$\Rightarrow dx = 2a \tan \theta \sec^2 \theta d\theta$$

$$= \int \sin^{-1}\left(\sqrt{\frac{a \tan^2 \theta}{a + a \tan^2 \theta}}\right) 2a \tan \theta \sec^2 \theta d\theta$$

$$= 2a \int \sin^{-1}(\sin \theta) \tan \theta \sec^2 \theta d\theta$$

$$= 2a \int \theta \tan \theta \sec^2 \theta d\theta$$

$$\text{Let } \tan \theta = t$$

$$\Rightarrow \sec^2 \theta d\theta = dt$$

$$= 2a \int t \tan^{-1} t dt$$

$$= 2a \left[\tan^{-1} t \int t dt - \int \left(\frac{1}{1+t^2} \right) \frac{t^2}{2} dt \right]$$

$$= 2a \left[\frac{t^2}{2} \tan^{-1} t - \frac{1}{2} \int \left(1 - \frac{1}{1+t^2} \right) dt \right]$$

$$= a [t^2 \tan^{-1} t - t + \tan^{-1} t] + c$$

$$= a [(t^2 + 1) \tan^{-1} t - t] + c$$

$$= a [(\tan^2 \theta + 1) \theta - \tan \theta] + c$$

$$= a \left[\left(\frac{x}{a} + 1 \right) \tan^{-1} \sqrt{\frac{x}{a}} - \sqrt{\frac{x}{a}} \right] + c$$

328. The given integral is

$$\int \log(1+x) dx$$

$$= \log(1+x) \int dx - \int \frac{x}{x+1} dx$$

$$= x \log(1+x) - \int \left(1 - \frac{1}{x+1} \right)$$

$$= x \log(1+x) - (x - \log|x+1|) + c$$

329. The given integral is

$$\int \frac{x - \sin x}{1 - \cos x} dx$$

$$= \int \frac{x dx}{1 - \cos x} - \int \frac{\sin x}{1 - \cos x} dx$$

$$= \int \frac{x dx}{2 \sin^2(x/2)} - \int \frac{2 \sin(x/2) \cos(x/2)}{2 \sin^2(x/2)} dx$$

$$= \frac{1}{2} \int x \operatorname{cosec}^2(x/2) dx - \int \cot\left(\frac{x}{2}\right) dx$$

$$= \frac{1}{2} \left[x \int \operatorname{cosec}^2(x/2) dx + 2 \int \cot(x/2) dx \right]$$

$$- \int \cot(x/2) dx + c$$

$$= \frac{1}{2} (-2x \cot(x/2)) + \int \cot(x/2) dx$$

$$- \int \cot(x/2) dx + c$$

$$= -\cot(x/2) + c$$

330. The given integral is

$$\int \sin \sqrt{x} dx$$

Let

$$x = t^2$$

$$dx = 2t dt$$

$$= 2 \int t \sin t dt$$

$$= 2 \left[t \int \sin t dt + \int \cos t dt + c \right]$$

$$= 2 [-t \cos t + \sin t] + c$$

$$= 2 [-\sqrt{x} \cos \sqrt{x} + \sin \sqrt{x}] + c$$

331. The given integral is

$$\int x \log(1+x) dx$$

$$= \log(1+x) \int x dx - \frac{1}{2} \int \frac{x^2}{(x+1)} dx$$

$$= \frac{x^2}{2} \log(1+x) - \frac{1}{2} \int \frac{(x^2-1)+1}{(x+1)} dx$$

$$= \frac{x^2}{2} \log(1+x) - \frac{1}{2} \int \left(x-1 + \frac{1}{x+1} \right) dx$$

$$= \frac{x^2}{2} \log(1+x) - \frac{1}{2} \left(\frac{x^2}{2} - x + \log|x+1| \right) + c$$

332. The given integral is

$$\int (\sin^{-1} x)^2 dx$$

$$= (\sin^{-1} x)^2 \int dx - \int \frac{2x \sin^{-1} x}{\sqrt{1-x^2}} dx$$

$$= x(\sin^{-1} x)^2 - 2 \int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$$

$$= x(\sin^{-1} x)^2 - 2 \int \theta \sin \theta d\theta, \text{ Let } \sin^{-1} x = \theta$$

$$\frac{x}{\sqrt{1-x^2}} dx = d\theta$$

$$= x(\sin^{-1} x)^2 - 2 \int \theta \sin \theta d\theta$$

$$= x(\sin^{-1} x)^2 - 2 [\theta \sin \theta d\theta + \theta \cos \theta d\theta] + c$$

$$= x(\sin^{-1} x)^2 - 2 [-\theta \cos \theta d\theta + \cos \theta]$$

$$= x(\sin^{-1} x)^2 - 2 [-\sqrt{1-x^2} \sin^{-1} x + x] + c$$

$$= x(\sin^{-1} x)^2 + 2 [\sqrt{1-x^2} \sin^{-1} x - x] + c$$

333. The given integral is

$$\int \frac{x - \sin x}{1 - \cos x} dx$$

$$= \int \frac{x}{1 - \cos x} dx - \int \frac{\sin x}{1 - \cos x} dx$$

$$= \int \frac{x}{2 \sin^2(x/2)} dx - \int \frac{2 \sin(x/2) \cos(x/2)}{2 \sin^2(x/2)} dx$$

$$\begin{aligned}
&= \frac{1}{2} \int x \operatorname{cosec}^2\left(\frac{x}{2}\right) dx - \int \cot\left(\frac{x}{2}\right) dx \\
&= \frac{1}{2} \left[x \int \operatorname{cosec}^2\left(\frac{x}{2}\right) dx - 1 \cdot \left(-2 \cot\left(\frac{x}{2}\right)\right) dx \right] \\
&\quad - \int \cot\left(\frac{x}{2}\right) dx \\
&= \frac{1}{2} \left[x \left(-2 \cot\left(\frac{x}{2}\right)\right) + \int \left(2 \cot\left(\frac{x}{2}\right)\right) dx \right] \\
&\quad - \int \cot\left(\frac{x}{2}\right) dx + c \\
&= -x \cot\left(\frac{x}{2}\right) + c
\end{aligned}$$

334. The given integral is

$$\begin{aligned}
&\int x \sin^3 x dx \\
&= x \int \sin^3 x dx - \int \left(\frac{\cos^3 x}{3} - \cos x\right) dx \\
&= x \left(\frac{\cos^3 x}{3} - \cos x\right) - \int \left(\frac{\cos^3 x}{3} - \cos x\right) dx \\
&= x \left(\frac{\cos^3 x}{3} - \cos x\right) - \frac{1}{3} \left(\sin x - \frac{\sin^3 x}{3}\right) \\
&\quad + \sin x + c
\end{aligned}$$

335. The given integral is

$$\begin{aligned}
&\int x \left(\frac{\sec 2x - 1}{\sec 2x + 1}\right) dx \\
&= \int x \left(\frac{1 - \cos 2x}{1 + \cos 2x}\right) dx \\
&= \int x \tan^2 x dx \\
&= \int x (\sec^2 x - 1) dx \\
&= \int x \sec^2 x dx - \int dx \\
&= x \int \sec^2 x dx - \int \tan x dx - x + c \\
&= x \tan x - \log |\sec x| - x + c
\end{aligned}$$

336. The given integral is

$$\begin{aligned}
&\int \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx \\
&= \int (2 \tan^{-1} x) dx \\
&= 2 \int \tan^{-1} x dx \\
&= 2 \left[\tan^{-1} x \int dx - \int \frac{x}{1+x^2} dx \right] \\
&= 2 \left[x \tan^{-1} x - \frac{1}{2} \log |1+x^2| \right] + c
\end{aligned}$$

337. The given integral is

$$\int \tan^{-1}\left(\frac{2x}{1-x^2}\right) dx$$

$$\begin{aligned}
&= \int (2 \tan^{-1} x) dx \\
&= 2 \int \tan^{-1} x dx \\
&= 2 \left[\tan^{-1} x \int dx - \int \frac{x}{1+x^2} dx \right] \\
&= 2 \left[x \tan^{-1} x - \frac{1}{2} \log |1+x^2| \right] + c
\end{aligned}$$

338. The given integral is

$$\begin{aligned}
&\int \frac{x^3 \sin^{-1}(x^2)}{\sqrt{1-x^4}} dx \\
&= \frac{1}{2} \int \frac{2x \sin^{-1}(x^2) \times x^2}{\sqrt{1-x^4}} dx \\
&\quad \text{Let } \sin^{-1}(x^2) = t \\
&\quad \Rightarrow \frac{2x}{\sqrt{1-x^4}} dx = dt \\
&= \frac{1}{2} \int t \sin t dt \\
&= \int [t \int \sin t dt + \int \cos t dt] + c \\
&= \frac{1}{2} [-t \cos t + \sin t] + c \\
&= \frac{1}{2} [-\sin^{-1}(-x^2) \sqrt{1-x^4+x^2}] + c
\end{aligned}$$

339. The given integral is

$$\begin{aligned}
&\int \frac{\sec x (2 + \sec x)}{(1 + 2 \sec x)^2} dx \\
&= \int \frac{2 \cos x + 1}{(\cos x + 2)^2} dx \\
&= \int \frac{\cos x (\cos x + 2) + \sin x}{(\cos x + 2)^2} dx \\
&= \int \frac{\cos x}{(\cos x + 2)} dx + \int \frac{\sin x}{(\cos x + 2)^2} dx \\
&= \frac{1}{(\cos x + 2)} \int \cos x dx - \int \frac{\sin x}{(\cos x + 2)^2} dx \\
&\quad + \int \frac{\sin x}{(\cos x + 2)^2} dx + c \\
&= \frac{\sin x}{\cos x + 2} + c
\end{aligned}$$

340. The given integral is

$$\begin{aligned}
&\int \frac{1-x}{e^x+x} dx \\
&= \int \frac{(1+e^x) - (x+e^x)}{(e^x+x)} dx
\end{aligned}$$

$$= \int \left(\frac{(1+e^x)}{(e^x+x)} - 1 \right) dx$$

$$= \log|(e^x+x)| - x + c$$

341. We have,

$$\int e^x(\sin x + \cos x) dx$$

We know that,

$$\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$$

Here, $f(x) = \sin x \Rightarrow f'(x) = \cos x$

Thus,

$$\int e^x(\sin x + \cos x) dx = e^x \sin x + c$$

342. We have,

$$\int \frac{x e^x}{(x+1)^2} dx$$

$$= \int \frac{e^x(x+1-1)}{(x+1)^2} dx$$

$$= \int e^x \left(\frac{x+1}{(x+1)^2} - \frac{1}{x+1} \right) dx$$

$$= - \int e^x \left(\frac{1}{x+1} - \frac{1}{(x+1)^2} \right) dx$$

$$= - \frac{e^x}{x+1} + c$$

343. We have,

$$\int e^x \left(\frac{2 + \sin 2x}{1 + \cos 2x} \right) dx$$

$$= \int e^x \left(\frac{2}{1 + \cos 2x} + \frac{\sin 2x}{1 + \cos 2x} \right) dx$$

$$= \int e^x \left(\frac{2}{2 \cos^2 x} + \frac{2 \sin x \cos x}{2 \cos^2 x} \right) dx$$

$$= \int e^x (\sec^2 x + \tan x) dx$$

$$= \int e^x (\tan x + \sec^2 x) dx$$

$$= e^x \tan x + c$$

344. We have,

$$\int e^x \left(\frac{1+x+x^3}{(1+x^2)^{3/2}} \right) dx$$

$$= \int e^x \left(\frac{1+x(1+x^2)}{(1+x^2)^{3/2}} \right) dx$$

$$= \int e^x \left(\frac{1}{(1+x^2)^{3/2}} + \frac{x}{(1+x^2)^{1/2}} \right) dx$$

$$= \int e^x \left(\frac{x}{(1+x^2)^{1/2}} + \frac{1}{(1+x^2)^{3/2}} \right) dx$$

$$= \frac{e^x x}{(1+x^2)^{1/2}} + c$$

345. We have,

$$\int e^x \left(\frac{x^2+1}{(x+1)^2} \right) dx$$

$$= \int e^x \left(\frac{x^2-1+2}{(x+1)^2} \right) dx$$

$$= \int e^x \left(\frac{(x+1)(x-1)+2}{(x+1)^2} \right) dx$$

$$= \int e^x \left(\frac{(x-1)}{(x+1)} + \frac{2}{(x+1)^2} \right) dx$$

$$= e^x \left(\frac{(x-1)}{(x+1)} \right) + c$$

346. The given integral is

$$\int e^x \left\{ \frac{1}{x} - \frac{1}{x^2} \right\} dx = \int e^x \left(\frac{1}{x} + \left(-\frac{1}{x^2} \right) \right) dx$$

$$= \frac{e^x}{x} + c$$

347. The given integral is

$$\int e^x \left(\frac{x}{(x+1)^2} \right) dx$$

$$= \int e^x \left(\frac{(x+1)-1}{(x+1)^2} \right) dx$$

$$= \int e^x \left(\frac{1}{(x+1)} - \frac{1}{(x+1)^2} \right) dx$$

$$= \int \frac{e^x}{(x+1)} + c$$

348. The given integral is

$$\int e^x \left\{ \frac{1 - \sin x}{1 - \cos x} \right\} dx$$

$$= \int e^x \left(\frac{1}{1 - \cos x} - \frac{\sin x}{1 - \cos x} \right) dx$$

$$= \int e^x \left(\frac{1}{2 \sin^2(x/2)} - \frac{2 \sin(x/2) \cos(x/2)}{2 \sin^2(x/2)} \right) dx$$

$$= \int e^x \left(\frac{1}{2} \operatorname{cosec}^2(x/2) - \cot(x/2) \right) dx$$

$$= \int e^x \left(\cot(x/2) + \left(-\frac{1}{2} \operatorname{cosec}^2(x/2) \right) \right) dx$$

$$= -e^x \cot\left(\frac{x}{2}\right) + c$$

349. The given integral is

$$\begin{aligned}
& \int e^x \left\{ \frac{2 + \sin 2x}{1 + \cos 2x} \right\} dx \\
&= \int e^x \left(\frac{2}{1 + \cos 2x} + \frac{\sin 2x}{1 + \cos 2x} \right) dx \\
&= \int e^x \left(\frac{2}{2 \cos^2 x} + \frac{2 \sin x \cos x}{2 \cos^2 x} \right) dx \\
&= \int e^x (\sec^2 x + \tan x) dx \\
&= \int e^x (\tan x + \sec^2 x) dx \\
&= e^x \tan x + c
\end{aligned}$$

350. The given integral is

$$\begin{aligned}
\int e^x \left(\frac{x^2 + 1}{(x + 1)^2} \right) dx &= \int e^x \left(\frac{(x^2 - 1) + 2}{(x + 1)^2} \right) dx \\
&= \int e^x \left(\frac{(x - 1)}{(x + 1)} + \frac{2}{(x + 1)^2} \right) dx \\
&= e^x \left(\frac{(x - 1)}{(x + 1)} \right) + c
\end{aligned}$$

351. The given integral is

$$\begin{aligned}
& \int \frac{\log x}{(1 + \log x)^2} dx \\
&= \int \frac{t}{(t + 1)^2} dt \quad \text{Let } \log x = t \\
&\Rightarrow dt = \frac{dx}{e^{\log x}} \\
&= \int \frac{[(t + 1) - 1]e^t}{(t + 1)^2} dt \\
&= \int e^t \left[\frac{1}{t + 1} + \left(-\frac{1}{(t + 1)^2} \right) \right] dt \\
&= \frac{e^t}{t + 1} + c \\
&= \frac{x}{\log x + 1} + c
\end{aligned}$$

352. The given integral is

$$\int e^x \left(\log x + \frac{1}{x} \right) dx = e^x \log x + c$$

353. The given integral is

$$\begin{aligned}
& \int e^x \left(\frac{x - 1}{(x + 1)^3} \right) dx \\
&= \int e^x \left(\frac{(x + 1) - 2}{(x + 1)^3} \right) dx \\
&= \int \left[e^x \frac{1}{(x + 1)^2} + \left(-\frac{2}{(x + 1)^3} \right) \right] dx
\end{aligned}$$

$$= \frac{e^x}{(x + 1)^2} + c$$

354. The given integral is

$$\begin{aligned}
& \int \left\{ \frac{1}{\log x} - \frac{1}{(\log x)^2} \right\} dx \\
&= \int e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt, \quad \text{Let } t = \log x \\
&\Rightarrow dt = \frac{dx}{e^{\log x}} \\
&= \frac{e^t}{t} + c \\
&= \frac{x}{\log x} + c
\end{aligned}$$

355. The given integral is

$$\begin{aligned}
& \int e^x \left(\frac{1 - x}{1 + x^2} \right)^2 dx \\
&= \int e^x \left(\frac{(1 - x)^2}{(1 + x^2)^2} \right) dx \\
&= \int e^x \left(\frac{(1 + x^2 - 2x)}{(1 + x^2)^2} \right) dx \\
&= \int e^x \left[\frac{1}{(1 + x^2)} + \left(\frac{-2x}{(1 + x^2)^2} \right) \right] dx \\
&= \frac{e^x}{(1 + x^2)} + c
\end{aligned}$$

356. The given integral is

$$\begin{aligned}
& \int e^x \left(\frac{1 - x}{1 + x} \right)^2 dx \\
&= \int e^x \left(\frac{(1 - x)^2}{(1 + x)^2} \right) dx \\
&= \int e^x \left(\frac{(1 + x^2) - 2x}{(1 + x)^2} \right) dx \\
&= \int e^x \left(\frac{x^2 - 1 - 2(x - 1)}{(x + 1)^2} \right) dx \\
&= \int e^x \left(\frac{x^2 - 1}{(x + 1)^2} - \frac{2(x - 1)}{(x + 1)^2} \right) dx \\
&= \int e^x \left(\frac{x^2 - 1}{(x + 1)^2} - \frac{2(x - 1)}{(x + 1)^2} \right) dx \\
&= \int e^x \left(\left(\frac{x - 1}{x + 1} \right) - \frac{2(x - 1)}{(x + 1)^2} \right) dx \\
&= \int e^x \left(\left(\frac{x - 1}{x + 1} \right) - \frac{2(x + 1 - 2)}{(x + 1)^2} \right) dx \\
&= \int e^x \left(\left(\frac{x - 3}{x + 1} \right) - \frac{4}{(x + 1)^2} \right) dx
\end{aligned}$$

$$= \int e^x \left(\frac{x-3}{x+1} \right) dx + c$$

357. The given integral is

$$\begin{aligned} & \int e^x \left(\frac{x^2+1}{(1+x)^2} \right) dx \\ &= \int e^x \left(\frac{x^2-1+2}{(x+1)^2} \right) dx \\ &= \int e^x \left(\frac{x^2-1}{(x+1)^2} + \frac{2}{(x+1)^2} \right) dx \\ &= \int e^x \left(\left(\frac{x-1}{x+1} \right) + \frac{2}{(1+x)^2} \right) dx \\ &= e^x \left(\frac{x-1}{x+1} \right) + c \end{aligned}$$

358. The given integral is

$$\begin{aligned} & \int e^{\cos^{-1}x} \left(\frac{(x+1) + \sqrt{1-x^2}}{(x+1)^2 \sqrt{1-x^2}} \right) dx \\ & \quad \text{Let } \cos^{-1}x = t \\ & \quad \Rightarrow \frac{dx}{\sqrt{1-x^2}} = -dt \\ &= -\int e^t \left(\frac{1 + \cos t + \sin t}{(1 + \cos t)^2} \right) dt \\ &= -\int e^t \left(\frac{1}{(1 + \cos t)} + \frac{\sin t}{(1 + \cos t)^2} \right) dt \\ &= -\frac{e^t}{1 + \cos t} + c \\ &= -\frac{e^{\cos^{-1}x}}{1+x} + c \end{aligned}$$

359. We have,

$$\begin{aligned} & \int e^x (2\sec^2 x - 1) \tan x dx \\ &= \int e^x (2\sec^2 x \tan x - \tan x) dx \\ &= \int e^x [(\sec^2 x - \tan x) + (2\sec^2 x \tan x - \sec^2 x)] dx \\ &= \int e^x (\sec^2 x - \tan x) + c \end{aligned}$$

360. We have,

$$\begin{aligned} & \int e^x (\log(\sec x + \tan x) + \sec x) dx \\ &= e^x \log(\sec x + \tan x) + c \end{aligned}$$

361. We have,

$$\begin{aligned} & \int e^{\sin x} \left(\frac{x \cos^3 x - \sin x}{\cos^2 x} \right) dx \\ &= \int e^{\sin x} (x \cos x - \sec x \tan x) dx \end{aligned}$$

$$= \int e^{\sin x} \left(\frac{x - \tan x}{\cos^2 x} \right) \cos x dx$$

Let $\sin x = t$

$$\Rightarrow \cos x dx = dt$$

$$\begin{aligned} &= \int e^t \left(\sin^{-1} t - \frac{t}{(1-t^2)^{3/2}} \right) dt \\ &= \int e^t \left(\left(\sin^{-1} t + \frac{1}{\sqrt{1-t^2}} \right) - \left(\frac{1}{\sqrt{1-t^2}} + \frac{t}{(1-t^2)^{3/2}} \right) \right) dt \\ &= e^t \left(\sin^{-1} t + \frac{1}{\sqrt{1-t^2}} \right) + c \\ &= e^{\sin x} (x + \sec x) + c \end{aligned}$$

$$362. \int e^x \left(\log x + \frac{1}{x^2} \right) dx$$

$$\begin{aligned} &= \int e^x \left(\left(\log x - \frac{1}{x} \right) + \left(\frac{1}{x} + \frac{1}{x^2} \right) \right) dx \\ &= e^x \left(\log x - \frac{1}{x} \right) + c \end{aligned}$$

$$363. \int \left(\log(\log x) + \frac{1}{(\log x)^2} \right) dx$$

$$\begin{aligned} &= \int e^t \left(\log t + \frac{1}{t^2} \right) dt, \quad \text{Let } t = \log x \\ &\Rightarrow dt = \frac{dx}{e^{\log x}} \end{aligned}$$

$$\begin{aligned} &= \int e^t \left(\left(\log t - \frac{1}{t} \right) + \left(\frac{1}{t} + \frac{1}{t^2} \right) \right) dt \\ &= e^t \left(\log t - \frac{1}{t} \right) + c \\ &= x \left(\log(\log x) - \frac{1}{\log x} \right) + c \end{aligned}$$

$$364. \int e^x \left(\frac{x^4 + 2}{(1+x^2)^{5/2}} \right) dx$$

$$\begin{aligned} &= \int e^x \left(\frac{(1+x^2)^2 + 1 - 2x^2}{(1+x^2)^{5/2}} \right) dx \\ &= \int e^x \left(\left(\frac{1}{\sqrt{1+x^2}} - \frac{x}{(1-x^2)^{3/2}} \right) + \left(\frac{x}{(1+x^2)^{3/2}} + \frac{1-2x^2}{(1+x^2)^{5/2}} \right) \right) dx \\ &= e^x \left(\frac{1}{\sqrt{1+x^2}} + \frac{x}{(1-x^2)^{3/2}} \right) + c \end{aligned}$$

$$365. \int \frac{e^{\sin x} (x \cos^3 x - \sin x)}{\cos^2 x} dx$$

$$= \int e^{\sin x} (x \cos x - \sec x \tan x) dx$$

$$\begin{aligned}\text{Let } \sin x &= t \\ \Rightarrow \cos x dx &= dt\end{aligned}$$

$$\begin{aligned}\Rightarrow dx &= \frac{dt}{\cos x} = \frac{dt}{\sqrt{1-t^2}} \\ &= \int e^t \left(\sin^{-1} t - \frac{t}{(1-t^2)^{3/2}} \right) dt \\ &= \int e^t \left(\sin^{-1} t + \frac{1}{\sqrt{1-t^2}} - \frac{1}{\sqrt{1-t^2}} - \frac{t}{(1-t^2)^{3/2}} \right) dt \\ &= e^t \left(\sin^{-1} t - \frac{1}{\sqrt{1-t^2}} \right) + c \\ &= e^{\sin x} (x - \sec x) + c\end{aligned}$$

366. We have,

$$\begin{aligned}&\int e^x \left(\frac{x^3 - x - 2}{(x^2 + 1)^2} \right) dx \\ &= \int e^x \left(\frac{(x^3 + x) + (x^2 + 1) + (1 - 2x^2) - 2x}{(x^2 + 1)^2} \right) dx \\ &= \int e^x \left(\frac{x(x^2 + 1) + (x^2 + 1) + (1 - 2x^2) - 2x}{(x^2 + 1)^2} \right) dx \\ &= \int e^x \left(\left\{ \frac{x}{(x^2 + 1)} + \frac{1}{(x^2 + 1)} \right\} \right. \\ &\quad \left. + \left\{ \frac{1 - x^2}{(x^2 + 1)^2} + \frac{-2x}{(x^2 + 1)^2} \right\} \right) dx \\ &= \int e^x \left(\frac{x}{(x^2 + 1)} + \frac{1}{(x^2 + 1)} \right) + c \\ &= e^x \left(\frac{x + 1}{(x^2 + 1)} \right) + c\end{aligned}$$

367. We have,

$$\int e^{3x} (3 \sin x + \cos x) dx = e^{3x} \sin x + c$$

368. We have,

$$\begin{aligned}&e^{2x} (\sec^2 x + 2 \tan x) dx \\ &= \int e^{2x} (2 \tan x + \sec^2 x) dx \\ &= e^{2x} 2 \tan x + c\end{aligned}$$

369. We have,

$$\begin{aligned}&\int e^{2x} \left(\frac{2 \sin 4x - 4}{1 - \cos 4x} \right) dx \\ &= \int e^x \left(\frac{2 \sin 4x}{1 - \cos 4x} - \frac{2}{1 - \cos 4x} \right) dx\end{aligned}$$

$$\begin{aligned}&= \int e^{2x} \left(\frac{4 \sin 2x \cos 2x}{2 \sin^2 2x} - \frac{4}{2 \sin^2 2x} \right) dx \\ &= \int e^{2x} (2 \cot 2x + (-2 \operatorname{cosec}^2 2x)) dx \\ &= e^{2x} \cot 2x + c\end{aligned}$$

370. We have,

$$\begin{aligned}&\int e^{2x} (-\sin x + 2 \cos x) dx \\ &= \int e^{2x} [2 \cos x + (-\sin x)] dx \\ &= e^{2x} 2 \cos x + c\end{aligned}$$

371. We have,

$$\begin{aligned}&\int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx \\ &= \int e^{2x} \left(\frac{1 + \sin 2x}{1 + \cos 2x} \right) dx \\ &= \int e^{2x} \left(\frac{1}{1 + \cos 2x} + \frac{\sin 2x}{1 + \cos 2x} \right) dx \\ &= \int e^{2x} \left(\frac{1}{1 + 2 \cos^2 x} + \frac{2 \sin x \cos x}{2 \cos^2 x} \right) dx \\ &= \int e^{2x} \left(\frac{1}{2} \sec^2 x + \tan x \right) dx \\ &= \frac{1}{2} \int e^{2x} (\sec^2 x + 2 \tan x) dx \\ &= \frac{1}{2} \int e^{2x} (2 \tan x + \sec^2 x) dx \\ &= \frac{e^{2x} 2 \tan x}{2} + c\end{aligned}$$

372. We have,

$$\begin{aligned}&\int e^{-\frac{x}{2}} \left(\frac{\sqrt{1 - \sin x}}{1 + \cos x} \right) dx \\ &= \int e^{-\frac{x}{2}} \left(\frac{\cos(x/2) - \sin(x/2)}{2 \cos^2(x/2)} \right) dx \\ &= \int e^{-\frac{x}{2}} \left(\frac{1}{2} \sec \left(\frac{x}{2} \right) - \frac{1}{2} \sec \left(\frac{x}{2} \right) \tan \left(\frac{x}{2} \right) \right) dx \\ &= \int e^{-\frac{x}{2}} \left(-\frac{1}{2} \sec \left(\frac{x}{2} \right) + \frac{1}{2} \sec \left(\frac{x}{2} \right) \tan \left(\frac{x}{2} \right) \right) dx \\ &= \int e^{-\frac{x}{2}} \sec \left(\frac{x}{2} \right) + c\end{aligned}$$

373. We have,

$$\begin{aligned}&\int e^{2x} [2 \times \log(\sec x + \tan x) + \sec x] dx \\ &= e^{2x} (\log(\sec x + \tan x)) + c\end{aligned}$$

$$= \left(x\sqrt{x^2 - \frac{1}{4}} - \frac{1}{4} \log \left| x + \sqrt{x^2 - \frac{1}{4}} \right| \right) + c$$

387. We have,

$$\begin{aligned} & \int \sqrt{x^2 + 2x + 3} \, dx \\ &= \int \sqrt{(x^2 + 2x + 1) + 2} \, dx \\ &= \int \sqrt{(x+1)^2 + (\sqrt{2})^2} \, dx \\ &= \log \left| (x+1) + \sqrt{x^2 + 2x + 3} \right| + c \end{aligned}$$

388. We have,

$$\begin{aligned} & \int \sqrt{3 - 4x - x^2} \, dx \\ &= \int \sqrt{-(x^2 + 4x - 3)} \, dx \\ &= \int \sqrt{-(x^2 + 2)^2 - 7} \, dx \\ &= \int \sqrt{7 - (x+2)^2} \, dx \\ &= \frac{(x+2)}{2} \sqrt{3 - 4x - x^2} + \frac{7}{2} \sin^{-1} \left(\frac{x+2}{\sqrt{7}} \right) + c \end{aligned}$$

387. We have,

$$\begin{aligned} & \int \sqrt{x^2 + 2x + 3} \, dx \\ &= \int \sqrt{(x+1)^2 + (\sqrt{2})^2} \, dx \\ &= \left(\frac{x+1}{2} \right) \sqrt{x^2 + 2x + 3} \\ &\quad + \frac{1}{2} \log \left| (x+1) + \sqrt{x^2 + 2x + 3} \right| + c \end{aligned}$$

388. We have,

$$\begin{aligned} & \int \sqrt{3 - 4x - x^2} \, dx \\ &= \int \sqrt{-(x^2 + 4x - 3)} \, dx \\ &= \int \sqrt{-(x+2)^2 - 7} \, dx \\ &= \int \sqrt{7 - (x+2)^2} \, dx \\ &= \frac{(x+2)}{2} \sqrt{3 - 4x - x^2} + \frac{7}{2} \sin^{-1} \left(\frac{x+2}{\sqrt{7}} \right) + c \end{aligned}$$

389. We have,

$$\begin{aligned} & \int \sqrt{2ax - x^2} \, dx \\ &= \int \sqrt{-(x^2 - 2ax)} \, dx \end{aligned}$$

$$\begin{aligned} &= \int \sqrt{-(x-a)^2 - a^2} \, dx \\ &= \int \sqrt{a^2 - (x-a)^2} \, dx \\ &= \frac{(x-a)}{2} \sqrt{2ax - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x-a}{a} \right) + c \end{aligned}$$

390. We have,

$$\begin{aligned} & \int \sqrt{2ax + x^2} \, dx \\ &= \int \sqrt{(x+a)^2 - a^2} \, dx \\ &= \frac{(x+a)}{2} \sqrt{x^2 + 2ax} \\ &\quad + \frac{a^2}{2} \log \left| (x+a) + \sqrt{x^2 + 2ax} \right| + c \end{aligned}$$

391. We have,

$$\begin{aligned} & \int \sqrt{x - 4x^2} \, dx \\ &= \int \sqrt{\left(\frac{1}{4}\right)^2 - 4\left(x - \frac{1}{8}\right)^2} \, dx \\ &= \int \sqrt{\left(\frac{1}{4}\right)^2 - 4\left(2x - \frac{1}{4}\right)^2} \, dx \\ &= \frac{1}{2} \int \sqrt{\left(\frac{1}{4}\right)^2 - t^2} \, dt, \quad \text{Let } \left(2x - \frac{1}{4}\right) = t \\ &\quad \Rightarrow 2dx = dt \\ &= \frac{1}{2} \int \sqrt{\left(\frac{1}{4}\right)^2 - t^2} \, dt \\ &= \frac{t}{4} \sqrt{\frac{1}{16} - t^2} + \frac{1}{32} \sin^{-1}(4t) + c \\ &= \frac{\left(2x - \frac{1}{4}\right)}{4} \sqrt{\frac{1}{16} - \left(2x - \frac{1}{4}\right)^2} \\ &\quad + \frac{1}{32} \sin^{-1} \left[4\left(2x - \frac{1}{4}\right) \right] + c \end{aligned}$$

392. We have,

$$\begin{aligned} & \int (2x+1) \sqrt{x^2 + 3x + 4} \, dx \\ &= \int ((2x+3) - 2) \sqrt{x^2 + 3x + 4} \, dx \\ &= \int ((2x+3)) \sqrt{x^2 + 3x + 4} \, dx \\ &\quad - 2 \int \sqrt{x^2 + 3x + 4} \, dx \\ &\quad \text{Let } x^2 + 3x + 4 = t^2 \\ &\quad \Rightarrow (2x+3)dx = 2t \, dt \\ &= \int (2t)t \cdot dt - 2 \int \sqrt{\left(x + \frac{3}{2}\right)^2 + \left(4 - \frac{9}{4}\right)} \, dx \\ &\quad 62 \end{aligned}$$

374. We have,

$$\begin{aligned}\int e^x \sin 3x dx &= \frac{e^x}{1^2 + 3^2} (1 \cdot \sin 3x - 3 \cos 3x) + c \\ &= \frac{e^x}{10} (\sin 3x - 3 \cos 3x) + c\end{aligned}$$

375. We have,

$$\begin{aligned}\int e^{4x} \cos 3x dx &= \frac{e^{4x}}{4^2 + 3^2} (4 \cos 3x + 3 \sin 3x) + c \\ &= \frac{e^{4x}}{25} (4 \cos 3x + 3 \sin 3x) + c\end{aligned}$$

376. We have,

$$\begin{aligned}\int e^{2x} \sin 3x dx &= \frac{e^{2x}}{(4+9)} (2 \sin 3x - 3 \cos 3x) + c \\ &= \frac{e^{2x}}{13} (2 \sin 3x - 3 \cos 3x) + c\end{aligned}$$

377. We have,

$$\begin{aligned}\int e^{-x} \cos x dx &= \frac{e^{-x}}{(1+1)} (-\cos x + \sin x) + c \\ &= \frac{e^{-x}}{2} (-\cos x + \sin x) + c\end{aligned}$$

378. We have,

$$\begin{aligned}\int e^{-x} \cos(3x+4) dx &= \frac{e^{-x}}{3(4+1)} [2 \cos(3x+4) + \sin(3x+4)] + c\end{aligned}$$

379. We have,

$$\begin{aligned}\int e^x \cos^2 x dx &= \frac{1}{2} \int e^x (2 \cos^2 x) dx \\ &= \frac{1}{2} \int e^x (1 + \cos 2x) dx \\ &= \frac{1}{2} \int (e^x + e^x \cos 2x) dx \\ &= \frac{1}{2} \left[e^x + \frac{e^x}{5} (\cos 2x + 2 \sin 2x) \right] + c\end{aligned}$$

380. We have,

$$\int \frac{1}{x^3} \sin(\log x) dx$$

$$\begin{aligned}&= \int e^{-2t} \sin t dt, & \text{Let } t = \log x \\ &\Rightarrow dt = \frac{dx}{e^{\log x}}\end{aligned}$$

$$\begin{aligned}&= \int e^{-2t} \sin t dt \\ &= \frac{e^{-2t}}{5} (-2 \sin t - \cos t) + c \\ &= -\frac{e^{-2t}}{5} (-2 \sin t + \cos t) + c \\ &= \frac{e^{-2 \log x}}{5} [2 \sin(\log x) + \cos(\log x)] + c \\ &= -\frac{1}{5x^2} [2 \sin(\log x) + \cos(\log x)] + c\end{aligned}$$

381. We have,

$$\begin{aligned}\int \sqrt{4-x^2} dx &= \frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1}\left(\frac{x}{2}\right) + c \\ &= \frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1}\left(\frac{x}{2}\right) + c\end{aligned}$$

382. We have,

$$\begin{aligned}\int \sqrt{1-9x^2} dx &= 3 \int \sqrt{\left(\frac{1}{3}\right)^2 - x^2} dx \\ &= 3 \left(\frac{x}{2} \sqrt{\frac{1}{9} - x^2} - \frac{1}{2} \sin^{-1}\left(\frac{x}{1/3}\right) \right) + c \\ &= 3 \left(\frac{x}{2} \sqrt{\frac{1}{9} - x^2} - \frac{1}{18} \sin^{-1}\left(\frac{x}{1/3}\right) \right) + c\end{aligned}$$

383. We have,

$$\int \sqrt{x^2+1} dx = \frac{x}{2} \sqrt{x^2+1} + \frac{1}{2} \log|x + \sqrt{x^2+1}| + c$$

384. We have,

$$\begin{aligned}\int \sqrt{3x^2+1} dx &= \sqrt{3} \int \sqrt{x^2 + \frac{1}{3}} dx \\ &= \sqrt{3} \left(\frac{x}{2} \sqrt{x^2 + \frac{1}{3}} + \frac{1/3}{2} \log \left| x + \sqrt{x^2 + \frac{1}{3}} \right| \right) + c\end{aligned}$$

385. We have,

$$\begin{aligned}\int \sqrt{x^2-9} dx &= \frac{x}{2} \sqrt{x^2-9} - \frac{9}{2} \log|x + \sqrt{x^2-9}| + c\end{aligned}$$

386. We have,

$$\begin{aligned}\int \sqrt{4x^2-1} dx &= 2 \int \sqrt{x^2 - \left(\frac{1}{x}\right)^2} dx \\ &= 2 \left(\frac{x}{2} \sqrt{x^2 - \frac{1}{4}} - \frac{1/4}{2} \log \left| x + \sqrt{x^2 - \frac{1}{4}} \right| \right) + c\end{aligned}$$

$$\begin{aligned}
&= 2 \int t^2 dt - 2 \int \sqrt{\left(x + \frac{3}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2} dx \\
&= 2 \left(\frac{t^3}{3}\right) - 2 \left[\frac{1}{2} \left(x + \frac{3}{2}\right) \sqrt{x^2 + 3x + 4} \right. \\
&\quad \left. + \frac{7}{4} \log \left| \left(x + \frac{3}{2}\right) + \sqrt{x^2 + 3x + 4} \right| \right] + c \\
&= \frac{2}{3} (x^2 + 3x + 4)^{3/2}
\end{aligned}$$

$$\begin{aligned}
&-2 \left[\frac{1}{2} \left(x + \frac{3}{2}\right) \sqrt{x^2 + 3x + 4} \right. \\
&\quad \left. + \frac{7}{4} \log \left| \left(x + \frac{3}{2}\right) + \sqrt{x^2 + 3x + 4} \right| \right] + c
\end{aligned}$$

392. We have,

$$\begin{aligned}
&\int (2x + 1) \sqrt{x^2 + 3x + 4} dx \\
&= \int (2x + 3 - 2) \sqrt{x^2 + 3x + 4} dx \\
&= \int (2x + 3) \sqrt{x^2 + 3x + 4} dx - 2 \int \sqrt{x^2 + 3x + 4} dx \\
&\quad \text{Let } x^2 + 3x + 4 = t^2 \\
&\quad \Rightarrow (2x + 3) dx = 2t dt
\end{aligned}$$

$$\begin{aligned}
&= 2 \int t^2 dt - 2 \int \sqrt{\left(x + \frac{3}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2} dx \\
&= \frac{2(x^2 + 3x + 4)^{3/2}}{3} - \left(x + \frac{3}{2}\right) \sqrt{x^2 + 3x + 4} \\
&\quad - \left[\frac{7}{4} \log \left| \left(x + \frac{3}{2}\right) + \sqrt{x^2 + 3x + 4} \right| \right] + c
\end{aligned}$$

393. We have,

$$\begin{aligned}
&\int (x - 5) \sqrt{x^2 + x} dx \\
&= \frac{1}{2} \int (2x - 10) \sqrt{x^2 + x} dx \\
&= \frac{1}{2} \int ((2x + 1) - 11) \sqrt{x^2 + x} dx \\
&= \frac{1}{2} \int (2x + 1) \sqrt{x^2 + x} dx \\
&\quad - \frac{11}{2} \int \sqrt{\left(x + \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} dx
\end{aligned}$$

and then you do.

394. We have,

$$\begin{aligned}
&\int (3x - 2) \sqrt{x^2 + x + 1} dx \\
&= 3 \int \left(x - \frac{2}{3}\right) \sqrt{x^2 + x + 1} dx \\
&= \frac{3}{2} \int \left(2x - \frac{4}{3}\right) \sqrt{x^2 + x + 1} dx
\end{aligned}$$

$$\begin{aligned}
&= \frac{3}{2} \int \left(\left(2x + \frac{4}{3}\right) - \frac{7}{3} \right) \sqrt{x^2 + x + 1} dx \\
&= \frac{3}{2} \int \left(\left(2x + \frac{4}{3}\right) - \frac{7}{3} \right) \sqrt{x^2 + x + 1} dx \\
&\quad + \frac{7}{2} \int \sqrt{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} dx
\end{aligned}$$

and then you do.

395. We have,

$$\begin{aligned}
&\int (4x + 1) \sqrt{x^2 - x - 2} dx \\
&= 2 \int \left(2x + \frac{1}{2}\right) \sqrt{x^2 - x - 2} dx \\
&= 2 \int \left((2x - 1) + \frac{3}{2}\right) \sqrt{x^2 - x - 2} dx \\
&= 2 \int \left((2x - 1) + \frac{3}{2}\right) \sqrt{x^2 - x - 2} dx \\
&\quad + 3 \int \sqrt{(x - 1)^2 - \left(\frac{3}{2}\right)^2} dx
\end{aligned}$$

and then you do it.

396. We have,

$$\begin{aligned}
&\int x \sqrt{1 + x - x^2} dx \\
&= \frac{1}{2} \int (2x) \sqrt{1 + x - x^2} dx \\
&= -\frac{1}{2} \int (-2x) \sqrt{1 + x - x^2} dx \\
&= -\frac{1}{2} \int ((-2x) - 1) \sqrt{1 + x - x^2} dx \\
&= -\frac{1}{2} \int (1 - 2x) \sqrt{1 + x - x^2} dx \\
&\quad + \frac{1}{2} \int \sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2} dx
\end{aligned}$$

and then you do it.

Partial Fractions

397. We have,

$$\begin{aligned}
&\int \frac{(2x + 1) dx}{(x + 2)(x + 3)} \\
&\Rightarrow \frac{(2x + 1)}{(x + 2)(x + 3)} = \frac{A}{(x + 2)} + \frac{B}{(x + 3)} \\
&\Rightarrow \frac{(2x + 1)}{(x + 2)(x + 3)} = \frac{A(x + 3) + B(x + 2)}{(x + 2)(x + 3)} \\
&\Rightarrow 2x + 1 = A(x + 3) + B(x + 2) \\
&\text{When } x = -2, A = -4 + 1 = -3
\end{aligned}$$

and $x = -3, B = -6 + 1 = -5$

$$\begin{aligned}\text{Thus, } \int \frac{(2x+1)dx}{(x+2)(x+3)} \\ &= -3 \int \frac{dx}{x+2} - 5 \int \frac{dx}{x+3} \\ &= -3 \log|x+2| - 5 \log|x+3| + c\end{aligned}$$

398. We have,

$$\begin{aligned}\int \frac{dx}{(x-1)(x-2)} \\ &= \int \left(\frac{1}{x-2} - \frac{1}{x-1} \right) dx \\ &= \log|x-2| - \log|x-1| + c \\ &= \log \left| \frac{x-2}{x-1} \right| + c\end{aligned}$$

399. We have,

$$\begin{aligned}\int \frac{dx}{(x+1)(x+2)(x+3)} \\ &= \frac{1}{2} \int \frac{((x+3) - (x+1))}{(x+1)(x+2)(x+3)} \\ &= \frac{1}{2} \int \frac{dx}{(x+1)(x+2)} - \frac{1}{2} \int \frac{dx}{(x+2)(x+3)} \\ &= \frac{1}{2} \int \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx - \frac{1}{2} \int \left(\frac{1}{x+2} - \frac{1}{x+3} \right) dx \\ &= \frac{1}{2} \log \left| \frac{x+1}{x+2} \right| - \frac{1}{2} \log \left| \frac{x+2}{x+3} \right| + c \\ &= \frac{1}{2} \log \left| \left(\frac{x+1}{x+2} \right) \left(\frac{x+3}{x+2} \right) \right|\end{aligned}$$

400. We have,

$$\begin{aligned}\int \frac{dx}{(x+1)(x+2)(x+3)(x+4)} \\ &= \frac{1}{3} \int \frac{((x+4) - (x+1))dx}{(x+1)(x+2)(x+3)(x+4)} \\ &= \frac{1}{3} \int \frac{dx}{(x+1)(x+2)(x+3)} \\ &\quad - \frac{1}{3} \int \frac{dx}{(x+1)(x+2)(x+3)} \\ &= -\frac{1}{6} \int \frac{((x+4) - (x+2))dx}{(x+2)(x+3)(x+4)} \\ &\quad + \frac{x}{(x^2+k)^{n-1}} + 2(n-1) \int \left(\frac{1}{(x^2+k)^{n-1}} - \frac{k}{(x^2+k)^n} \right) dx \\ &= \frac{1}{6} \int \frac{dx}{(x+2)(x+3)} - \frac{1}{6} \int \frac{dx}{(x+3)(x+4)}\end{aligned}$$

$$\begin{aligned}&+ \frac{1}{6} \int \frac{dx}{(x+2)(x+3)} - \frac{1}{6} \int \frac{dx}{(x+1)(x+2)} \\ &= \frac{1}{6} \log \left| \left(\frac{x+2}{x+3} \right) \left(\frac{x+4}{x+3} \right) \right| \\ &\quad + \frac{1}{6} \log \left| \left(\frac{x+2}{x+3} \right) \left(\frac{x+2}{x+1} \right) \right| + c\end{aligned}$$

401. We have,

$$\begin{aligned}\int \frac{x-1}{(x+1)(x-2)} dx \\ &= \int \frac{(x+1) - 2}{(x+1)(x-2)} dx \\ &= \int \frac{dx}{(x-2)} - \int \frac{2}{(x+1)(x-2)} dx \\ &= \int \frac{dx}{(x-2)} - 6 \int \left(\frac{1}{x-2} - \frac{1}{x+1} \right) dx \\ &= \log|x-2| - 6 \log|x-2| + 6 \log|x+1| + c \\ &= 6 \log|x+1| - 5 \log|x-2| + c\end{aligned}$$

402. We have,

$$\begin{aligned}\int \frac{2x-1}{(x-1)(x+2)(x-3)} dx \\ \text{Now } \frac{(2x-1)}{(x-1)(x+2)(x-3)} \\ &= \frac{A}{(x-1)} + \frac{B}{(x+2)} + \frac{C}{(x-3)} \\ (2x-1) &= A(x+2)(x-3) \\ &\quad + B(x-1)(x-3) + C(x-1)(x+2)\end{aligned}$$

When $x = 1$, then $A = -1/6$

$x = -2$, then $B = -1/3$

and $x = 3$, then $C = 1/2$

Thus, the given integral reduces to

$$\begin{aligned}-\frac{1}{6} \int \frac{dx}{(x-1)} + \frac{1}{3} \int \frac{dx}{(x+2)} + \frac{1}{2} \int \frac{dx}{(x-3)} \\ &= -\frac{1}{6} \log|x-1| + \frac{1}{3} \log|x+2| + \frac{1}{2} \log|x-3| + c\end{aligned}$$

403. We have,

$$\begin{aligned}\int \frac{x^3}{(x-1)(x-2)} dx \\ &= \int \frac{x^3}{x^2 - 3x + 2} dx\end{aligned}$$

$$\begin{aligned}
&= \int \left(x - 3 + \frac{7x - 6}{x^2 - 3x + 2} \right) dx \\
&= \int \left(x - 3 + \frac{7x - 6}{(x-1)(x-2)} \right) dx \\
&= \frac{x^2}{2} - 3x + \int \left(\frac{7x - 6}{(x-1)(x-2)} \right) dx
\end{aligned}$$

Now, $\frac{7x - 6}{(x-1)(x-2)} = \frac{A}{(x-1)} + \frac{B}{(x-2)}$

$$(7x - 6) = A(x - 2) + B(x - 1)$$

When $x = 1$, then $A = -1$

and $x = 2$, then $B = 8$

Thus, the given integral reduces to

$$\begin{aligned}
&\left(\frac{x^2}{2} - 3x \right) + \int \left(\frac{-1}{(x-1)} + \frac{8}{(x-2)} \right) dx \\
&= \left(\frac{x^2}{2} - 3x \right) - \log|(x-1)| + 8 \log|x-2| + c
\end{aligned}$$

404. We have,

$$\int \frac{2x}{(x^2 + 1)(x^2 + 2)} dx$$

$$\begin{aligned}
\text{Let } x^2 &= t \\
\Rightarrow 2x dx &= dt
\end{aligned}$$

Thus, the given integral reduces to

$$\begin{aligned}
&2 \int \frac{dt}{(t+1)(t+2)} \\
&= 2 \int \left(\frac{1}{t+1} - \frac{1}{t+2} \right) dt \\
&= 2 \log \left| \frac{(t+1)}{(t+2)} \right| + c \\
&= 2 \log \left| \frac{(x^2 + 1)}{(x^2 + 2)} \right| + c
\end{aligned}$$

405. We have,

$$\int \frac{\cos \theta}{(2 + \sin \theta)(3 + \sin \theta)} d\theta$$

$$\begin{aligned}
\text{Let } \sin \theta &= t \\
\Rightarrow \cos \theta d\theta &= dt
\end{aligned}$$

Thus, the given integral reduces to

$$\begin{aligned}
&\int \frac{dt}{(t+2)(t+3)} \\
&= \int \left(\frac{1}{t+2} - \frac{1}{t+3} \right) dt
\end{aligned}$$

$$= \log \left| \frac{t+2}{t+3} \right| + c$$

$$= \log \left| \frac{\sin \theta + 2}{\sin \theta + 3} \right| + c$$

406. We have,

$$\begin{aligned}
&\int \frac{(1 - \cos x)}{\cos x(1 + \cos x)} dx \\
&= \int \frac{(1 + \cos x - 2 \cos x)}{\cos x(1 + \cos x)} dx \\
&= \int \frac{dx}{\cos x} - 2 \int \frac{dx}{(1 + \cos x)} \\
&= \int \sec x dx - 2 \int \left(\frac{1 - \cos x}{\sin^2 x} \right) dx \\
&= \int \sec x dx - 2 \int (1 - \operatorname{cosec}^2 x - \operatorname{cosec} x \cos x) dx \\
&= \log |\sec x + \tan x| - 2(\operatorname{cosec} x - \cos x) + c
\end{aligned}$$

407. We have,

$$\begin{aligned}
&\int \frac{dx}{\sin x - \sin 2x} \\
&= \int \frac{dx}{\sin x - 2 \sin x \cos x} \\
&= \int \frac{dx}{\sin x(1 - 2 \cos x)} \\
&= \int \frac{\sin x dx}{\sin^2 x(1 - 2 \cos x)} \\
&= \int \frac{\sin x dx}{(1 - \cos^2 x)(1 - 2 \cos x)} \\
&= \int \frac{dt}{(1 - t^2)(1 - 2t)}, \text{ where } \cos x = t \\
&= - \int \frac{dt}{(t-1)(t+1)(2t-1)}
\end{aligned}$$

Now, $\frac{1}{(t-1)(t+1)(2t-1)}$

$$\frac{A}{(t-1)} + \frac{B}{(t+1)} + \frac{C}{(2t-1)}$$

$$-1 = A(t+1)(2t-1) + B(t-1)(2t-1) + C(t^2-1)$$

Put $t = 1$, then $A = -1/2$

$t = -1$, then $B = 1/10$

$t = 1/2$, then $C = 4/3$

The given integral reduces to

$$-\frac{1}{2} \int \frac{dt}{(t-1)} + \frac{1}{10} \int \frac{dt}{t+1} + \frac{4}{3} \int \frac{dt}{(2t-1)}$$

$$\begin{aligned}
&= -\frac{1}{2} \log|t-1| + \frac{1}{10} \log|t+1| \\
&\quad + \frac{4}{3} \log|2t-1| + c \\
&= -\frac{1}{2} \log|\cos x-1| + \frac{1}{10} \log|\cos x+1| \\
&\quad + \frac{4}{3} \log|2\cos x-1| + c
\end{aligned}$$

408. We have,

$$\int \frac{dx}{(x+1)(x+1)^2}$$

$$\text{Now, } \frac{1}{(x+1)(x-1)^2} = \frac{A}{(x+1)} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2}$$

$$\begin{aligned}
\Rightarrow \frac{1}{(x+1)(x-1)^2} &= \frac{A(x-1)^2 + B(x^2-1) + C(x-1)}{(x+1)(x-1)^2} \\
\Rightarrow 1 &= A(x-1)^2 + B(x^2-1) + C(x+1)
\end{aligned}$$

$$\text{When } x = -1, A = 1/4$$

$$x = 1, C = 1/2$$

$$\text{and } x = 0, A - B + C = 1$$

$$\Rightarrow B = 3/4 - 1 = -1/4$$

$$\text{Thus, } \int \frac{dx}{(x+1)(x+1)^2}$$

$$\begin{aligned}
&= \frac{1}{4} \int \frac{dx}{(x+1)} - \frac{1}{4} \int \frac{dx}{(x-1)} + \frac{1}{2} \int \frac{dx}{(x-1)^2} \\
&= \frac{1}{4} \log|x+1| - \frac{1}{4} \log|x-1| - \frac{1}{2(x-1)} + c
\end{aligned}$$

409. We have,

$$\int \frac{2x+1}{(x+2)(x-3)^2} dx$$

$$\begin{aligned}
&\frac{2x+1}{(x+2)(x-3)^2} \\
&= \frac{A}{(x+2)} + \frac{B}{(x-3)} + \frac{C}{(x-3)^2}
\end{aligned}$$

$$\begin{aligned}
&\frac{2x+1}{(x+2)(x-3)^2} \\
&= \frac{A(x-3)^2 + B(x^2-9) + C(x+2)}{(x+2)(x-3)^2}
\end{aligned}$$

$$\Rightarrow 2x+1 = A(x-3)^2 + B(x^2-9) + C(x+2)$$

$$\text{When } x = -2, A = -3$$

$$\text{Put } x = 3, C = 7/5$$

$$\text{and } x = 0, 9A - 6B + 2C = 1$$

$$\Rightarrow 6B = 14/5 - 27 - 1$$

$$\Rightarrow B = -21/5$$

$$\text{Thus, } \frac{2x+1}{(x+2)(x-3)^2} dx$$

$$\begin{aligned}
&= -3 \int \frac{dx}{x+2} - \frac{21}{5} \int \frac{dx}{x-3} + \frac{7}{5} \int \frac{dx}{(x-3)^2} \\
&= -3 \log|x+2| - \frac{21}{5} \log|x-3| - \frac{7}{5(x-3)} + c
\end{aligned}$$

415. We have,

$$\int \frac{(2x+3)dx}{(x+1)(x^2+4)}$$

$$\text{Now, } \frac{(2x+3)}{(x+1)(x^2+4)} = \frac{A}{(x+1)} + \frac{Bx+C}{(x^2+4)}$$

$$\Rightarrow 2x+3 = A(x^2+4) + Bx(x+1) + C(x+1)$$

$$\text{When } x = -1, A = 1/5$$

$$\text{and } x = 0, 4A + C = 3$$

$$\Rightarrow C = 3 - 4A = 3 - 4/5 = 11/5$$

$$\text{Also } x = 1, 5A + 2B + 2C = 5$$

$$\Rightarrow 2B = 5 - 5A - 2C = 5 - 1 - 22/5 = -2/5$$

$$\Rightarrow B = -1/5$$

$$\text{Thus, } \int \frac{(2x+3)dx}{(x+1)(x^2+4)}$$

$$\begin{aligned}
&= \frac{1}{5} \int \frac{dx}{x+1} - \frac{1}{5} \int \frac{xdx}{x^2+4} + \frac{11}{5} \int \frac{dx}{x^2+4} \\
&= \frac{1}{5} \log|x+1| - \frac{1}{10} \log|x^2+4| + \frac{1}{5} \tan^{-1}(x) + c
\end{aligned}$$

416. We have,

$$\int \frac{(3x-2)dx}{(x-1)(x^2+9)}$$

$$\text{Now, } \frac{(3x-2)}{(x-1)(x^2+9)} = \frac{A}{(x-1)} + \frac{Bx+C}{(x^2+9)}$$

$$\Rightarrow (3x-2) = A(x^2+9) + Bx(x-1) + C(x-1)$$

$$\text{When } x = 1, A = 1/10$$

$$\text{and } x = 0, 9A - C = -2$$

$$\Rightarrow C = 2 + 9A = 2 + 9/10 = 29/10$$

Also $x = -1, 10A + 2B - 2C = -5$

$$\Rightarrow 2B = 2C - 10A - 5 = 29/5 \cdot -1 - 5$$

$$\Rightarrow 2B = -1/5$$

$$\Rightarrow B = -1/10$$

Thus,
$$\int \frac{(3x-2)dx}{(x-1)(x^2+9)}$$

$$= \frac{1}{10} \int \frac{dx}{x-1} - \frac{1}{10} \int \frac{xdx}{x^2+9} + \frac{29}{10} \int \frac{dx}{x^2+9}$$

$$= \frac{1}{10} \log|x-1| - \frac{1}{20} \log|x^2+9|$$

$$+ \frac{29}{30} \tan^{-1}\left(\frac{x}{3}\right) + c$$

417. We have,

$$\int \frac{2x-1}{(x+1)(x^2+2)} dx$$

Now,
$$\frac{2x-1}{(x+1)(x^2+2)} = \frac{A}{(x+1)} + \frac{Bx+C}{(x^2+2)}$$

$$(2x-1) = A(x^2+2) + Bx(x+1) + C(x+1)$$

When $x = -1$, then $A = -1$

$x = 0$, then $C = 1$

and $x = 1$, then $B = 1$

Thus, the given integral reduces to

$$-\int \frac{dx}{(x+1)} + \int \frac{x}{(x^2+2)} dx + \int \frac{dx}{(x^2+2)}$$

$$= -\log|x+1| + \frac{1}{2} \log|x^2+2| + \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + c$$

418.
$$\int \frac{x}{(x+1)(x^2+4)} dx$$

Now,
$$\frac{x}{(x+1)(x^2+4)} = \frac{A}{(x+1)} + \frac{Bx+C}{(x^2+4)}$$

$$x = A(x^2+4) + Bx(x+1) + C(x+1)$$

Let $x = -1$, then $A = -1/5$

$x = 0$, then $C = 4/5$

and $x = 1$, then $B = 1/5$

Thus, the given integral reduces to

$$-\frac{1}{5} \int \frac{dx}{(x+1)} + \frac{1}{5} \int \frac{x}{x^2+4} + \frac{4}{5} \int \frac{dx}{x^2+4}$$

$$= -\frac{1}{5} \log|(x+1)| + \frac{1}{5} \log|x^2+4| + \frac{2}{5} \tan^{-1}\left(\frac{x}{2}\right) + c$$

419. Do yourself

420. Do yourself

421. We have,

$$\int \frac{x^2}{(x^2-1)(x^2+1)} dx$$

Put $x^2 = t$

Now,
$$\frac{t}{(t-1)(t+1)} = \frac{A}{(t-1)} + \frac{B}{(t+1)}$$

$$\Rightarrow \frac{t}{(t-1)(t+1)} = \frac{A(t+1) + B(t-1)}{(t-1)(t+1)}$$

$$\Rightarrow t = A(t+1) + B(t-1)$$

Put $t = 1, A = 1/2$

Put $t = -1, A = -1/2$

Thus,
$$\int \frac{x^2}{(x^2-1)(x^2+1)} dx$$

$$= \frac{1}{2} \int \frac{dx}{x^2-1} - \frac{1}{2} \int \frac{dx}{x^2+1}$$

$$= \frac{1}{4} \log \left| \frac{x-1}{x+1} \right| - \frac{1}{2} \tan^{-1}(x) + c$$

426. We have,

$$\int \frac{(x^2+3)(x^2+1)}{(x^2-1)(x^2+2)} dx$$

Put $x^2 = t$

Thus,
$$\frac{(t+3)(t+1)}{(t-1)(t+2)} = \frac{t^2+4t+c}{t^2+t-2}$$

$$= \left(1 + \frac{-3t+5}{t^2+t-2}\right) = \left(1 + \frac{-3t+5}{(t+1)(t-2)}\right)$$

Now,
$$\frac{-3t+5}{(t+1)(t-2)} = \frac{A}{t-1} + \frac{B}{t-2}$$

$$\Rightarrow (-3t+5) = A(t-2) + B(t-1)$$

When $t = 1$, then $A = -2$

and $t = 2$, then $B = -1$

Therefore, the given integral reduces to

$$\int dx - 2 \int \frac{dx}{x^2-1} - \int \frac{dx}{x^2-2}$$

$$= x - \log \left| \frac{x-1}{x+1} \right| - \frac{1}{2\sqrt{2}} \log \left| \frac{x-\sqrt{2}}{x+\sqrt{2}} \right| + c$$

427. We have,

$$\int \frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} dx$$

Put $x^2 = t$

Now,
$$\frac{(t+1)(t+2)}{(t+3)(t+4)} = \frac{t^2+3t+2}{t^2+7t+12}$$